$$\frac{\Gamma(x) = \tau}{\Gamma \vdash x : \tau}\ [\text{U-Var}] \qquad \frac{}{\Gamma \vdash () : \mathrm{unit}}\ [\text{U-Unit}] \qquad \frac{\Gamma \vdash t : \tau}{\Gamma \vdash \mathrm{ann}_\ell(t) : \tau}\ [\text{U-Ann}]$$

$$\frac{\Gamma \vdash t_1 : \tau_1 \qquad \Gamma \vdash t_2 : \tau_2}{\Gamma \vdash (t_1, t_2) : \tau_1 \times \tau_2}\ [\text{U-Pair}] \qquad \frac{\Gamma \vdash t : \tau_1 \times \tau_2}{\Gamma \vdash \mathrm{proj}_i(t) : \tau_i}\ [\text{U-Proj}]$$

$$\frac{\Gamma \vdash t : \tau_1}{\Gamma \vdash \mathrm{inl}_{\tau_2}(t) : \tau_1 + \tau_2}\ [\text{U-Inl}] \qquad \frac{\Gamma \vdash t : \tau_2}{\Gamma \vdash \mathrm{inr}_{\tau_1}(t) : \tau_1 + \tau_2}\ [\text{U-Inr}]$$

$$\frac{\Gamma \vdash t : \tau_1 + \tau_2 \qquad \Gamma, x : \tau_1 \vdash t_1 : \tau \qquad \Gamma, y : \tau_2 \vdash t_2 : \tau}{\Gamma \vdash \textbf{case}\ t\ \textbf{of}\ \{\mathrm{inl}(x) \to t_1; \mathrm{inr}(y) \to t_2\} : \tau}\ [\text{U-Case}]$$

$$\frac{\Gamma \vdash t_1 : \tau_1 \to \tau_2 \qquad \Gamma \vdash t_2 : \tau_1}{\Gamma \vdash t_1\ t_2 : \tau_2}\ [\text{U-App}] \qquad \frac{\Gamma, x : \tau_1 \vdash t : \tau_2}{\Gamma \vdash \lambda x : \tau_1.t : \tau_1 \to \tau_2}\ [\text{U-Abs}]$$

$$\frac{\Gamma, x : \tau \vdash t : \tau}{\Gamma \vdash \mu x : \tau.t : \tau}\ [\text{U-Fix}] \qquad \frac{\Gamma \vdash t_1 : \tau_1 \qquad \Gamma \vdash t_2 : \tau_2}{\Gamma \vdash \mathrm{seq}\ t_1\ t_2 : \tau_2}\ [\text{U-Seq}]$$

Fig. 13: Underlying type system ($\Gamma \vdash_{\mathrm{t}} t : \tau$)

A Addendum Higher-Ranked Annotation Polymorphic Dependency Analysis

In this document we provide proofs of all theorems in the paper

Higher-Ranked Annotation Polymorphic Dependency Analysis

by

Fabian Thorand and Jurriaan Hage

that was accepted to ESOP 2020, Dublin.

It includes additional definitions, examples and all the necessary auxiliary lemma's, some of which are also listed (without proof) in the published paper. We have ensured that numbering of results is consistent with the paper.

We now provide the underlying type system of the source language in figure 13. It is unsurprising, but will sometimes be referred to in the upcoming lemma's and theorems.

In the proofs the terminology can be slightly different from that of the paper. In particular, here we may use the term "effect" to refer to dependency terms, the term "type and effect" as synonymous to "annotated type and depencency", and "type and effect environment" as synonymous to "annotated type environment".

B Proofs of Section 3

The complete set of sorting rules for $\lambda^{\sqcup}$ is given in figure 14.

Additionally, the well-sorted judgment provides some insight into the free variables of dependency terms.

$$\frac{\Sigma(\beta) = \kappa}{\Sigma \vdash_{\mathrm{s}} \beta : \kappa}\ [\text{S-Var}] \qquad \frac{\ell \in \mathcal{L}}{\Sigma \vdash_{\mathrm{s}} \ell : \star}\ [\text{S-Lat}]$$

$$\frac{\Sigma \vdash_{\mathrm{s}} \xi_1 : \kappa \qquad \Sigma \vdash_{\mathrm{s}} \xi_2 : \kappa}{\Sigma \vdash_{\mathrm{s}} \xi_1 \sqcup \xi_2 : \kappa}\ [\text{S-Join}]$$

$$\frac{\Sigma, \beta : \kappa_1 \vdash_{\mathrm{s}} \xi : \kappa_2}{\Sigma \vdash_{\mathrm{s}} \lambda\beta :: \kappa_1 . \xi : \kappa_1 \Rightarrow \kappa_2}\ [\text{S-Abs}] \qquad \frac{\Sigma \vdash_{\mathrm{s}} \xi_1 : \kappa_1 \Rightarrow \kappa_2 \qquad \Sigma \vdash_{\mathrm{s}} \xi_2 : \kappa_1}{\Sigma \vdash_{\mathrm{s}} \xi_1\ \xi_2 : \kappa_2}\ [\text{S-App}]$$

Fig. 14: $\lambda^{\sqcup}$-calculus: sorting rules ($\Sigma \vdash_{\mathrm{s}} \xi : \kappa$)

Lemma 1. *If we have $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$, then* $\mathrm{fav}(\xi) \subseteq \mathrm{dom}(\Sigma)$.

Proof. By induction on $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$.

[S-Var] We have $\xi = \beta$ for some variable β and $\Sigma(\beta) = \kappa$. Hence, $\beta \in \mathrm{dom}(\Sigma)$ and $\mathrm{fav}(\xi) = \{\beta\} \subseteq \mathrm{dom}(\Sigma)$.

[S-Abs] We have $\xi = \lambda\beta_1 :: \kappa_1.\xi'$ and $\kappa = \kappa_1 \Rightarrow \kappa_2$ for some β_1, κ_1, κ_2 and ξ' such that $\Sigma, \beta_1 :: \kappa_1 \vdash_{\mathrm{s}} \xi' : \kappa_2$ holds. By induction, $\mathrm{fav}(\xi') \subseteq \mathrm{dom}(\Sigma, \beta_1 :: \kappa_1)$. But then, $\mathrm{fav}(\xi) = \mathrm{fav}(\xi') \setminus \{\beta_1\} \subseteq \mathrm{dom}(\Sigma)$.

[S-App] We have $\xi = \xi_1\ \xi_2$ such that $\Sigma \vdash_{\mathrm{s}} \xi_1 : \kappa_2 \Rightarrow \kappa$ and $\Sigma \vdash_{\mathrm{s}} \xi_2 : \kappa_2$ hold for some κ_2. By induction, $\mathrm{fav}(\xi_1) \subseteq \mathrm{dom}(\Sigma)$ and $\mathrm{fav}(\xi_2) \subseteq \mathrm{dom}(\Sigma)$. Thus, $\mathrm{fav}(\xi) = \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2) \subseteq \mathrm{dom}(\Sigma)$.

[S-Join] Analogous to the previous case.

[S-Lat] We have $\xi = \ell$ for some $\ell \in \mathcal{L}$. Hence, $\mathrm{fav}(\xi) = \emptyset \subseteq \mathrm{dom}(\Sigma)$.

The following lemma allows us to modify unrelated bindings in the sort environment without harming well-sortedness.

Lemma 2. *Let ξ be a $\lambda^{\sqcup}$-term of sort κ and let Σ denote a sort environment such that $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$. Then, for all Σ' such that for all $\beta \in \mathrm{fav}(\xi)$ we have $\Sigma(\beta) = \Sigma'(\beta)$, $\Sigma' \vdash_{\mathrm{s}} \xi : \kappa$ also holds.*

Proof. By induction on the derivation tree of $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$.

[S-Var] By definition of the rule, $\xi = \beta'$ for some variable β'. The premise is $\Sigma(\beta') = \kappa$. By assumption $\Sigma'(\beta') = \Sigma(\beta')$ since $\beta' \in \mathrm{fav}(\beta')$. Therefore, $\Sigma' \vdash_{\mathrm{s}} \beta' : \kappa$ holds by [S-Var].

[S-Abs] By definition of the rule, $\xi = \lambda\beta' :: \kappa_1.\xi'$ for some β', κ_1 and ξ' and further $\kappa = \kappa_1 \Rightarrow \kappa_2$ for some κ_2. The premise is $\Sigma, \beta' :: \kappa_1 \vdash_{\mathrm{s}} \xi' : \kappa_2$.
By definition, we have $\mathrm{fav}(\xi) = \mathrm{fav}(\xi') \setminus \{\beta'\}$. Therefore, $(\Sigma', \beta' :: \kappa_1)(\beta) = (\Sigma, \beta' :: \kappa_1)(\beta)$ for all $\beta \in \mathrm{fav}(\xi')$. This means we can apply the induction hypothesis in order to get $\Sigma', \beta' :: \kappa_1 \vdash_{\mathrm{s}} \xi' : \kappa_2$. By [S-Abs], we have $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$.

[S-App] By definition of the rule, $\xi = \xi_1\ \xi_2$. The premises are $\Sigma \vdash_{\mathrm{s}} \xi_1 : \kappa' \Rightarrow \kappa$ and $\Sigma \vdash_{\mathrm{s}} \xi_2 : \kappa'$ for some κ'.
Since $\mathrm{fav}(\xi_1\ \xi_2) = \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$, we have in particular that $\Sigma(\beta) = \Sigma'(\beta)$ for all $\beta \in \mathrm{fav}(\xi_1)$. Hence we can apply the induction hypothesis and get

$\Sigma' \vdash_s \xi_1 : \kappa' \Rightarrow \kappa$. Analogously, we can conclude $\Sigma' \vdash_s \xi_2 : \kappa'$. Lastly, $\Sigma' \vdash_s \xi_1\ \xi_2 : \kappa$ holds by [S-App].

[S-Join] Can be proven analogously to the previous case by propagating the environments.

[S-Lat] Trivial, because [S-Lat] can be used with any environment.

Lastly, we show that well-sorted substitutions preserve the well-sortedness of dependency terms. A substitution is a map from variables to terms usually denoted by the letter θ. The application of a substitution θ to a term ξ is written $\theta\xi$ and replaces all free variables in ξ that are also in the domain of ξ with the corresponding terms they are mapped to. A concrete substitution replacing the variables $\beta_1, \dots, \beta_n$ with terms $\xi_1, \dots, \xi_n$ is written $[\xi_1/\beta_1, \dots, \xi_n/\beta_n]$.

The following lemma says that well-sorted substitutions preserve the well-sortedness of dependency terms.

Lemma 3. *Let ξ be a dependency term, let κ be a sort, let Σ, Σ' be sort environments such that $\Sigma, \Sigma' \vdash_s \xi : \kappa$ and let θ be a substitution such that $\Sigma \vdash_s \theta(\beta) : \Sigma'(\beta)$ for all $\beta \in \mathrm{dom}(\theta) = \mathrm{dom}(\Sigma')$. Then $\Sigma \vdash_s \theta\xi : \kappa$.*

Proof. By induction on the derivation of $\Sigma \vdash_s \xi : \kappa$.

[S-Var] We have $\xi = \beta$ and $(\Sigma, \Sigma')(\beta) = \kappa$. If $\beta \in \mathrm{dom}(\theta) = \mathrm{dom}(\Sigma')$, we have $\Sigma \vdash_s \theta\xi : \Sigma'(\beta)$ by assumption. If $\beta \notin \mathrm{dom}(\theta) = \mathrm{dom}(\Sigma')$, $\theta\xi = \xi$, and therefore $\Sigma \vdash_s \theta\xi : \kappa$ by [S-Var].

[S-Abs] We have $\xi = \lambda\beta :: \kappa_1.\xi'$, $\kappa = \kappa_1 \Rightarrow \kappa_2$ and the premise is $\Sigma, \Sigma', \beta : \kappa_1 \vdash_s \xi' : \kappa_2$. We can assume without loss of generality that β is distinct from any variables in the contexts Σ and Σ', simply by renaming them. Then, the above statement is equivalent to $\Sigma, \beta : \kappa_1, \Sigma' \vdash_s \xi' : \kappa_2$ by lemma 2 and $\theta(\lambda\beta :: \kappa_1.\xi') = \lambda\beta :: \kappa_1.\theta\xi'$.
Also by lemma 2, $\Sigma, \beta : \kappa_1 \vdash_s \theta(\beta) : \Sigma'(\beta)$ for all $\beta \in \mathrm{dom}(\theta)$. By induction, $\Sigma, \beta : \kappa_1 \vdash_s \theta\xi' : \kappa_2$. Hence, $\Sigma' \vdash_s \theta\xi : \kappa$ by [S-Abs].

The remaining cases simply propagate the substitution.

Given this notion of compatibility between the sort environment and the environment used in the denotational semantics, we can now establish the well-definedness of the semantics:

Lemma 4. *Given a term ξ of sort κ, a sort environment Σ and a* compatible *environment $\rho : \mathbf{AnnVar} \to_{\text{fin}} \bigcup\{V_\kappa \mid \kappa \in \mathbf{AnnSort}\}$ such that $\Sigma \vdash_s \xi : \kappa$ holds, then*

1. *$[\![\xi]\!]_{\_}$ is monotone in the environment ρ, i.e. for all environments ρ_1 such that $\rho \sqsubseteq \rho_1$ (seen as a pointwise extension lattice), $[\![\xi]\!]_\rho \sqsubseteq [\![\xi]\!]_{\rho_1}$.*
2. *$[\![\xi]\!]_\rho$ is well-defined,*
3. *$[\![\xi]\!]_\rho \in V_\kappa$.*

Proof. Proof by induction on the derivation tree of $\Sigma \vdash_s \xi : \kappa$.

[S-Var] By definition of the rule, we have $\xi = \beta$ and $\Sigma(\beta) = \kappa$.

1. Let $\rho \sqsubseteq \rho_1$ be arbitrary. Then $[\![\beta]\!]_\rho = \rho(\beta) \sqsubseteq \rho_1(\beta) = [\![\beta]\!]_{\rho_1}$ by assumption.
2. Since ρ is compatible with Σ, and $\beta \in \mathrm{dom}(\Sigma)$, $[\![\beta]\!]_\rho$ is well-defined.
3. By the same argument $[\![\beta]\!]_\rho \in V_{\Sigma(\beta)}$.

[S-Abs] By definition of the rule, we have $\xi = \lambda\beta :: \kappa_1.\xi'$ and $\kappa = \kappa_1 \Rightarrow \kappa_2$. The premise is $\Sigma, \beta :: \kappa_1 \vdash_\mathrm{s} \xi' : \kappa_2$.

1. Let $\rho \sqsubseteq \rho_1$ be arbitrary, then
$$\begin{aligned}[\![\lambda\beta :: \kappa_1.\xi']\!]_\rho =& \lambda v \in V_{\kappa_2}.\, [\![\xi']\!]_{\rho[\beta\mapsto v]} \\ \sqsubseteq& \lambda v \in V_{\kappa_2}.\, [\![\xi']\!]_{\rho_1[\beta\mapsto v]} = [\![\lambda\beta :: \kappa_1.\xi']\!]_{\rho_1}\end{aligned}$$
2. By induction, $[\![\xi']\!]_{\rho'}$ is well-defined for all ρ' compatible with $\Sigma, \beta :: \kappa_1$. Therefore, $[\![\lambda\beta :: \kappa_1.\xi']\!]_\rho$ is well-defined as well.
3. By induction, $[\![\xi']\!]_{\rho'} \in V_{\kappa_2}$ for all ρ' compatible with $\Sigma, \beta : \kappa_1$. Furthermore, since $[\![\xi']\!]_{\_}$ is monotone, so is the function $\lambda v \in V_{\kappa_1}.\, [\![\xi']\!]_{\rho[\beta\mapsto v]}$. In order to see why, let $x, y \in V_{\kappa_2}$ be arbitrary such that $x \sqsubseteq y$,
$$\begin{aligned}(\lambda v \in V_{\kappa_1}.\, [\![\xi']\!]_{\rho[\beta\mapsto v]})(x) &= [\![\xi']\!]_{\rho[\beta\mapsto x]} \\ &\sqsubseteq [\![\xi']\!]_{\rho[\beta\mapsto y]} = (\lambda v \in V_{\kappa_1}.\, [\![\xi']\!]_{\rho[\beta\mapsto v]})(y).\end{aligned}$$
Consequently, $[\![\lambda\beta :: \kappa_1.\xi']\!]_\rho \in V_{\kappa_1\Rightarrow\kappa_2}$.

[S-App] By definition of the rule, we have $\xi = \xi_1\ \xi_2$. The premises are $\Sigma \vdash_\mathrm{s} \xi_1 : \kappa_1 \Rightarrow \kappa$ and $\Sigma \vdash_\mathrm{s} \xi_2 : \kappa_1$.

1. Let $\rho \sqsubseteq \rho_1$ be arbitrary. By induction, $[\![\xi_1]\!]_\rho$ is a monotone function. Hence,
$$[\![\xi_1\ \xi_2]\!]_\rho = [\![\xi_1]\!]_\rho ([\![\xi_2]\!]_\rho) \sqsubseteq [\![\xi_1]\!]_\rho ([\![\xi_2]\!]_{\rho_1}) \sqsubseteq [\![\xi_1]\!]_{\rho_1} ([\![\xi_2]\!]_{\rho_1}) = [\![\xi_1\ \xi_2]\!]_{\rho_1}.$$
The first and last step hold by definition of the denotational semantics, the first inequality is due to the monotone function, the second inequality due to the pointwise extension of inequality.
2. By induction $[\![\xi_1]\!]_\rho \in V_{\kappa_1\Rightarrow\kappa}$ and $[\![\xi_2]\!]_\rho \in V_{\kappa_1}$, therefore the function application is well-defined.
3. Since by the previous argument $[\![\xi_1]\!]_\rho$ is a function from V_{κ_1} to V_κ, we know $[\![\xi_1\ \xi_2]\!]_\rho \in V_\kappa$.

[S-Join] By definition of the rule, we have $\xi = \xi_1 \sqcup \xi_2$. The premises are $\Sigma \vdash_\mathrm{s} \xi_1 : \kappa$ and $\Sigma \vdash_\mathrm{s} \xi_2 : \kappa$.

1. Let $\rho \sqsubseteq \rho_1$ be arbitrary. Then
$$\begin{aligned}[\![\xi_1 \sqcup \xi_2]\!]_\rho &= [\![\xi_1]\!]_\rho \sqcup [\![\xi_2]\!]_\rho \\ &\sqsubseteq [\![\xi_1]\!]_{\rho_1} \sqcup [\![\xi_2]\!]_{\rho_1} = [\![\xi_1 \sqcup \xi_2]\!]_{\rho_1}.\end{aligned}$$
2. By induction, $[\![\xi_1]\!]_\rho \in V_\kappa$ and $[\![\xi_2]\!]_\rho \in V_\kappa$, therefore the join operation is well-defined.
3. By the previous argument, we have $[\![\xi_1]\!]_\rho \sqcup [\![\xi_2]\!]_\rho \in V_\kappa$.

[S-Lat] By definition of the rule, we have $\xi = \ell$ and $\kappa = \star$.

1. Let $\rho \sqsubseteq \rho_1$ be arbitrary, then

$$[\![\ell]\!]_\rho = \ell \sqsubseteq \ell = [\![\ell]\!]_{\rho_1} .$$

2. This case is trivially well-defined.
3. By definition, $\ell \in V_\star$.

The following lemma shows that computing the denotation of a dependency term after a substitution can always be replaced by simply choosing an appropriate environment when evaluating the original term.

Lemma 5. *For all sort environments Σ, Σ' with disjoint domains, environments ρ compatible with Σ, dependency terms ξ and substitutions $\theta : \mathrm{dom}(\Sigma') \to$* **AnnTm** *such that $\Sigma, \Sigma' \vdash_{\mathrm{s}} \xi : \kappa$ and $\Sigma \vdash_{\mathrm{s}} \theta(\beta') : \Sigma'(\beta')$ for all $\beta' \in \mathrm{dom}(\theta)$, we have $[\![\theta\xi]\!]_\rho = [\![\xi]\!]_{\rho'}$ for the environment ρ' compatible with Σ, Σ' given by*

$$\rho'(\beta) = \begin{cases} \rho(\beta) & \textit{if } \beta \in \mathrm{dom}(\Sigma) \\ [\![\theta(\beta)]\!]_\rho & \textit{otherwise} \end{cases}$$

Proof. By induction on ξ.

$\xi = \beta$ By definition of ρ', $[\![\theta\beta]\!]_\rho = \rho'(\beta) = [\![\beta]\!]_{\rho'}$.
$\xi = \lambda\beta :: \kappa_1.\xi'$ We have

$$[\![\theta\lambda\beta :: \kappa_1.\xi']\!]_\rho = \lambda v \in V_{\kappa_1}.\, [\![\theta\xi']\!]_{\rho[\beta\mapsto v]} = \lambda v \in V_{\kappa_1}.\, [\![\xi']\!]_{\rho''} = [\![\lambda\beta :: \kappa_1.\xi']\!]_{\rho'}$$

The first and third step use the definition of the denotational semantics while the second step follows by induction, giving us a suitable ρ''. We denote by $\rho \upharpoonright_X$ the restriction of the environment to the variables in a set X. Then, we have $\rho' = \rho'' \upharpoonright_{\mathrm{dom}(\rho'')\setminus\{\beta\}}$. Without loss of generality we assume that the variable β bound in the abstraction is fresh.

$\xi = \xi_1\ \xi_2$ We apply the induction hypothesis to both subterms. Since we have the same sort environment and substitution in both cases, the environment ρ' is also the same. We get

$$[\![\theta(\xi_1\ \xi_2)]\!]_\rho = [\![\theta\xi_1]\!]_\rho\,([\![\theta\xi_2]\!]_\rho) = [\![\xi_1]\!]_{\rho'}\,([\![\xi_2]\!]_{\rho'}) = [\![\xi_1\ \xi_2]\!]_{\rho'}$$

$\xi = \xi_1 \sqcup \xi_2$ Similar to the previous case.
$\xi = \ell$ Trivial, as the denotation of a constant is independent of the environment.

It remains to show that the environment ρ' is compatible with Σ, Σ'. Let $\beta \in \mathrm{dom}(\rho')$ be arbitrary. If $\beta \in \mathrm{dom}(\Sigma)$, we have $\rho'(\beta) = \rho(\beta) \in V_{\Sigma(\beta)}$ because ρ is compatible with Σ. If $\beta \in \mathrm{dom}(\Sigma')$, we have $\rho'(\beta) = [\![\theta(\beta)]\!]_\rho \in V_{\Sigma'(\beta)}$ because $\Sigma \vdash_{\mathrm{s}} \theta(\beta) : \Sigma'(\beta)$ and by lemma 4.

We will also need the following fact that subsumption is preserved by certain substitutions.

Lemma 6. *Let Σ and Σ' be sort environments with disjoint domains and let $\theta_1 : \mathrm{dom}(\Sigma') \to \mathbf{AnnTm}$ and $\theta_2 : \mathrm{dom}(\Sigma') \to \mathbf{AnnTm}$ be substitutions such that $\Sigma \vdash_{\mathrm{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$, $\Sigma \vdash_{\mathrm{s}} \theta_1(\beta) : \Sigma'(\beta)$ and $\Sigma \vdash_{\mathrm{s}} \theta_2(\beta) : \Sigma'(\beta)$ for all $\beta \in \mathrm{dom}(\Sigma')$.*

Let ξ_1 and ξ_2 be dependency terms such that $\Sigma, \Sigma' \vdash_{\mathrm{s}} \xi_1 : \kappa$ and $\Sigma, \Sigma' \vdash_{\mathrm{s}} \xi_2 : \kappa$ for some sort κ.

If we have $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$ we also have $\Sigma \vdash_{\mathrm{sub}} \theta_1\xi_1 \sqsubseteq \theta_2\xi_2$.

Proof. We have to show that $[\![\theta_1\xi_1]\!]_\rho \sqsubseteq [\![\theta_2\xi_2]\!]_\rho$ for all environments ρ compatible with Σ.

Applying lemma 5 to ξ_1 and ξ_2 we get two environments ρ_1 and ρ_2 such that $[\![\theta_1\xi_1]\!]_\rho = [\![\xi_1]\!]_{\rho_1}$ and $[\![\theta_2\xi_2]\!]_\rho = [\![\xi_2]\!]_{\rho_2}$.

Moreover, we know $\rho_1(\beta) \sqsubseteq \rho_2(\beta)$ for all $\beta \in \mathrm{dom}(\rho_1) = \mathrm{dom}(\rho_2)$. In order to see why, let $\beta \in \mathrm{dom}(\rho_1)$ be arbitrary. If $\beta \in \mathrm{dom}(\Sigma)$, we have $\rho_1(\beta) = \rho(\beta) = \rho_2(\beta)$. If $\beta \in \mathrm{dom}(\Sigma')$, we have $\rho_1(\beta) = [\![\theta_1(\beta)]\!]_\rho \sqsubseteq [\![\theta_2(\beta)]\!]_\rho = \rho_2(\beta)$ due to the fact that $\Sigma \vdash_{\mathrm{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$.

Now we can derive $[\![\theta_1\xi_1]\!]_\rho = [\![\xi_1]\!]_{\rho_1} \sqsubseteq [\![\xi_1]\!]_{\rho_2} \sqsubseteq [\![\xi_2]\!]_{\rho_2} = [\![\theta_2\xi_2]\!]_\rho$. The first and last step simply use lemma 5. The second step uses the fact that the denotation function is monotone in the environment parameter (by lemma 4). The third step holds by $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$.

We can omit variables that do not occur free in a term from the environment when computing its denotation.

Lemma 7. *Let ξ be a dependency term, and let Σ be a context such that $\Sigma \vdash_{\mathrm{s}} \xi : \kappa$ for some κ. Let ρ be an environment compatible with Σ. Let $X \supseteq \mathrm{fav}(\xi)$ be a set of variables.*

Then, $[\![\xi]\!]_\rho = [\![\xi]\!]_{\rho\restriction_X}$.

Proof. By induction on ξ.

$\xi = \beta$ We have $X \supseteq \mathrm{fav}(\beta) = \{\beta\}$. Clearly, $[\![\beta]\!]_\rho = \rho(\beta) = \rho\restriction_X (\beta) = [\![\beta]\!]_{\rho\restriction_X}$.

$\xi = \lambda\beta :: \kappa.\xi'$ Using the induction hypothesis, we can show

$$\begin{aligned}[\![\lambda\beta :: \kappa.\xi']\!]_\rho &= \lambda v \in V_\kappa.\, [\![\xi']\!]_{\rho[\beta\mapsto v]} = \lambda v \in V_\kappa.\, [\![\xi']\!]_{\rho[\beta\mapsto v]\restriction_{X\cup\{\beta\}}} \\ &= \lambda v \in V_\kappa.\, [\![\xi']\!]_{\rho\restriction_X[\beta\mapsto v]} = [\![\lambda\beta :: \kappa.\xi']\!]_{\rho\restriction_X}.\end{aligned}$$

$\xi = \xi_1\ \xi_2$ Since $X \supseteq \mathrm{fav}(\xi_1\ \xi_2) = \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$, we have

$$[\![\xi_1\ \xi_2]\!]_\rho = [\![\xi_1]\!]_\rho([\![\xi_2]\!]_\rho) = [\![\xi_1]\!]_{\rho\restriction_X}([\![\xi_2]\!]_{\rho\restriction_X}) = [\![\xi_1\ \xi_2]\!]_{\rho\restriction_X}.$$

$\xi = \xi_1 \sqcup \xi_2$ Since $X \supseteq \mathrm{fav}(\xi_1 \sqcup \xi_2) = \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$, we have

$$[\![\xi_1 \sqcup \xi_2]\!]_\rho = [\![\xi_1]\!]_\rho \sqcup [\![\xi_2]\!]_\rho = [\![\xi_1]\!]_{\rho\restriction_X} \sqcup [\![\xi_2]\!]_{\rho\restriction_X} = [\![\xi_1 \sqcup \xi_2]\!]_{\rho\restriction_X}.$$

$\xi = \ell$ Trivial, as $[\![\ell]\!]_\rho = \ell$ regardless of the environment.

Additionally, it is possible to change the context of subsumption statements under certain conditions.

Lemma 8. *Let ξ_1 and ξ_2 be dependency terms and Σ a sort environment such that $\Sigma \vdash_{\mathrm{s}} \xi_1 : \kappa$, $\Sigma \vdash_{\mathrm{s}} \xi_2 : \kappa$ and $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$. Then, $\Sigma' \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$ holds for all Σ' such that for all $\beta \in \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$ we have $\Sigma(\beta) = \Sigma'(\beta)$.*

Proof. In order to prove $\Sigma' \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$, we need to show that $[\![\xi_1]\!]_\rho \sqsubseteq [\![\xi_2]\!]_\rho$ holds for all environments ρ compatible with Σ'.

First, we prove that the assignments of the variables that are not free in a certain term have no effect on its denotation.

Let ρ' be an arbitrary environment compatible with Σ'. We define an environment ρ with domain $\mathrm{dom}(\Sigma)$ by

$$\rho(\beta) = \begin{cases} \rho'(\beta) & \text{if } \beta \in \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2) \\ \bot & \text{otherwise} \end{cases}$$

where $\bot$ denotes the least element of the corresponding lattice $V_{\Sigma(\beta)}$. Clearly, $\rho\restriction_{\mathrm{fav}(\xi_1)} = \rho'\restriction_{\mathrm{fav}(\xi_1)}$ and $\rho\restriction_{\mathrm{fav}(\xi_2)} = \rho'\restriction_{\mathrm{fav}(\xi_2)}$ by definition.
Claim: ρ is compatible with Σ.
Proof: Let $\beta \in \mathrm{dom}(\Sigma)$ be an arbitrary variable. If $\beta \in \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$, then $\rho(\beta) = \rho'(\beta) \in V_{\Sigma'(\beta)}$ by lemma 4. Since $\Sigma(\beta) = \Sigma'(\beta)$, $\rho(\beta) \in V_{\Sigma(\beta)}$.

Otherwise, if $\beta \notin \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\xi_2)$, then we have $\rho(\beta) = \bot \in V_{\Sigma(\beta)}$ by definition. ■

Using lemma 7, we can derive

$$\begin{aligned} [\![\xi_1]\!]_{\rho'} &= [\![\xi_1]\!]_{\rho'\restriction_{\mathrm{fav}(\xi_1)}} = [\![\xi_1]\!]_{\rho\restriction_{\mathrm{fav}(\xi_1)}} = [\![\xi_1]\!]_{\rho} \\ \sqsubseteq [\![\xi_2]\!]_{\rho} &= [\![\xi_2]\!]_{\rho\restriction_{\mathrm{fav}(\xi_2)}} = [\![\xi_2]\!]_{\rho'\restriction_{\mathrm{fav}(\xi_2)}} = [\![\xi_2]\!]_{\rho'} \end{aligned}$$

where the inequality follows from $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$ and the above claim, stating that ρ is compatible with Σ.

C Proofs for Section 4

Well-formedness is defined formally in 15. As the following lemma demonstrates, we can modify the environment in a well-formedness judgment in a similar way to what we have shown for sorting judgments in lemma 2.

Lemma 9. *Let $\widehat{\tau}$ be an annotated type and Σ a sort environment such that $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}$. Then, for all Σ' such that for all $\beta \in \mathrm{fav}(\widehat{\tau})$ we have $\Sigma(\beta) = \Sigma'(\beta)$, $\Sigma' \vdash_{\mathrm{wft}} \widehat{\tau}$ holds.*

Proof. This follows directly by induction using lemma 2 on the well-sortedness judgments.

Lemma 10. *If two target types $\widehat{\tau}_1$ and $\widehat{\tau}_2$ have the same shape, then $\lfloor\widehat{\tau}_1\rfloor = \lfloor\widehat{\tau}_2\rfloor$.*

Proof. By induction on $\widehat{\tau}_1$.

$\widehat{\tau}_1 = \widehat{\mathrm{unit}}$ Then $\widehat{\tau}_2 = \widehat{\mathrm{unit}}$ as well, and clearly $\lfloor\widehat{\tau}_1\rfloor = \lfloor\widehat{\tau}_2\rfloor$.

$$\frac{\Sigma, \beta :: \kappa \vdash_{\text{wft}} \widehat{\tau}}{\Sigma \vdash_{\text{wft}} \forall \beta :: \kappa . \widehat{\tau}} \text{[W-FORALL]}$$

$$\frac{}{\Sigma \vdash_{\text{wft}} \widehat{\text{unit}}} \text{[W-UNIT]}$$

$$\frac{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1 \qquad \Sigma \vdash_{\text{s}} \xi_1 : \star \qquad \Sigma \vdash_{\text{wft}} \widehat{\tau}_2 \qquad \Sigma \vdash_{\text{s}} \xi_2 : \star}{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle} \text{[W-SUM]}$$

$$\frac{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1 \qquad \Sigma \vdash_{\text{s}} \xi_1 : \star \qquad \Sigma \vdash_{\text{wft}} \widehat{\tau}_2 \qquad \Sigma \vdash_{\text{s}} \xi_2 : \star}{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle} \text{[W-PROD]}$$

$$\frac{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1 \qquad \Sigma \vdash_{\text{s}} \xi_1 : \star \qquad \Sigma \vdash_{\text{wft}} \widehat{\tau}_2 \qquad \Sigma \vdash_{\text{s}} \xi_2 : \star}{\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle} \text{[W-ARR]}$$

Fig. 15: Well-formedness of annotated types ($\Sigma \vdash_{\text{wft}} \widehat{\tau}$)

$\widehat{\tau}_1 = \forall \beta\ \kappa\ \widehat{\tau}'_1$ Then $\widehat{\tau}_2 = \forall \beta\ \kappa\ \widehat{\tau}'_2$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape. By induction, $\lfloor \widehat{\tau}'_1 \rfloor = \lfloor \widehat{\tau}'_2 \rfloor$. Therefore, $\lfloor \widehat{\tau}_1 \rfloor = \lfloor \widehat{\tau}_2 \rfloor$.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ Then $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape, as well as $\widehat{\tau}''_1$ and $\widehat{\tau}''_2$. By induction, $\lfloor \widehat{\tau}'_1 \rfloor = \lfloor \widehat{\tau}'_2 \rfloor$ and $\lfloor \widehat{\tau}''_1 \rfloor = \lfloor \widehat{\tau}''_2 \rfloor$. Therefore, $\lfloor \widehat{\tau}_1 \rfloor = \lfloor \widehat{\tau}_2 \rfloor$.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}''_1\langle\xi''_1\rangle$ Analogous to the previous case.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$ Analogous to the previous case.

Similar to lemma 3 for $\lambda^{\sqcup}$, the following lemma shows that substitutions satisfying certain conditions preserve well-formedness of annotated types.

Lemma 11. *Let $\widehat{\tau}$ be an annotated type and let Σ, Σ' be sort environments such that $\Sigma, \Sigma' \vdash_{\text{wft}} \widehat{\tau}$ and let θ be a substitution such that $\Sigma \vdash_{\text{s}} \theta(\beta) : \Sigma'(\beta)$ for all $\beta \in \text{dom}(\theta) = \text{dom}(\Sigma')$. Then $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}$.*

Proof. By induction on the derivation tree of $\Sigma, \Sigma' \vdash_{\text{wft}} \theta$.

[W-FORALL] We have $\widehat{\tau} = \forall\beta{::}\kappa.\widehat{\tau}'$ and the premise $\Sigma, \Sigma', \beta{::}\kappa \vdash_{\text{wft}} \widehat{\tau}'$. Without loss of generality, we can assume that β is distinct from all variables in the environments Σ, Σ' by renaming. We can then reorder the context to $\Sigma, \beta :: \kappa, \Sigma' \vdash_{\text{wft}} \widehat{\tau}'$ by lemma 9 and $\theta\widehat{\tau} = \forall \beta :: \kappa.\theta\widehat{\tau}'$.
Also by lemma 9, $\Sigma, \beta :: \kappa \vdash_{\text{s}} \theta(\beta') : \Sigma'(\beta')$ for all $\beta' \in \text{dom}(\theta) = \text{dom}(\Sigma')$.
By induction, $\Sigma, \beta :: \kappa \vdash_{\text{wft}} \theta\widehat{\tau}'$. Hence, $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}$ by [W-FORALL].

[W-UNIT] Trivial, since $\theta\widehat{\text{unit}} = \widehat{\text{unit}}$.

[W-SUM] We have $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ such that $\Sigma, \Sigma' \vdash_{\text{wft}} \widehat{\tau}_1$, $\Sigma, \Sigma' \vdash_{\text{wft}} \widehat{\tau}_2$, $\Sigma, \Sigma' \vdash_{\text{s}} \xi_1 : \star$ and $\Sigma, \Sigma' \vdash_{\text{s}} \xi_2 : \star$ hold.
By lemma 3, $\Sigma \vdash_{\text{s}} \theta\xi_1 : \star$ and $\Sigma \vdash_{\text{s}} \theta\xi_2 : \star$. By induction, $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}_1$ and $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}_2$. Thus, we can derive $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ by [W-SUM].

The cases for [W-PROD] and [W-ARR] can be proven analogously.

To define subtyping we need the following auxiliary relation: two annotated types $\widehat{\tau}_1$ and $\widehat{\tau}_2$ *have the same shape* if

1. $\widehat{\tau}_1 = \widehat{\text{unit}}$ and $\widehat{\tau}_2 = \widehat{\text{unit}}$, or
2. $\widehat{\tau}_1 = \forall\beta\ \kappa\ \widehat{\tau}'_1$, $\widehat{\tau}_2 = \forall\beta\ \kappa\ \widehat{\tau}'_2$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape, or
3. $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$, $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape and $\widehat{\tau}''_1$ and $\widehat{\tau}''_2$ have the same shape, or
4. $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}''_1\langle\xi''_1\rangle$, $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle \times \widehat{\tau}''_2\langle\xi''_2\rangle$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape and $\widehat{\tau}''_1$ and $\widehat{\tau}''_2$ have the same shape, or
5. $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$, $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle \to \widehat{\tau}''_2\langle\xi''_2\rangle$ and $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape and $\widehat{\tau}''_1$ and $\widehat{\tau}''_2$ have the same shape.

The following lemma shows that the subtyping relation implies that the two involved types have the same shape.

Lemma 12. *Let Σ be a sort environment, and let $\widehat{\tau}_1$ and $\widehat{\tau}_2$ be well-formed annotated types. If $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$, then $\widehat{\tau}_1$ and $\widehat{\tau}_2$ have the same shape.*

Proof. By induction on $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$.

[Sub-Refl] Since $\widehat{\tau}_1 = \widehat{\tau}_2$, they clearly have the same shape.

[Sub-Trans] There is a $\widehat{\tau}_3$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_2$ hold. By induction, $\widehat{\tau}_1$ and $\widehat{\tau}_3$ have the same shape, and $\widehat{\tau}_3$ and $\widehat{\tau}_2$ do. Therefore, $\widehat{\tau}_1$ and $\widehat{\tau}_2$ also have the same shape.

[Sub-Forall] We have $\widehat{\tau}_1 = \forall\beta\ \kappa.\widehat{\tau}'_1$ and $\widehat{\tau}_2 = \forall\beta\ \kappa.\widehat{\tau}'_2$ such that $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$ holds. By induction, $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape, thus $\widehat{\tau}_1$ and $\widehat{\tau}_2$ also have the same shape.

[Sub-Sum] We have $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ and $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$ hold. By induction, $\widehat{\tau}'_1$ and $\widehat{\tau}'_2$ have the same shape, as well as $\widehat{\tau}''_1$ and $\widehat{\tau}''_2$. Therefore, $\widehat{\tau}_1$ and $\widehat{\tau}_2$ also have the same shape.

The cases for [Sub-Prod] and [Sub-Arr] can be proven analogously to [Sub-Sum].

The following lemma allows us to draw conclusions about the subtyping relation between parts of two types when there is a subtyping relation between the whole types.

Lemma 13. *1. If we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}'_2\langle\xi'_2\rangle$, then we also have $\Sigma \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}_1$, $\Sigma \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi_1$, $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}'_2$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi'_2$.*
2. If we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle$, then we also have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}'_1$, $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi'_1$, $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}'_2$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi'_2$.
3. If we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}'_2\langle\xi'_2\rangle$, then we also have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}'_1$, $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi'_1$, $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}'_2$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi'_2$.
4. If we have $\Sigma \vdash_{\text{sub}} \forall\beta :: \kappa.\widehat{\tau}_1 \leqslant \forall\beta :: \kappa.\widehat{\tau}_2$, then we also have $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$.

Proof. We prove the first statement by induction on the derivation of $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}'_2\langle\xi'_2\rangle$.

[Sub-Refl] We have $\widehat{\tau}_1 = \widehat{\tau}'_1$, $\widehat{\tau}_2 = \widehat{\tau}'_2$, $\xi_1 = \xi'_1$ and $\xi_2 = \xi'_2$. The conclusions hold by reflexivity.

[Sub-Trans] There are $\widehat{\tau}$, $\widehat{\tau}'$, ξ, ξ' such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}\langle\xi\rangle \to \widehat{\tau}'\langle\xi'\rangle$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}\langle\xi\rangle \to \widehat{\tau}'\langle\xi'\rangle \leqslant \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}'_2\langle\xi'_2\rangle$. The conclusions follow by induction and transitivity.

[Sub-Arr] The conclusions follow directly from the premises of the rule at hand.

All other cases can be ruled out because they do not apply to arrow types.

The remaining statements about sum types, product types and quantification follow analogously.

Similar to the previous kinds of judgments, we can reorder the environment of subtyping judgments under certain conditions.

Lemma 14. *Let $\widehat{\tau}_1$ and $\widehat{\tau}_2$ be annotated types and Σ a sort environment such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$. Then, for all Σ' such that for all $\beta \in \text{fav}(\widehat{\tau}_1 \cup \widehat{\tau}_2)$ we have $\Sigma(\beta) = \Sigma'(\beta)$, $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ holds.*

Proof. By induction on $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$.

[Sub-Refl] We have $\widehat{\tau}_1 = \widehat{\tau}_2$. Since we implicitly assume the well-formedness of this type, we also have $\Sigma' \vdash_{\text{wft}} \widehat{\tau}_1$ by lemma 9. Thus, we can derive $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ by [Sub-Refl].

[Sub-Trans] There is a $\widehat{\tau}_3$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_2$ hold. By induction, $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$ and $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_2$. Therefore, $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ holds by [Sub-Trans].

[Sub-Forall] We have $\widehat{\tau}_1 = \forall\beta :: \kappa.\widehat{\tau}'_1$ and $\widehat{\tau}_2 = \forall\beta :: \kappa.\widehat{\tau}'_2$ such that $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$ holds.
By definition, $\text{fav}(\widehat{\tau}'_1) \subseteq \text{fav}(\widehat{\tau}_1) \cup \{\beta\}$ and $\text{fav}(\widehat{\tau}'_2) \subseteq \text{fav}(\widehat{\tau}_2) \cup \{\beta\}$. Since additionally, $(\Sigma, \beta :: \kappa)(\beta) = \kappa = (\Sigma', \beta :: \kappa)(\beta)$, we can apply the induction hypothesis. This results in $\Sigma', \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$, from which we can derive $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ by [Sub-Forall].

[Sub-Sum] We have $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ and $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$, $\Sigma \vdash_{\text{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$, $\Sigma \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi'_2$ and $\Sigma \vdash_{\text{sub}} \xi''_1 \sqsubseteq \xi''_2$ hold.
By induction, $\Sigma' \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$ and $\Sigma' \vdash_{\text{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$. By lemma 8, $\Sigma' \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi'_2$ and $\Sigma' \vdash_{\text{sub}} \xi''_1 \sqsubseteq \xi''_2$. In conclusion, we have $\Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ by [Sub-Sum].

[Sub-Prod] Analogous to the previous case.

[Sub-Arr] Analogous to the previous cases, except taking into account the contravariance of the argument types and effects.

We also show that substitutions preserve the subtyping relation of annotated types.

Lemma 15. *Let $\widehat{\tau}_1$, $\widehat{\tau}_2$ be annotated types, let Σ and Σ' be sort environments and let $\theta : \text{dom}(\Sigma') \to$ **AnnTm** be a substitution.*

If we have $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$, then we also have $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_2$.

Proof. By induction on the derivation of $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$.

[Sub-Refl] We have $\widehat{\tau}_1 = \widehat{\tau}_2$ and the type $\theta\widehat{\tau}_1$ is well-formed under Σ by lemma 11 (because $\widehat{\tau}_1$ is implicitly assumed to be well-formed), hence we have $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_2$ by [Sub-Refl].

[Sub-Trans] There is some $\widehat{\tau}_3$ such that $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$ and $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_2$ hold. By induction, we get $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_3$ and $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_3 \leqslant \theta\widehat{\tau}_2$. Thus, $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_2$ can be derived using transitivity.

[Sub-Forall] We have $\widehat{\tau}_1 = \forall\beta :: \kappa.\widehat{\tau}_1'$ and $\widehat{\tau}_2 = \forall\beta :: \kappa.\widehat{\tau}_2'$ such that $\Sigma, \Sigma', \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$ holds.
We assume without loss of generality that β does not occur in the environment Σ' by appropriate renaming.
By lemma 14, we have $\Sigma, \beta :: \kappa, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$. Applying the induction hypothesis results in $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \theta\widehat{\tau}_1' \leqslant \theta\widehat{\tau}_2'$. We can derive $\Sigma \vdash_{\text{sub}} \forall\beta :: \kappa.\theta\widehat{\tau}_1' \leqslant \forall\beta :: \kappa.\theta\widehat{\tau}_2'$ by [Sub-Forall]. By assumption, β is not in the domain of θ, hence $\theta\forall\beta :: \kappa.\widehat{\tau}_1' = \forall\beta :: \kappa.\theta\widehat{\tau}_1'$ and similarly $\theta\forall\beta :: \kappa.\widehat{\tau}_2' = \forall\beta :: \kappa.\theta\widehat{\tau}_2'$.
This completes the proof.

[Sub-Sum] We have $\widehat{\tau}_1 = \widehat{\tau}_1'\langle\xi_1'\rangle + \widehat{\tau}_1''\langle\xi_1''\rangle$ and $\widehat{\tau}_2 = \widehat{\tau}_2'\langle\xi_2'\rangle + \widehat{\tau}_2''\langle\xi_2''\rangle$ such that $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$, $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1'' \leqslant \widehat{\tau}_2''$, $\Sigma \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_2'$ and $\Sigma \vdash_{\text{sub}} \xi_1'' \sqsubseteq \xi_2''$ hold.
Applying lemma 6 using θ as both the left and the right substitution results in $\Sigma \vdash_{\text{sub}} \theta\xi_1' \sqsubseteq \theta\xi_2'$ as well as $\Sigma \vdash_{\text{sub}} \theta\xi_1'' \sqsubseteq \theta\xi_2''$. By induction, we get $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1' \leqslant \theta\widehat{\tau}_2'$ and $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1'' \leqslant \theta\widehat{\tau}_2''$.
This lets us derive $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_2$ using [Sub-Sum].

[Sub-Prod] Analogous to the previous case.

[Sub-Arr] We have $\widehat{\tau}_1 = \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_1''\langle\xi_1''\rangle$ and $\widehat{\tau}_2 = \widehat{\tau}_2'\langle\xi_2'\rangle \to \widehat{\tau}_2''\langle\xi_2''\rangle)$ such that $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_2' \leqslant \widehat{\tau}_1'$, $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1'' \leqslant \widehat{\tau}_2''$, $\Sigma \vdash_{\text{sub}} \xi_2' \sqsubseteq \xi_1'$ and $\Sigma \vdash_{\text{sub}} \xi_1'' \sqsubseteq \xi_2''$ hold.
Applying lemma 6 using θ as both the left and the right substitution results in $\Sigma \vdash_{\text{sub}} \theta\xi_2' \sqsubseteq \theta\xi_1'$ as well as $\Sigma \vdash_{\text{sub}} \theta\xi_1'' \sqsubseteq \theta\xi_2''$. By induction, we get $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_2' \leqslant \theta\widehat{\tau}_1'$ and $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1'' \leqslant \theta\widehat{\tau}_2''$.
This lets us derive $\Sigma \vdash_{\text{sub}} \theta\widehat{\tau}_1 \leqslant \theta\widehat{\tau}_2$ using [Sub-Arr].

Lemma 16. *If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t} : \widehat{\tau} \mathbin{\&} \xi$ then we also have $\lfloor\widehat{\Gamma}\rfloor \vdash_{\text{t}} \lfloor\widehat{t}\rfloor : \lfloor\widehat{\tau}\rfloor$.*

Proof. By induction on the derivation of $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t} : \widehat{\tau} \mathbin{\&} \xi$.

[T-Var] We have $\widehat{\Gamma}(x) = \widehat{\tau} \& \xi$. By definition $\lfloor\widehat{\Gamma}\rfloor(x) = \lfloor\widehat{\tau}\rfloor$ and $\lfloor x\rfloor = x$. Hence, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\text{t}} x : \lfloor\widehat{\tau}\rfloor$ holds by [U-Var].

[T-Unit] Since $\lfloor\widehat{\text{unit}}\rfloor = \text{unit}$ and $\lfloor()\rfloor = ()$, we have $\lfloor\widehat{\Gamma}\rfloor \vdash_{\text{t}} () : \text{unit}$ by [U-Unit].

[T-Sub] There are $\widehat{\tau}'$ and ξ' such that $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t} : \widehat{\tau}' \mathbin{\&} \xi'$, $\Sigma \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau}$ and $\Sigma \vdash_{\text{sub}} \xi' \sqsubseteq \xi$. By induction, we have $\lfloor\widehat{\Gamma}\rfloor \vdash_{\text{t}} \lfloor\widehat{t}\rfloor : \lfloor\widehat{\tau}'\rfloor$ and by lemmas 10 and 12, we have $\lfloor\widehat{\tau}'\rfloor = \lfloor\widehat{\tau}\rfloor$.

[T-AnnAbs] We have $\widehat{t} = \Lambda\beta :: \kappa.\widehat{t'}$ and $\widehat{\tau} = \forall\beta :: \kappa.\widehat{\tau}'$. Moreover, $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}' \mathbin{\&} \xi$ holds. By induction, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\text{t}} \lfloor\widehat{t'}\rfloor : \lfloor\widehat{\tau}'\rfloor$ holds. Furthermore, $\lfloor\widehat{\tau}\rfloor = \lfloor\forall\beta :: \kappa.\widehat{\tau}'\rfloor = \lfloor\widehat{\tau}'\rfloor$ and $\lfloor\widehat{t}\rfloor = \lfloor\Lambda\beta :: \kappa.\widehat{t'}\rfloor = \lfloor\widehat{t'}\rfloor$ hold by definition.

[T-AnnApp] We have $\widehat{t} = \widehat{t'}\ \langle \xi' \rangle$, $\widehat{\tau} = [\xi' \ / \ \beta]\widehat{\tau}'$ and $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t'} : \forall \beta :: \kappa.\widehat{\tau}' \,\&\, \xi$. Since the substitution only affects annotations, we have $\lfloor [\xi' \ / \ \beta]\widehat{\tau}' \rfloor = \lfloor \widehat{\tau}' \rfloor$. By definition, $\lfloor \forall \beta :: \kappa.\widehat{\tau}' \rfloor = \lfloor \widehat{\tau}' \rfloor$ and $\lfloor \widehat{t'}\ \langle \xi' \rangle \rfloor = \lfloor \widehat{t'} \rfloor$. By induction, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\mathrm{t}} \lfloor \widehat{t'} \rfloor : \lfloor \widehat{\tau}' \rfloor$.

The remaining cases follow analogously.

The following lemma is used in various proofs making assumptions about the structure of a typing derivation.

Lemma 17. *If we have a derivation $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \& \xi$, then there is a derivation of the statement that ends with exactly one application of* [T-Sub].

Proof. By induction on $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$.

[T-Sub] We have

$$\frac{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau}' \,\&\, \xi' \qquad \Sigma \vdash_{\mathrm{sub}} \widehat{\tau}' \leqslant \widehat{\tau} \qquad \Sigma \vdash_{\mathrm{sub}} \xi' \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi} \text{[T-Sub]}$$

and by induction, there is a derivation for $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau}' \,\&\, \xi'$ ending with [T-Sub]:

$$\frac{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau}'' \,\&\, \xi'' \qquad \Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'' \leqslant \widehat{\tau}' \qquad \Sigma \vdash_{\mathrm{sub}} \xi'' \sqsubseteq \xi'}{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau}' \,\&\, \xi'} \text{[T-Sub]}$$

But then we can derive

$$\frac{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau}'' \,\&\, \xi'' \qquad \dfrac{\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'' \leqslant \widehat{\tau}' \qquad \Sigma \vdash_{\mathrm{sub}} \widehat{\tau}' \leqslant \widehat{\tau}}{\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'' \leqslant \widehat{\tau}} \qquad \Sigma \vdash_{\mathrm{sub}} \xi'' \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi}$$

by [Sub-Trans] and the transitivity of subsumption.

In all other cases, we can derive

$$\frac{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi \qquad \Sigma \vdash_{\mathrm{sub}} \widehat{\tau} \leqslant \widehat{\tau} \qquad \Sigma \vdash_{\mathrm{sub}} \xi \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi} \text{[T-Sub]}$$

by reflexivity.

the following lemma allows us to deconstruct the typing judgment of an annotated term.

Lemma 18. *If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \mathrm{ann}_{\ell}(t) : \widehat{\tau} \,\&\, \xi$, then we also have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$ and $\Sigma \vdash_{\mathrm{sub}} \ell \sqsubseteq \xi$.*

Proof. By lemma 17, there is a derivation for $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{ann}_\ell(t) : \widehat{\tau} \,\&\, \xi$ of the following form.

$$\dfrac{\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau}' \,\&\, \xi' \qquad \Sigma \vdash_{\text{sub}} \ell \sqsubseteq \xi'}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{ann}_\ell(t) : \widehat{\tau}' \,\&\, \xi'}\,[\text{T-ANN}] \quad \begin{array}{c}\Sigma \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau} \\ \Sigma \vdash_{\text{sub}} \xi' \sqsubseteq \xi\end{array}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{ann}_\ell(t) : \widehat{\tau} \,\&\, \xi}\,[\text{T-SUB}]$$

But then, we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ using rule [T-Sub], and we have $\Sigma \vdash_{\text{sub}} \ell \sqsubseteq \xi$ by transitivity.

The existence of a typing derivation also provides information about the free variables in a target term.

Lemma 19. *If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$, then $\text{ftv}(t) \subseteq \text{dom}(\widehat{\Gamma})$.*

Proof. By induction on the derivation of $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$. In particular, whenever we encounter a variable in t, that variable is also part of the context $\widehat{\Gamma}$. Also, whenever bound variables are removed from the context (as in the [T-Abs] and [T-Case] rules), this is reflected by the definition of ftv.

Similar to the well-sortedness judgments for dependency terms we can show that all variables in the context that are not free in the term have no effect on its typing judgment.

Lemma 20. *Let t be a target term, $\widehat{\tau} \,\&\, \xi$ an annotated type and depencency pair, Σ a sort environment and $\widehat{\Gamma}, \widehat{\Gamma'}$ two type and effect environments.*

If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ and for all $x \in \text{ftv}(t)$, $\widehat{\Gamma}(x) = \widehat{\Gamma'}(x)$, then $\Sigma \mid \widehat{\Gamma'} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$.

Proof. By induction on $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$, similar to the previous reordering proofs.

Unsurprisingly, the same holds for annotations variables.

Lemma 21. *Let t be a target term, $\widehat{\tau} \,\&\, \xi$ a type and effect pair, Σ and Σ' sort environments and $\widehat{\Gamma}$ two type and effect environments.*

If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ and for all $\beta \in \text{fav}(t)$, $\Sigma(x) = \Sigma'(x)$, then $\Sigma' \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$.

Proof. By induction on $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$, again similar to previous proofs for the other judgments.

$$C ::= \text{proj}_i(\Box) \mid \Box\ t_2 \mid \Box\ \langle\xi\rangle \mid \text{ann}_\ell(\Box) \mid \text{seq}\ \Box\ t_2 \mid \textbf{case}\ \Box\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\}$$

Fig. 16: Evaluation contexts

D Proofs of Section 5

We start by defining a small-step operational semantics for the target language based on the call-by-name semantics employed by [16] for exception terms. Sometimes it is necessary to perform reduction on a sub-term, e.g. on the scrutinee of a case expression. Those circumstances are captured by contexts C (see figure 16) corresponding to terms with exactly one hole $\Box$. We denote by $C[t]$ the term obtained by filling the single hole in C with t.

The reduction rules of the small-step semantics are shown in figure 17. It is an extension of the standard rules for a call-by-name evaluation of a functional language, in particular β-reduction is provided by [E-Abs] and deconstruction of data constructors by [E-Proj], [E-CaseInl] and [E-CaseInr]. The rule for fixpoints [E-Fix] unrolls the term one level at a time. Similar to β-reduction for term variables, rule [E-AnnAbs] implements β-reduction for annotation variables.

The [E-Context] rule allows us to evaluate sub-terms where it is required for progressing the overall evaluation, and only there. This means that it is, for example, not possible to perform reduction on the argument of a constructor. The reason is that the semantics is designed to evaluate terms to WHNF, but not further.

Annotation terms $\text{ann}_\ell(t)$ are always preserved by the semantics, in the sense that a term that is enclosed by an annotation always remains enclosed by an annotation at least as large. The rules moving annotations outwards are called *lifting* rules. There is also a rule [E-JoinAnn] which merges two adjacent annotation terms. Since the lattice values are joined, the resulting annotation is never smaller than any of the original ones. All these rules require the annotated term to be a value already in order to make the reduction deterministic.

Theorem 1 (Progress). *If $\emptyset \mid \emptyset \vdash_{\text{te}} t : \widehat{\tau}\ \&\ \xi$, then either $t \in \mathbf{Nf}$ or there is a t' such that $t \to t'$.*

Proof. By induction on annotated terms t.

$t = x$ By lemma 19, $\{x\} = \text{ftv}(x) \subseteq \emptyset$, contradiction. Thus, this case cannot happen.

$t = \lambda x : \widehat{\tau}'\ \&\ \xi'.t_1$ Clearly, $t \in \mathbf{Nf}$.

$t = t_1\ t_2$ We apply the induction hypothesis to t_1 and distinguish two cases. If $t_1 \in \mathbf{Nf}$, then either $t_1 \in \mathbf{Nf}'$ or $t_1 = \text{ann}_\ell(v')$. Suppose the former holds, then $t_1 = \lambda x : \widehat{\tau}' \& \xi'.t_1'$ as this is the only possibility such that t is well-typed. But then, we can apply [E-Abs] and get $(\lambda x : \widehat{\tau}' \& \xi'.t_1')\ t_2 \to [t_2/x]t_1'$. In the latter case, we can apply [E-LiftApp], and get $(\text{ann}_\ell(v'))\ t_2 \to \text{ann}_\ell(v'\ t_2)$.

$$\frac{}{(\lambda x : \widehat{\tau}\,\&\,\xi.t_1)\ t_2 \to [t_2 \,/\, x]t_1}\ [\text{E-ABS}] \qquad \frac{}{(\Lambda\beta :: \kappa.t')\ \langle\xi\rangle \to [\xi \,/\, \beta]t'}\ [\text{E-ANNABS}]$$

$$\frac{}{\mu x : \widehat{\tau}\,\&\,\xi.t \to [\mu x : \widehat{\tau}\,\&\,\xi.t \,/\, x]t}\ [\text{E-FIX}] \qquad \frac{}{\mathrm{proj}_i(t_1, t_2) \to t_i}\ [\text{E-PROJ}]$$

$$\frac{}{\textbf{case}\ \mathrm{inl}_\tau(t')\ \textbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \to [t' \,/\, x]t_2}\ [\text{E-CASEINL}]$$

$$\frac{}{\textbf{case}\ \mathrm{inr}_\tau(t')\ \textbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \to [t' \,/\, y]t_3}\ [\text{E-CASEINR}]$$

$$\frac{v' \in \mathbf{Nf'}}{\mathrm{seq}\ v'\ t_2 \to t_2}\ [\text{E-SEQ}] \qquad \frac{t \to t'}{C[t] \to C[t']}\ [\text{E-CONTEXT}]$$

$$\frac{v' \in \mathbf{Nf'}}{(\mathrm{ann}_\ell(v'))\ t_2 \to \mathrm{ann}_\ell(v'\ t_2)}\ [\text{E-LIFTAPP}]$$

$$\frac{v' \in \mathbf{Nf'}}{(\mathrm{ann}_\ell(v'))\ \langle\xi\rangle \to \mathrm{ann}_\ell(v'\ \langle\xi\rangle)}\ [\text{E-LIFTANNAPP}]$$

$$\frac{v' \in \mathbf{Nf'}}{\mathrm{proj}_i(\mathrm{ann}_\ell(v')) \to \mathrm{ann}_\ell(\mathrm{proj}_i(v'))}\ [\text{E-LIFTPROJ}]$$

$$\frac{v' \in \mathbf{Nf'}}{\begin{array}{l}\textbf{case}\ (\mathrm{ann}_\ell(v'))\ \textbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \\ \to \mathrm{ann}_\ell(\ \textbf{case}\ v'\ \textbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\})\end{array}}\ [\text{E-LIFTCASE}]$$

$$\frac{v' \in \mathbf{Nf'}}{\mathrm{seq}\ (\mathrm{ann}_\ell(v'))\ t_2 \to \mathrm{ann}_\ell(\mathrm{seq}\ v'\ t_2)}\ [\text{E-LIFTSEQ}]$$

$$\frac{v' \in \mathbf{Nf'}}{\mathrm{ann}_{\ell_1}(\mathrm{ann}_{\ell_2}(v')) \to \mathrm{ann}_{\ell_1 \sqcup \ell_2}(v')}\ [\text{E-JOINANN}]$$

Fig. 17: Small-step semantics ($t_1 \to t_2$)

When the above is not the case, we get a reduction $t_1 \to t_1'$ and we can apply [E-Context], resulting in

$$\frac{t_1 \to t_1'}{t_1\ t_2 \to t_1'\ t_2}$$

$t = (t_1, t_2)$ Clearly, $t \in \mathbf{Nf}$.
$t = \mathrm{proj}_i(t')$ We apply the induction hypothesis to t' and distinguish two cases. If $t' \in \mathbf{Nf}$, then either $t' \in \mathbf{Nf}'$ or $t' = \mathrm{ann}_\ell(v')$. Suppose the former holds, then $t' = (t_1, t_2)$ as this is the only possibility such that t is well-typed. But then, we can apply [E-Proj] and get $\mathrm{proj}_i(t_1, t_2) \to t_i$. In the latter case, we can apply [E-LiftProj] and get $\mathrm{proj}_i(\mathrm{ann}_\ell(v')) \to \mathrm{ann}_\ell(\mathrm{proj}_i(v'))$.
When the above is not the case, we get a reduction $t' \to t''$ and we can apply [E-Context], resulting in

$$\frac{t' \to t''}{\mathrm{proj}_i(t') \to \mathrm{proj}_i(t'')}$$

$t = \mathrm{inl}_\tau(t')$ Clearly, $t \in \mathbf{Nf}$.
$t = \mathrm{inr}_\tau(t')$ Clearly, $t \in \mathbf{Nf}$.
$t = \mathbf{case}\ t_1\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\}$ We apply the induction hypothesis to t_1 and distinguish two cases.
If $t_1 \in \mathbf{Nf}$, then either $t_1 \in \mathbf{Nf}'$ or $t_1 = \mathrm{ann}_\ell(v')$. In the former case, we have either $t_1 = \mathrm{inl}_{\tau_2}(t_1')$ or $t_1 = \mathrm{inr}_{\tau_1}(t_1')$ as these are the only possibilities for t to be well-typed. But then, we either have

$$\mathbf{case}\ \mathrm{inl}_{\tau_2}(t_1')\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \to [t_1' \,/\, x]t_2$$

by [E-CaseInl] or

$$\mathbf{case}\ \mathrm{inr}_{\tau_1}(t_1')\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \to [t_1' \,/\, y]t_3$$

by [E-CaseInr]. In the latter case, $t_1 = \mathrm{ann}_\ell(v')$, we have

$$\begin{aligned}&(\mathbf{case}\ (\mathrm{ann}_\ell(v'))\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\})\\ \to&\mathrm{ann}_\ell(\ \mathbf{case}\ v'\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\})\end{aligned}$$

by [E-LiftCase].
If the above does not hold, we get a reduction $t_1 \to t_1'$ and we have

$$\frac{t_1 \to t_1'}{\mathbf{case}\ t_1\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\} \to\ \mathbf{case}\ t_1'\ \mathbf{of}\ \{\mathrm{inl}(x) \to t_2; \mathrm{inr}(y) \to t_3\}}$$

by [E-Context].
$t = \mu x : \widehat{\tau}' \,\&\, \xi'.t'$ We can apply [E-Fix] and get $\mu x : \widehat{\tau}' \,\&\, \xi'.t' \to [\mu x : \widehat{\tau}' \,\&\, \xi'.t' \,/\, x]t'$.

$t = \text{seq } t_1\ t_2$ We apply the induction hypothesis to t_1 and distinguish two cases. If $t_1 \in \mathbf{Nf}$, then either $t_1 \in \mathbf{Nf'}$ or $t_1 = \text{ann}_\ell(v')$. In the former case, we have $\text{seq } t_1\ t_2 \to t_2$ by [E-Seq], in the latter case $\text{seq } (\text{ann}_\ell(v'))\ t_2 \to \text{ann}_\ell(\text{seq } v'\ t_2)$ by [E-LiftSeq].

If the above does not hold, then there is a reduction $t_1 \to t_1'$ and we have

$$\frac{t_1 \to t_1'}{\text{seq } t_1\ t_2 \to \text{seq } t_1'\ t_2}$$

by [E-Context].

$t = \text{ann}_\ell(t')$ We apply the induction hypothesis to t' and distinguish two cases. If $t' \in \mathbf{Nf}$, then either $t' \in \mathbf{Nf'}$ or $t' = \text{ann}_{\ell'}(v')$. In the former case, $t \in \mathbf{Nf}$. In the latter case, we have $v' \in \mathbf{Nf'}$ and $\text{ann}_\ell(\text{ann}_{\ell'}(v')) \to \text{ann}_{\ell \sqcup \ell'}(v')$ by [E-AnnJoin].

If the above does not hold, then there is a reduction $t' \to t''$ and we have

$$\frac{t' \to t''}{\text{ann}_\ell(t') \to \text{ann}_\ell(t'')}$$

by [E-Context].

$t = \Lambda\beta :: \kappa.t'$ Clearly, $t \in \mathbf{Nf}$.

$t = t'\ \langle\xi\rangle$ We apply the induction hypothesis to t' and distinguish two cases. If $t' \in \mathbf{Nf}$, then either $t' \in \mathbf{Nf'}$ or $t' = \text{ann}_\ell(v')$. Suppose the former holds, then $t' = \Lambda\beta :: \kappa.t''$ as this is the only possibility such that t is well-typed. But then, we can apply [E-AnnAbs] and get $(\Lambda\beta :: \kappa.t'')\ \langle\xi\rangle \to [\xi\ /\ \beta]t''$. In the latter case, we can apply [E-LiftAnnApp] and get $(\text{ann}_\ell(v'))\ \langle\xi\rangle \to \text{ann}_\ell(v'\ \langle\xi\rangle)$.

When the above is not the case, we get a reduction $t' \to t''$, and we can apply [E-Context], resulting in

$$\frac{t' \to t''}{t'\ \langle\xi\rangle \to t''\ \langle\xi\rangle}$$

The following lemma is needed in the proof of subject reduction. It states that substituting a variable in a well-typed term with a closed term of the right type again results in a well-typed term that no longer depends on the variable.

Lemma 22. *If* $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ *and* $\emptyset \mid \emptyset \vdash_{\text{te}} t' : \widehat{\tau}' \,\&\, \xi'$, *then* $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t'\ /\ x]t : \widehat{\tau} \,\&\, \xi$.

Proof. By induction on $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$.

[T-Var] If $t = x$, we have $\widehat{\tau} = \widehat{\tau}'$ and $\xi = \xi'$ and therefore $\emptyset \mid \emptyset \vdash_{\text{te}} [t'/x]x{:}\widehat{\tau}\&\xi$. By extending the context (see lemma 20), we arrive at $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t'/x]t{:}\widehat{\tau}\&\xi$. If $t \neq x$, then $[t'\ /\ x]t = t$. As t is a different variable than x, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ by removing the binding for x from the context in the assumption (allowed by lemma 20).

[T-Unit] We have $[t' \,/\, x]t = [t' \,/\, x]() = ()$. Since t does not contain a free occurrence of x, the result can be inferred by applying lemma 20 to the assumption.

[T-Abs] We have $t = \lambda x_1 : \widehat{\tau}_1 \& \xi_1 . t_2$ for some t_2 and $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}' \& \xi', x_1{:}\widehat{\tau}_1 \& \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi_2$ such that $\widehat{\tau} = \widehat{\tau}_1\langle \xi_1 \rangle \to \widehat{\tau}_2\langle \xi_2 \rangle$ and $\xi = \xi_2$.
If $x_1 = x$, then $[t' \,/\, x]\lambda x : \widehat{\tau}_1 \,\&\, \xi_1 . t_2 = \lambda x : \widehat{\tau}_1 \,\&\, \xi_1 . t_2$. Moreover, we can replace the context $(\widehat{\Gamma}, x{:}\widehat{\tau}' \& \xi', x_1{:}\widehat{\tau}_1 \& \xi_1)$ in the premise with $(\widehat{\Gamma}, x_1{:}\widehat{\tau}_1 \& \xi_1)$ because the older binding of x is shadowed by the newer one. Then we get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$ by [T-Abs].
If $x_1 \neq x$, then we can swap the bindings in the context of the premise and get $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \& \xi_1, x : \widehat{\tau}' \& \xi' \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \& \xi_2$ by lemma 20. We can then apply the induction hypothesis in order to get $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \& \xi_1 \vdash_{\text{te}} [t' / x]t_2 : \widehat{\tau}_2 \& \xi_2$. Finally, we apply [T-Abs], arriving at $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1 . [t' \,/\, x]t_2 : \widehat{\tau} \,\&\, \xi$. As $x_1 \neq x$, $[t' \,/\, x]\lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1 . t_2 = \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1 . [t' \,/\, x]t_2$, hence the proof is complete.

[T-App] We have $t = t_1 \; t_2$ and some type and effect pair $\widehat{\tau}_2 \,\&\, \xi_2$ such that $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_2\langle \xi_2 \rangle \to \widehat{\tau}\langle \xi \rangle \,\&\, \xi$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi_2$ hold.
By induction, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_2\langle \xi_2 \rangle \to \widehat{\tau}\langle \xi \rangle \,\&\, \xi$ and $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_2 : \widehat{\tau}_2 \,\&\, \xi_2$. Hence, we can also derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x](t_1 \; t_2) : \widehat{\tau} \,\&\, \xi$.

[T-Pair] We have $t = (t_1, t_2)$, $\xi = \bot$ and $\widehat{\tau} = \widehat{\tau}_1\langle \xi_1 \rangle \times \widehat{\tau}_2\langle \xi_2 \rangle$ for some $\widehat{\tau}_1$, ξ_1, $\widehat{\tau}_2$ and ξ_2 such that $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}' \& \xi' \vdash_{\text{te}} t_1{:}\widehat{\tau}_1 \& \xi_1$ and $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}' \& \xi' \vdash_{\text{te}} t_2{:}\widehat{\tau}_2 \& \xi_2$ hold.
By induction, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t'/x]t_1{:}\widehat{\tau}_1 \& \xi_1$ and $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t'/x]t_2{:}\widehat{\tau}_2 \& \xi_2$. Hence, $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x](t_1, t_2) : \widehat{\tau}_1\langle \xi_1 \rangle \times \widehat{\tau}_2\langle \xi_2 \rangle \,\&\, \xi$ also holds.

[T-Proj] We have $t = \text{proj}_i(t_1)$ and some $\widehat{\tau}_1, \widehat{\tau}_2, \xi_1, \xi_2$ such that $\widehat{\tau} = \widehat{\tau}_i$, $\xi = \xi_i$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1\langle \xi_1 \rangle \times \widehat{\tau}_2\langle \xi_2 \rangle \,\&\, \xi_i$.
By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t'/x]t_1{:}\widehat{\tau}_1\langle \xi_1 \rangle \times \widehat{\tau}_2\langle \xi_2 \rangle \& \xi_i$. Using [T-Proj], we can infer $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]\text{proj}_i(t_1) : \widehat{\tau} \,\&\, \xi$.

[T-Inl] We have $t = \text{inl}_\tau(t_1)$, $\xi = \bot$ and $\widehat{\tau} = \widehat{\tau}_1\langle \xi_1 \rangle + \widehat{\tau}_2 \; \xi_2$ such that $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ holds and $\lfloor \widehat{\tau}_2 \rfloor = \tau$.
By induction, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ and by [T-Inl] we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]\text{inl}_\tau(t_1) : \widehat{\tau} \,\&\, \xi$.

[T-Inr] Analogous to the previous case.

[T-Case] We have $t = \textbf{case}\; t_1 \;\textbf{of}\; \{\text{inl}(x_1) \to t_2; \text{inl}(x_2) \to t_3\}$ and $\widehat{\tau}_1$, $\widehat{\tau}_2$, ξ_1 and ξ_2 such that $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1\langle \xi_1 \rangle + \widehat{\tau}_2\langle \xi_2 \rangle \,\&\, \xi$, $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi', x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau} \,\&\, \xi$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi', x_2 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\text{te}} t_3 : \widehat{\tau} \,\&\, \xi$ hold.
By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1\langle \xi_1 \rangle + \widehat{\tau}_2\langle \xi_2 \rangle \,\&\, \xi$.
If $x = x_1 = x_2$, we have $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \& \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau} \& \xi$ and $\Sigma \mid \widehat{\Gamma}, x_2 : \widehat{\tau}_2 \& \xi_2 \vdash_{\text{te}} t_3 : \widehat{\tau} \,\&\, \xi$ by lemma 20. Moreover, $[t' \,/\, x]\; \textbf{case}\; t_1 \;\textbf{of}\; \{\text{inl}(x) \to t_2; \text{inr}(x) \to t_3\} = \textbf{case}\; ([t' \,/\, x]t_1) \;\textbf{of}\; \{\text{inl}(x) \to t_2; \text{inr}_x() \to t_3\}$ because x is bound in the subterms. Hence, we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$ by [T-Case].
If $x = x_1 \neq x_2$, we again have $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau} \,\&\, \xi$, and we have $\Sigma \mid \widehat{\Gamma}, x_2 : \widehat{\tau}_2 \,\&\, \xi_2, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_3 : \widehat{\tau} \,\&\, \xi$ by lemma 20. By induction, we

get $\Sigma \mid \widehat{\Gamma}, x_2 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\mathrm{te}} [t' \,/\, x]t_3 : \widehat{\tau} \,\&\, \xi$. As $[t' \,/\, x]$ **case** t_1 **of** $\{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to t_3\} =$ **case** $([t' \,/\, x]t_1)$ **of** $\{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to [t' \,/\, x]t_3\}$ in this case, we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$ by [T-Case].
The case where $x = x_2 \neq x_1$ can be handled analogously. Similarly, if $x \neq x_1$ and $x \neq x_2$, we apply the induction hypothesis to both branches of the case expression.

[T-Ann] We have $t = \mathrm{ann}_\ell(t_1)$, $\Sigma \vdash_{\mathrm{sub}} \ell \sqsubseteq \xi$ and $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}'\&\xi' \vdash_{\mathrm{te}} t_1{:}\widehat{\tau}\&\xi$. By induction, $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t'/x]t{:}\widehat{\tau}\&\xi$ holds. Then, $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t'/x]\mathrm{ann}_\ell(t_1){:}\widehat{\tau}\&\xi$ can be inferred by [T-Ann].

[T-Fix] We have $t = \mu x_1 : \widehat{\tau} \,\&\, \xi.t_1$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi', x_1 : \widehat{\tau} \,\&\, \xi \vdash_{\mathrm{te}} t_1 : \widehat{\tau} \,\&\, \xi$. If $x = x_1$, then $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau} \,\&\, \xi \vdash_{\mathrm{te}} t_1 : \widehat{\tau} \,\&\, \xi$ because the newer binding in the context shadows the old one. Since in this case, $[t' \,/\, x]\mu x_1 : \widehat{\tau} \,\&\, \xi.t_1 = \mu x_1 : \widehat{\tau} \,\&\, \xi.t_1$, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$ simply by applying [T-Fix].
If $x \neq x_1$, then we can use lemma 20 to infer $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau} \,\&\, \xi, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau} \,\&\, \xi$. By induction, we have $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau} \,\&\, \xi \vdash_{\mathrm{te}} [t' \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi$. As $[t' \,/\, x]\mu x_1 : \widehat{\tau} \,\&\, \xi.t_1 = \mu x_1 : \widehat{\tau} \,\&\, \xi.[t' \,/\, x]t_1$, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$ by [T-Fix].

[T-Seq] We have $t = \mathrm{seq}\ t_1\ t_2$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi$ for some $\widehat{\tau}_1$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_2 : \widehat{\tau} \,\&\, \xi$. By induction and [T-Seq], we can infer $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]\mathrm{seq}\ t_1\ t_2 : \widehat{\tau} \,\&\, \xi$.

[T-Sub] We have $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t : \widehat{\tau}_1 \,\&\, \xi_1$ for some $\widehat{\tau}_1 \,\&\, \xi_1$ such that $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}$ and $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi$ hold. By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau}_1 \,\&\, \xi_1$ and by [T-Sub] we get $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$.

[T-AnnAbs] We have $t = \Lambda\beta :: \kappa.t_1$ and $\widehat{\tau} = \forall\beta :: \kappa.\widehat{\tau}_1$ such that $\Sigma, \beta :: \kappa \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi$ holds.
By induction, we get $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi$ and by [T-AnnAbs], $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \Lambda\beta :: \kappa.[t' \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi$. Since $[t' \,/\, x]\Lambda\beta :: \kappa.t_1 = \Lambda\beta :: \kappa.[t' \,/\, x]t_1$, this completes the proof.

[T-AnnApp] We have $t = t_1\ \langle\xi_1\rangle$ and $\widehat{\tau} = [\xi_1/\beta]\widehat{\tau}_1$ such that $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}'\&\xi' \vdash_{\mathrm{te}} t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi$ holds.
By induction, we get $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t'/x]t_1{:}\forall\beta{::}\kappa.\widehat{\tau}_1\&\xi$. Applying [T-AnnApp] results in $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t_1\ \langle\xi_1\rangle : [\xi_1 \,/\, \beta]\widehat{\tau}_1 \,\&\, \xi$. Since $[t' \,/\, x](t_1\ \langle\xi_1\rangle) = [t' \,/\, x]t_1\ \langle\xi\rangle$, this completes the proof.

Similarly, substituting an annotation variable with an annotation term of the right sort in a well-typed target term results again in a well-typed term.

Lemma 23. *If we have* $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t} : \widehat{\tau} \,\&\, \xi$ *and* $\Sigma \vdash_{\mathrm{s}} \xi' : \kappa$*, then* $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\mathrm{te}} [\xi' \,/\, \beta]\widehat{t} : [\xi' \,/\, \beta]\widehat{\tau} \,\&\, [\xi' \,/\, \beta]\xi$.

Proof. By induction on $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t} : \widehat{\tau} \,\&\, \xi$.

[T-Var] We have $\widehat{t} = x$ and $\widehat{\Gamma}(x) = \widehat{\tau} \,\&\, \xi$. Then, $([\xi' \,/\, \beta]\widehat{\Gamma})(x) = [\xi' \,/\, \beta]\widehat{\tau} \,\&\, [\xi' \,/\, \beta]\xi$. Since $[\xi' \,/\, \beta]x = x$, we have $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\mathrm{te}} x : [\xi' \,/\, \beta]\widehat{\tau} \,\&\, [\xi' \,/\, \beta]\xi$ by [T-Var].

[T-Unit] We have $\widehat{t} = ()$, $\widehat{\tau} = \widehat{\text{unit}}$ and $\xi = \bot$. Since $[\xi' \,/\, \beta]\widehat{\text{unit}} = \widehat{\text{unit}}$ and $[\xi' \,/\, \beta]\bot = \bot$, the result follows trivially.

[T-Abs] We have $\widehat{t} = \lambda x : \widehat{\tau}_1 \,\&\, \xi_1.\widehat{t_1}$ and $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ such that $\Sigma, \beta :: \kappa \mid \widehat{\Gamma}, x : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_2 \,\&\, \xi_2$. By induction, $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma}, x : [\xi' \,/\, \beta]\widehat{\tau}_1 \,\&\, [\xi' \,/\, \beta]\xi_1 \vdash_{\text{te}} [\xi' \,/\, \beta]\widehat{t_1} : [\xi' \,/\, \beta]\widehat{\tau}_2 \,\&\, [\xi' \,/\, \beta]\xi_2$. From this, we can derive $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\lambda x : \widehat{\tau}_1 \,\&\, \xi_1.\widehat{t_1} : [\xi' \,/\, \beta](\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle) \,\&\, [\xi' \,/\, \beta]\xi_2$ by applying [T-Abs] and moving the substitutions outwards.

[T-Inl] We have $\widehat{t} = \text{inl}_{\lfloor\widehat{\tau}_2\rfloor}(\widehat{t'})$, $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ and $\xi = \bot$ for some $\widehat{t'}$, $\widehat{\tau}_1 \,\&\, \xi_1$, $\widehat{\tau}_2 \,\&\, \xi_2$ such that $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}_1 \,\&\, \xi_1$ holds.
By induction, $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\widehat{t'} : [\xi' \,/\, \beta]\widehat{\tau}_1 \,\&\, [\xi' \,/\, \beta]\xi_1$ holds. Using [T-Inl] and moving the substitutions outwards, $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\text{inl}_{\lfloor\widehat{\tau}_2\rfloor}(\widehat{t'}) : [\xi' \,/\, \beta](\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle) \,\&\, \bot$ can be derived.

[T-Ann] We have $\widehat{t} = \text{ann}_\ell(\widehat{t'})$ such that $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau} \& \xi$ and $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \ell \sqsubseteq \xi$.
Applying lemma 6 using $[\xi' \,/\, \beta]$ for both substitutions results in $\Sigma \vdash_{\text{sub}} [\xi' \,/\, \beta]\ell \sqsubseteq [\xi' \,/\, \beta]\xi$. But $[\xi' \,/\, \beta]\ell = \ell$, hence we can omit the substitution of the left term.
By induction, we get $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\widehat{t'} : [\xi' \,/\, \beta]\widehat{\tau} \,\&\, [\xi' \,/\, \beta]\xi$. We can derive the desired result by [T-Ann].

[T-Sub] We have $\widehat{\tau}_1$ and ξ_1 such that $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t} : \widehat{\tau}_1 \& \xi_1$, $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}$ and $\Sigma, \beta :: \kappa \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi$ hold.
By induction, $\Sigma \mid [\xi'/\beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi'/\beta]\widehat{t} : [\xi'/\beta]\widehat{\tau}_1 \& [\xi'/\beta]\xi_1$. Applying lemma 6 with $\theta_1 = \theta_2 = [\xi' \,/\, \beta]$ leads to $\Sigma \vdash_{\text{sub}} [\xi' \,/\, \beta]\xi_1 \sqsubseteq [\xi' \,/\, \beta]\xi$. By lemma 15, $\Sigma \vdash_{\text{sub}} [\xi'/\beta]\widehat{\tau}_1 \leqslant [\xi'/\beta]\widehat{\tau}$. Hence, $\Sigma \mid [\xi'/\beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi'/\beta]\widehat{t} : [\xi'/\beta]\widehat{\tau} \& [\xi'/\beta]\xi$ also holds.

[T-AnnAbs] We have $\widehat{t} = \Lambda\beta_1 :: \kappa_1.\widehat{t'}$ for some $\widehat{t'}$ and $\widehat{\tau} = \forall\beta_1 :: \kappa_1.\widehat{\tau}'$ for some $\widehat{\tau}'$ such that $\Sigma, \beta :: \kappa, \beta_1 :: \kappa_1 \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}' \,\&\, \xi$ and $\beta \notin \text{fav}(\widehat{\Gamma}) \cup \text{fav}(\xi)$.
We assume $\beta_1 \neq \beta$, as the substitution would not apply to $\widehat{t'}$ otherwise and the result follows trivially. By lemma 21, we then also have $\Sigma, \beta :: \kappa, \beta_1 :: \kappa_1 \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}' \,\&\, \xi$.
Applying the induction hypothesis results in $\Sigma, \beta_1 :: \kappa_1 \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\widehat{t'} : [\xi' \,/\, \beta]\widehat{\tau}' \,\&\, [\xi' \,/\, \beta]\xi$.
By lemma 1, $\text{fav}(\xi') \subseteq \text{dom}(\Sigma)$. Therefore, the condition $\beta \notin \text{fav}([\xi' \,/\, \beta]\widehat{\Gamma}) \cup \text{fav}([\xi' \,/\, \beta]\xi)$ is still fulfilled (as no occurrences of β could have been introduced). This allows us to derive $\Sigma \mid [\xi' \,/\, \beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi' \,/\, \beta]\widehat{t} : [\xi' \,/\, \beta]\widehat{\tau} \,\&\, [\xi' \,/\, \beta]\xi$ by [T-AnnAbs].

[T-AnnApp] We have $\widehat{t} = \widehat{t_1}\ \langle\xi_1\rangle$, $\widehat{\tau} = [\xi_1/\beta_1]\widehat{\tau}'$, $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \forall\beta_1 :: \kappa_1.\widehat{\tau}' \& \xi$ and $\Sigma, \beta :: \kappa \vdash_{\text{s}} \xi_1 : \kappa_1$. By lemma 3, $\Sigma \vdash_{\text{s}} [\xi' \,/\, \beta]\xi_1 : \kappa_1$.
Without loss of generality, we assume $\beta_1 \neq \beta$. Then $[\xi'/\beta]\forall\beta_1 :: \kappa_1.\widehat{\tau}' = \forall\beta_1 :: \kappa_1.[\xi'/\beta]\widehat{\tau}'$. By induction, $\Sigma \mid [\xi'/\beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi'/\beta]\widehat{t_1} : \forall\beta_1 :: \kappa_1.[\xi'/\beta]\widehat{\tau}' \& [\xi'/\beta]\xi$. Hence, we can derive $\Sigma \mid [\xi'/\beta]\widehat{\Gamma} \vdash_{\text{te}} [\xi'/\beta]\widehat{t_1}\ \langle[\xi'/\beta]\xi_1\rangle : [\xi_1/\beta_1][\xi'/\beta']\widehat{\tau}' \,\&\, [\xi'/\beta]\xi$ by [T-AnnApp]. As $\beta_1 \neq \beta$, $[\xi_1/\beta_1][\xi'/\beta']\widehat{\tau}' = [\xi'/\beta][\xi_1/\beta_1]\widehat{\tau}' = [\xi' \,/\, \beta]\widehat{\tau}$, this is what we needed to show.

The remaining cases can be proven similarly.

Now we can show subject reduction, i.e. the reduction of a well-typed term results in a term of the same type.

Theorem 2 (Subject Reduction). *If* $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$ *and there is a* t' *such that* $t \to t'$*, then* $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t' : \widehat{\tau} \,\&\, \xi$*.*

Proof. By induction on $t \to t'$.

[E-ABS] We have $t = (\lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1)\ t_2$ and $t' = [t_2 \,/\, x]t_1$. By applying lemma 17 and discarding the cases of non-matching rules, we can assume that the derivation of $\emptyset \mid \emptyset \vdash_{\mathrm{te}} (\lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1)\ t_2 : \widehat{\tau} \,\&\, \xi$ has the following form.

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1 : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_2 \qquad \emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}_1 \,\&\, \xi_1}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} (\lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1)\ t_2 : \widehat{\tau}_2 \,\&\, \xi_2}\,[\text{T-App}] \quad \emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\mathrm{sub}} \xi_2 \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} (\lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1)\ t_2 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

Similarly, the derivation for the lambda expression can be assumed to have the following form.

$$\dfrac{\dfrac{\emptyset \mid x : \widehat{\tau}_1' \,\&\, \xi_1' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_2' \,\&\, \xi_2'}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \lambda x : \widehat{\tau}_1' \,\&\, \xi_1'.t_1 : \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle \,\&\, \xi_2'}\,[\text{T-Abs}] \quad \emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle \leqslant \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \quad \emptyset \vdash_{\mathrm{sub}} \xi_2' \sqsubseteq \xi_2}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \lambda x : \widehat{\tau}_1 \,\&\, \xi_1.t_1 : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_2}\,[\text{T-Sub}]$$

By lemma 13, we also must have $\emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_1'$, $\emptyset \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_1'$, $\emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_2' \leqslant \widehat{\tau}_2$ and $\emptyset \vdash_{\mathrm{sub}} \xi_2' \sqsubseteq \xi_2$. Hence, we can derive $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}_1' \,\&\, \xi_1'$ using [T-SUB]. By lemma 22, we get $\emptyset \mid \emptyset \vdash_{\mathrm{te}} [t_2 \,/\, x]t_1 : \widehat{\tau}_2' \,\&\, \xi_2'$. Since we know $\emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_2' \leqslant \widehat{\tau}_2$ and $\emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}$ (and the same for the corresponding effect), we can derive $\emptyset \mid \emptyset \vdash_{\mathrm{te}} [t_2 \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi$.

[E-ANNABS] We have $t = (\Lambda\beta :: \kappa.t_1)\ \langle\xi'\rangle$ and $t' = [\xi' \,/\, \beta]t_1$. By lemma 17, there is a derivation

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \Lambda\beta :: \kappa.t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1 \qquad \emptyset \vdash_{\mathrm{s}} \xi' : \kappa}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} (\Lambda\beta :: \kappa.t_1)\ \langle\xi'\rangle : [\xi' \,/\, \beta]\widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-AnnApp}] \quad \emptyset \vdash_{\mathrm{sub}} [\xi' \,/\, \beta]\widehat{\tau}_1 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} (\Lambda\beta :: \kappa.t_1)\ \langle\xi'\rangle : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

And by the same lemma,

$$\dfrac{\dfrac{\beta :: \kappa \mid \emptyset \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_2 \,\&\, \xi_2 \qquad \beta \notin \mathrm{fav}(\emptyset) \cup \mathrm{fav}(\xi_2)}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \Lambda\beta :: \kappa.t_1 : \forall\beta :: \kappa.\widehat{\tau}_2 \,\&\, \xi_2}\,[\text{T-AnnAbs}] \quad \emptyset \vdash_{\mathrm{sub}} \forall\beta :: \kappa.\widehat{\tau}_2 \leqslant \forall\beta :: \kappa.\widehat{\tau}_1 \quad \emptyset \vdash_{\mathrm{sub}} \xi_2 \sqsubseteq \xi_1}{\emptyset \mid \emptyset \vdash_{\mathrm{te}} \Lambda\beta :: \kappa.t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-Sub}]$$

By lemma 13, $\beta{::}\kappa \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_1$. Since $\beta \notin \text{fav}(\xi_2)$ (and also $\beta \notin \text{fav}(\xi_1)$), we have $\beta{::}\kappa \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi_1$ by lemma 14. We can then derive $\beta{::}\kappa \mid \emptyset \vdash_{\text{te}} t_1{:}\widehat{\tau}_1 \& \xi_1$ by [T-Sub].

By lemma 23, we then have $\emptyset \mid \emptyset \vdash_{\text{te}} [\xi' \,/\, \beta]t_1 : [\xi' \,/\, \beta]\widehat{\tau}_1 \;\&\; [\xi' \,/\, \beta]\xi_1$. Applying [T-Sub] and noting that β is not a free variable of ξ_1 results in $\emptyset \mid \emptyset \vdash_{\text{te}} [\xi' \,/\, \beta]t_1 : \widehat{\tau} \& \xi$.

[E-Fix] We have $t = \mu x : \widehat{\tau}_1 \& \xi_1.t_1$ and $t' = [\mu x : \widehat{\tau}_1 \& \xi_1.t_1 \,/\, x]t_1$. By lemma 17, we have

$$\dfrac{\dfrac{\emptyset \mid x : \widehat{\tau}_1 \& \xi_1 \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \& \xi_1}{\emptyset \mid \emptyset \vdash_{\text{te}} \mu x : \widehat{\tau}_1 \& \xi_1.t_1 : \widehat{\tau}_1 \& \xi_1}\,[\text{T-Fix}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \mu x : \widehat{\tau}_1 \& \xi_1.t_1 : \widehat{\tau} \& \xi}\,[\text{T-Sub}]$$

Applying lemma 22 results in $\emptyset \mid \emptyset \vdash_{\text{te}} [\mu x : \widehat{\tau}_1 \& \xi_1.t_1 \,/\, x]t_1 {:} \widehat{\tau}_1 \& \xi_1$. Applying subtyping allows us to infer $\emptyset \mid \emptyset \vdash_{\text{te}} [\mu x : \widehat{\tau}_1 \& \xi_1.t_1 \,/\, x]t_1 : \widehat{\tau} \& \xi$.

[E-Proj] We have $t = \text{proj}_i(t_1, t_2)$ and $t' = t_i$. By applying lemma 17 twice, we get

$$\dfrac{\dfrac{\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} t_1 : \widehat{\tau}'_1 \& \xi'_1 \qquad \emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}'_2 \& \xi'_2}{\emptyset \mid \emptyset \vdash_{\text{te}} (t_1, t_2) : \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}'_2\langle\xi'_2\rangle \& \bot}\,[\text{T-Pair}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}'_2\langle\xi'_2\rangle \leqslant \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \quad \emptyset \vdash_{\text{sub}} \bot \sqsubseteq \xi_i}{\emptyset \mid \emptyset \vdash_{\text{te}} (t_1, t_2) : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \& \xi_i}\,[\text{T-Sub}]}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1, t_2) : \widehat{\tau}_i \& \xi_i}\,[\text{T-Proj}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_i \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1, t_2) : \widehat{\tau} \& \xi}\,[\text{T-Sub}]$$

By lemma 13, we have $\emptyset \vdash_{\text{sub}} \widehat{\tau}'_i \leqslant \widehat{\tau}_i$ and $\emptyset \vdash_{\text{sub}} \xi'_i \sqsubseteq \xi_i$. Hence, we can also derive $\emptyset \mid \emptyset \vdash_{\text{te}} t_i : \widehat{\tau} \& \xi$.

[E-CaseInl] We have $t = \textbf{case } \text{inl}_\tau(t_1) \textbf{ of } \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\}$ and $t' = [t_1 \,/\, x]t_2$. By lemma 17, we have

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{inl}_\tau(t_1) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \& \xi' \qquad \emptyset \mid x : \widehat{\tau}_1 \& \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau}' \& \xi' \quad \emptyset \mid y : \widehat{\tau}_2 \& \xi_2 \vdash_{\text{te}} t_3 : \widehat{\tau}' \& \xi'}{\emptyset \mid \emptyset \vdash_{\text{te}} t : \widehat{\tau}' \& \xi'}\,[\text{T-Case}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi' \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} t : \widehat{\tau} \& \xi}\,[\text{T-Sub}]$$

By the same lemma,

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} t_1 : \widehat{\tau}'_1 \& \xi'_1}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{inl}_\tau(t_1) : \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle \& \bot}\,[\text{T-Inl}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle \leqslant \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \quad \emptyset \vdash_{\text{sub}} \bot \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{inl}_\tau(t_1) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \& \xi}\,[\text{T-Sub}]$$

By lemma 13, $\emptyset \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_1$ and $\emptyset \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_1$. Then, $\emptyset \mid \emptyset \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ holds by [T-Sub].

By lemma 22, we get $\emptyset \mid \emptyset \vdash_{\text{te}} [t_1 \,/\, x]t_2 : \widehat{\tau}' \,\&\, \xi'$. Using [T-Sub], we can derive $\emptyset \mid \emptyset \vdash_{\text{te}} [t_1 \,/\, x]t_2 : \widehat{\tau} \,\&\, \xi$.

[E-CaseInr] Analogous to the previous case.

[E-Seq] We have $t = \text{seq } v' \ t'$ for any t'. By lemma 17, we have

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} v' : \widehat{\tau}_1 \,\&\, \xi' \qquad \emptyset \mid \emptyset \vdash_{\text{te}} t' : \widehat{\tau}_2 \,\&\, \xi'}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{seq } v' \ t' : \widehat{\tau}_2 \,\&\, \xi'}\,[\text{T-Seq}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi' \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{seq } v' \ t' : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

But then we can also derive $\emptyset \mid \emptyset \vdash_{\text{te}} t' : \widehat{\tau} \,\&\, \xi$ by [T-Sub].

[E-Context] We prove this case exemplarily for $C = \text{proj}_i(\Box)$, the other cases can be handled similarly. We have $t = \text{proj}_i(t_1)$ and $t' = \text{proj}_i(t_1')$ with $t_1 \to t_1'$. By lemma 17, there is a derivation

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} t_1 : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1) : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_i \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, $\emptyset \mid \emptyset \vdash_{\text{te}} t_1' : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i$ holds. Then

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} t_1' : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1') : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_i \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(t_1') : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

is also a valid derivation, completing the proof. The general idea for the remaining cases is to simply replace t_1 with t_1' in the derivation tree, as both share the same type and effect by induction.

[E-LiftApp] We have $t = \text{ann}_\ell(v') \ t_2$ and $t' = \text{ann}_\ell(v' \ t_2)$. By lemma 17, we have

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_2 \qquad \emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}_1 \,\&\, \xi_1}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') \ t_2 : \widehat{\tau}_2 \,\&\, \xi_2}\,[\text{T-App}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') \ t_2 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By lemma 18, we have $\emptyset \mid \emptyset \vdash_{\text{te}} v' : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_2$ and $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi_2$. Then we can also derive $\emptyset \mid \emptyset \vdash_{\text{te}} v' \ t_2 : \widehat{\tau} \,\&\, \xi$ analogously to the above tree, and we have $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$ by transitivity. Therefore, $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v' \ t_2) : \widehat{\tau} \,\&\, \xi$ holds by [T-Ann].

[E-LiftAnnApp] We have $t = \text{ann}_\ell(v')\ \langle\xi'\rangle$ and $t' = \text{ann}_\ell(v'\ \langle\xi'\rangle)$. By lemma 17, we have

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1 \qquad \emptyset \vdash_{\text{s}} \xi' : \kappa}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v')\ \langle\xi'\rangle : [\xi' \,/\, \beta]\widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-AnnApp}] \quad \begin{array}{c}\emptyset \vdash_{\text{sub}} [\xi' \,/\, \beta]\widehat{\tau}_1 \leqslant \widehat{\tau} \\ \emptyset \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi\end{array}}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v')\ \langle\xi'\rangle : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By lemma 18, we have $\emptyset \mid \emptyset \vdash_{\text{te}} v' : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1$ and $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi_1$. Then we can also derive $\emptyset \mid \emptyset \vdash_{\text{te}} v'\ \langle\xi'\rangle : \widehat{\tau} \,\&\, \xi$ analogously to the above tree, and we have $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$ by transitivity. Therefore, $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v'\ \langle\xi'\rangle) : \widehat{\tau} \,\&\, \xi$ holds by [T-Ann].

[E-LiftProj] We have $t = \text{proj}_i(\text{ann}_\ell(v'))$ and $t' = \text{ann}_\ell(\text{proj}_i(v'))$. By lemma 17,

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(\text{ann}_\ell(v')) : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \begin{array}{c}\emptyset \vdash_{\text{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \\ \emptyset \vdash_{\text{sub}} \xi_i \sqsubseteq \xi\end{array}}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(\text{ann}_\ell(v')) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

Similar to the preceding cases, we can derive $\emptyset \mid \emptyset \vdash_{\text{te}} \text{proj}_i(v') : \widehat{\tau} \,\&\, \xi$ and $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$ from the results of lemma 18. Therefore, $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(\text{proj}_i(v')) : \widehat{\tau} \,\&\, \xi$ must also hold.

[E-LiftCase] We have $t = \textbf{case}\ \text{ann}_\ell(v')\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\}$ and $t' = \text{ann}_\ell(\ \textbf{case}\ v'\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\})$. By lemma 17,

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi' \qquad \begin{array}{c}\emptyset \mid x : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} t_2 : \widehat{\tau}' \,\&\, \xi' \\ \emptyset \mid y : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\text{te}} t_3 : \widehat{\tau}' \,\&\, \xi'\end{array}}{\emptyset \mid \emptyset \vdash_{\text{te}}\ \textbf{case}\ \text{ann}_\ell(v')\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\} : \widehat{\tau}' \,\&\, \xi'}\,[\text{T-Case}] \quad \begin{array}{c}\emptyset \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau} \\ \emptyset \vdash_{\text{sub}} \xi' \sqsubseteq \xi\end{array}}{\emptyset \mid \emptyset \vdash_{\text{te}}\ \textbf{case}\ \text{ann}_\ell(v')\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\} : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

From the results of lemma 18 we can derive $\emptyset \mid \emptyset \vdash_{\text{te}}\ \textbf{case}\ v'\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\}{:}\widehat{\tau}\&\xi$ and $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$. Then, $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(\ \textbf{case}\ v'\ \textbf{of}\ \{\text{inl}(x) \to t_2; \text{inr}(y) \to t_3\}) : \widehat{\tau} \,\&\, \xi$ must also hold.

[E-LiftSeq] We have $t = \text{seq}\ (\text{ann}_\ell(v'))\ t_2$ and $t' = \text{ann}_\ell(\text{seq}\ v'\ t_2)$. By lemma 17,

$$\dfrac{\dfrac{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \widehat{\tau}_1 \,\&\, \xi' \qquad \emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi'}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{seq}\ (\text{ann}_\ell(v'))\ t_2 : \widehat{\tau}_2 \,\&\, \xi'}\,[\text{T-Seq}] \quad \begin{array}{c}\emptyset \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \\ \emptyset \vdash_{\text{sub}} \xi' \sqsubseteq \xi\end{array}}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{seq}\ (\text{ann}_\ell(v'))\ t_2 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

Again, we infer $\emptyset \mid \emptyset \vdash_{\text{te}} \text{seq}\ v'\ t_2 : \widehat{\tau} \,\&\, \xi$ and $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$ from the results of lemma 18. Then, $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(\text{seq}\ v'\ t_2) : \widehat{\tau} \,\&\, \xi$ must also hold.

[E-JoinAnn] We have $t = \text{ann}_{\ell_1}(\text{ann}_{\ell_2}(v'))$ and $t' = \text{ann}_{\ell_1 \sqcup \ell_2}(v')$. Applying lemma 18 twice results in $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_{\ell_2}(v') : \widehat{\tau} \,\&\, \xi$ and $\emptyset \mid \emptyset \vdash_{\text{te}} v' : \widehat{\tau} \,\&\, \xi$ with $\Sigma \vdash_{\text{sub}} \ell_1 \sqsubseteq \xi$ and $\Sigma \vdash_{\text{sub}} \ell_2 \sqsubseteq \xi$. Hence, we also have $\Sigma \vdash_{\text{sub}} \ell_1 \sqcup \ell_2 \sqsubseteq \xi$ and therefore

$$\frac{\emptyset \mid \emptyset \vdash_{\text{te}} v' : \widehat{\tau} \,\&\, \xi \qquad \emptyset \vdash_{\text{sub}} \ell_1 \sqcup \ell_2 \sqsubseteq \xi}{\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_{\ell_1 \sqcup \ell_2}(v') : \widehat{\tau} \,\&\, \xi} \text{[T-Ann]}$$

Another useful property is that our small-step semantics is deterministic, i.e. for every term there is just one unique term a reduction can lead to.

Lemma 24 (Determinism). *If we have $t_1 \to t_2$ and $t_1 \to t_2'$, then $t_2 = t_2'$.*

Proof. By induction on $t_1 \to t_2$.

[E-Abs] We have $t_1 = (\lambda x : \widehat{\tau}\&\xi.t_1')\ t$. If $t_1 \to t_2'$ also follows by [E-Abs], $t_2 = t_2'$ by definition. The only other rule with a matching head is [E-Context] with the context $\square\ t$. But then, there would be a reduction $\lambda x : \widehat{\tau} \,\&\, \xi.t_1' \to t_1''$. This is a contradiction because a λ-abstraction is a normal form.
The cases for [E-AnnAbs], [E-Proj], [E-CaseInl], [E-CaseInr] and [E-Seq] can be proven analogously to the previous one. The same applies to [E-LiftApp], [E-LiftAnnApp], [E-LiftProj], [E-LiftCase], [E-LiftSeq] and [E-JoinAnn].

[E-Fix] Only this rule applies, therefore $t_2 = t_2'$ by definition.

[E-Context] We have $t_1 = C[t]$ and $t_2 = C[t']$ and a reduction $t \to t'$. Therefore, t cannot be a normal form. This rules out all potentially matching rules for $t_1 \to t_2'$, except for [E-Context] where we have $t_2' = C[t'']$ and a reduction $t \to t''$. By induction, $t' = t''$ and therefore $t_2 = C[t'] = C[t''] = t_2'$.

In order to describe the evaluation of whole programs, we need to chain multiple individual reduction steps together.

Definition 1. *We write $t \to^* v$ if there is a finite sequence of terms $(t_i)_{0 \leqslant i \leqslant n}$ with $t_0 = t$ and $t_n = v \in \mathbf{Nf}$ and reductions $(t_i \to t_{i+1})_{0 \leqslant i < n}$ between them. If there is no such a sequence, this is denoted by $t \Uparrow$ and t is said to* diverge.

And lastly, if a term evaluates to an annotated value, this annotation is compatible with the effect that has been assigned to the term.

Theorem 3 (Semantic Soundness). *If we have $\emptyset \mid \emptyset \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$ and $t \to^* \text{ann}_\ell(v')$, then $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$.*

Proof. By corollary 1, we have $\emptyset \mid \emptyset \vdash_{\text{te}} \text{ann}_\ell(v') : \widehat{\tau} \,\&\, \xi$. By lemma 18, we then also have $\emptyset \mid \emptyset \vdash_{\text{te}} v' : \widehat{\tau} \,\&\, \xi$ and, in particular, $\emptyset \vdash_{\text{sub}} \ell \sqsubseteq \xi$.

We now give the formal definition for similarity of normal forms:

Definition 2. *We say two normal forms $v_1, v_2 \in \mathbf{Nf}$ are* similar*, if their top level constructors (and annotations, if present) match, i.e. if*

- $v_1 = \lambda x : \widehat{\tau} \,\&\, \xi.t_1$ *and* $v_2 = \lambda x : \widehat{\tau} \,\&\, \xi.t_2$ *for some* t_1, t_2, *or*
- $v_1 = \Lambda\beta :: \kappa.t_1$ *and* $v_2 = \Lambda\beta :: \kappa.t_2$ *for some* t_1, t_2, *or*
- $v_1 = ()$ *and* $v_2 = ()$, *or*
- $v_1 = \mathrm{inl}_\tau(t_1)$ *and* $v_2 = \mathrm{inl}_\tau(t_2)$ *for some* t_1, t_2, *or*
- $v_1 = \mathrm{inr}_\tau(t_1)$ *and* $v_2 = \mathrm{inr}_\tau(t_2)$ *for some* t_1, t_2, *or*
- $v_1 = (t_1, t_1')$ *and* $v_2 = (t_2, t_2')$ *for some* t_1, t_1', t_2, t_2', *or*
- $v_1 = \mathrm{ann}_\ell(v_1')$ *and* $v_2 = \mathrm{ann}_\ell(v_2')$ *and* v_1' *and* v_2' *(which cannot be annotations themselves) are similar.*

We also use $v_1 \simeq v_2$ *to denote the statement that* v_1 *and* v_2 *are similar.*

We first show noninterference for normal forms.

Lemma 25. *Let* t *be a target term such that* $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$ *and* $\emptyset \vdash_{\mathrm{sub}} \xi' \not\sqsubseteq \xi$.

If there is a t_1 *with* $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_1 : \widehat{\tau}' \,\&\, \xi'$ *such that* $[t_1 \,/\, x]t \in \mathbf{Nf}$, *then for all* t_2 *with* $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ *we have* $[t_2 \,/\, x]t \in \mathbf{Nf}$ *and* $[t_1 \,/\, x]t \simeq [t_2 \,/\, x]t$.

Proof. We first show that $t \neq x$ by contradiction. Suppose that $t = x$, then by lemma 17 there is a derivation

$$\dfrac{\dfrac{(x : \widehat{\tau}' \,\&\, \xi')(x) = \widehat{\tau}' \,\&\, \xi'}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} x : \widehat{\tau}' \,\&\, \xi'}\,[\text{T-Var}] \qquad \emptyset \vdash_{\mathrm{sub}} \widehat{\tau}' \leqslant \widehat{\tau} \qquad \emptyset \vdash_{\mathrm{sub}} \xi' \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} x : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

contradicting our assumption $\emptyset \vdash_{\mathrm{sub}} \xi' \not\sqsubseteq \xi$. A similar reasoning can be used to rule out the case $t = \mathrm{ann}_\ell(x)$.

This leaves us with either $t = v$ or $t = \mathrm{ann}_\ell(v)$ where v falls under one of the following cases.

$v = \lambda y : \widehat{\tau}_1 \,\&\, \xi_1.t'$ We assume without loss of generality that $x \neq y$. Then, $[t_1 \,/\, x]v \in \mathbf{Nf'}$ and $[t_2 \,/\, x]v \in \mathbf{Nf'}$, and we have $[t_1 \,/\, x]v \simeq [t_2 \,/\, x]v$.

$v = \Lambda\beta :: \kappa.t'$ We have $[t_1 \,/\, x]v = \Lambda\beta :: \kappa.[t_1 \,/\, x]t' \in \mathbf{Nf'}$ and $[t_2 \,/\, x]v = \Lambda\beta :: \kappa.[t_2 \,/\, x]t' \in \mathbf{Nf'}$, therefore $[t_1 \,/\, x]v \simeq [t_2 \,/\, x]v$.

$v = ()$ It is easy to see that $[t_1 \,/\, x]v = () \in \mathbf{Nf'}$ and $[t_2 \,/\, x]v = () \in \mathbf{Nf'}$, and therefore $[t_1 \,/\, x]v \simeq [t_2 \,/\, x]v$.

$v = \mathrm{inl}_\tau(t')$ We have $[t_1 \,/\, x]v = \mathrm{inl}_\tau([t_1 \,/\, x]t') \in \mathbf{Nf'}$ and $[t_2 \,/\, x]v = \mathrm{inl}_\tau([t_2 \,/\, x]t') \in \mathbf{Nf'}$, therefore $[t_1 \,/\, x]v \simeq [t_2 \,/\, x]v$.

$v = \mathrm{inr}_\tau(t')$ Analogous to the previous case.

$v = (t_1, t_2)$ Similar to the previous cases.

In each of the cases, we have $[t_1 / x]v, [t_2 / x]v \in \mathbf{Nf'}$ and $[t_1 / x]v \simeq [t_2 / x]v$, and therefore also $[t_1 / x]\mathrm{ann}_\ell(v) \simeq [t_2 / x]\mathrm{ann}_\ell(v)$. This lets us conclude $[t_2 / x]t \in \mathbf{Nf}$ and $[t_1 \,/\, x]t \simeq [t_2 \,/\, x]t$.

The following lemma is used to "recover" a more general typing statement from a type that is well-typed after a substitution.

Lemma 26. *If we have a term* t *such that for all* t' *with* $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t' : \widehat{\tau}' \,\&\, \xi'$ *we have* $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} [t' \,/\, x]t : \widehat{\tau} \,\&\, \xi$, *then* $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$.

Proof. By induction on t.

$t = x_1$ If $x_1 = x$, then $\widehat{\tau}' = \widehat{\tau}$, $\xi' = \xi$ and clearly $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi$. Otherwise, $[t' \,/\, x]x_1 = x_1$ and we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} x_1 : \widehat{\tau} \,\&\, \xi$ by assumption. Extending the context allows us to conclude $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} x_1 : \widehat{\tau} \,\&\, \xi$ by lemma 20.

$t = ()$ Since $[t' \,/\, x]() = ()$, we have $\widehat{\tau} = \widehat{\text{unit}}$ and therefore $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} () : \widehat{\text{unit}} \,\&\, \xi$ by [T-Unit] and [T-Sub].

$t = \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_2$ If $x = x_1$, then $[t' \,/\, x]t = t$ and the result follows by lemma 20. If $x \neq x_1$, then we have $[t' \,/\, x]t = \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_2$. By lemma 17, there is a derivation of the form

$$\cfrac{\cfrac{\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} [t' \,/\, x]t_2 : \widehat{\tau}_2 \,\&\, \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_2 : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\,[\text{T-Abs}] \quad \begin{array}{c}\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau} \\ \Sigma \vdash_{\text{sub}} \bot \sqsubseteq \xi\end{array}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_2 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we therefore have $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi_2$. We can swap the last two bindings in the context and infer $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_2 : \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot$ by [T-Abs], due to $x \neq x_1$. Lastly, applying rule [T-Sub] results in $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \lambda x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_2 : \widehat{\tau} \,\&\, \xi$.

$t = t_1\ t_2$ We have $[t' \,/\, x](t_1\ t_2) = [t' \,/\, x]t_1\ [t' \,/\, x]t_2$, and by lemma 17, there is a derivation

$$\cfrac{\cfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_2\langle\xi_2\rangle \to \widehat{\tau}_1\langle\xi_1\rangle \,\&\, \xi_1 \qquad \Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_2 : \widehat{\tau}_2 \,\&\, \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1\ [t' \,/\, x]t_2 : \widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-App}] \quad \begin{array}{c}\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau} \\ \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi\end{array}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1\ [t' \,/\, x]t_2 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

for any t' with $\emptyset \mid \emptyset \vdash_{\text{te}} t' : \widehat{\tau}' \,\&\, \xi'$.
By induction, we get $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_2\langle\xi_2\rangle \to \widehat{\tau}_1\langle\xi_1\rangle \,\&\, \xi_1$ and $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}'\&\xi' \vdash_{\text{te}} t_2{:}\widehat{\tau}_2\&\xi_2$. Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x{:}\widehat{\tau}'\&\xi' \vdash_{\text{te}} t_1\ t_2{:}\widehat{\tau}\&\xi$ by [T-App] and [T-Sub].

$t = (t_1, t_2)$ We have $[t' \,/\, x](t_1, t_2) = ([t' \,/\, x]t_1, [t' \,/\, x]t_2)$. By lemma 17, we get a derivation

$$\cfrac{\cfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1 \qquad \Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_2 : \widehat{\tau}_2 \,\&\, \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} ([t' \,/\, x]t_1, [t' \,/\, x]t_2) : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\,[\text{T-Pair}] \quad \begin{array}{c}\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau} \\ \Sigma \vdash_{\text{sub}} \bot \sqsubseteq \xi\end{array}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} ([t' \,/\, x]t_1, [t' \,/\, x]t_2) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we get $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi_2$. Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} (t_1, t_2) : \widehat{\tau} \,\&\, \xi$ by [T-Pair] and [T-Sub].

$t = \mathrm{proj}_i(t_1)$ We have $[t' \,/\, x]\mathrm{proj}_i(t_1) = \mathrm{proj}_i([t' \,/\, x]t_1)$. By lemma 17, there is a derivation

$$\dfrac{\dfrac{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} \mathrm{proj}_i([t' \,/\, x]t_1) : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \widehat{\tau}_i \vdash_{\mathrm{sub}} \widehat{\tau} \leqslant \cdot \quad \xi_i \vdash_{\mathrm{sub}} \xi \sqsubseteq \cdot}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} \mathrm{proj}_i([t' \,/\, x]t_1) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we have $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i$. Using [T-Proj] and [T-Sub], we can infer $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t : \widehat{\tau} \,\&\, \xi$.

$t = \mathrm{inl}_{\tau_2}(t_1)$ We have $[t' \,/\, x]\mathrm{inl}_\tau(t_1) = \mathrm{inl}_\tau([t' \,/\, x]t_1)$. By lemma 17, we get a derivation

$$\dfrac{\dfrac{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} \mathrm{inl}_\tau([t' \,/\, x]t_1) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\,[\text{T-Inl}] \quad \varSigma \vdash_{\mathrm{sub}} \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau} \quad \varSigma \vdash_{\mathrm{sub}} \bot \sqsubseteq \xi}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} \mathrm{inl}_\tau([t' \,/\, x]t_1) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we get $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$. Thus, we can derive $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{inl}_\tau(t_1) : \widehat{\tau} \,\&\, \xi$ by [T-Inl] and [T-Sub].

$t = \mathrm{inr}_{\tau_1}(t_2)$ Analogous to the previous case.

$t = \textbf{case } t_1 \textbf{ of } \{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to t_3\}$ If $x \neq x_1$ and $x \neq x_2$, then we have $[t' \,/\, x]t = \textbf{case } [t' \,/\, x]t_1 \textbf{ of } \{\mathrm{inl}(x_1) \to [t' \,/\, x]t_2; \mathrm{inr}(x_2) \to [t' \,/\, x]t_3\}$. By lemma 17, there is a derivation

$$\dfrac{\dfrac{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_3 \quad \begin{array}{c}\varSigma \mid \widehat{\varGamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\mathrm{te}} [t' \,/\, x]t_2 : \widehat{\tau}_3 \,\&\, \xi_3 \\ \varSigma \mid \widehat{\varGamma}, x_2 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\mathrm{te}} [t' \,/\, x]t_3 : \widehat{\tau}_3 \,\&\, \xi_3\end{array}}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} [t' \,/\, x]\ \textbf{case } t_1 \textbf{ of } \{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to t_3\} : \widehat{\tau}_3 \,\&\, \xi_3}\,[\text{T-Case}] \quad \varSigma \vdash_{\mathrm{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau} \quad \varSigma \vdash_{\mathrm{sub}} \xi_3 \sqsubseteq \xi}{\varSigma \mid \widehat{\varGamma} \vdash_{\mathrm{te}} [t' \,/\, x]\ \textbf{case } t_1 \textbf{ of } \{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to t_3\} : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we get $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_1 : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_3$, $\varSigma \mid \widehat{\varGamma}, x_1 : \widehat{\tau}_2 \,\&\, \xi_2, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_2 : \widehat{\tau}_3 \,\&\, \xi_3$ and $\varSigma \mid \widehat{\varGamma}, x_2 : \widehat{\tau}_2 \,\&\, \xi_2, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_3 : \widehat{\tau}_3 \,\&\, \xi_3$. Using lemma 20, we can reorder the contexts of the latter two typing statements, resulting in $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi', x_1 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\mathrm{te}} t_2 : \widehat{\tau}_3 \,\&\, \xi_3$ and $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi', x_2 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\mathrm{te}} t_3 : \widehat{\tau}_3 \,\&\, \xi_3$.

Now we can derive $\varSigma \mid \widehat{\varGamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \textbf{case } t_1 \textbf{ of } \{\mathrm{inl}(x_1) \to t_2; \mathrm{inr}(x_2) \to t_3\} : \widehat{\tau} \,\&\, \xi$ using [T-Case] and [T-Sub].

If $x = x_1$ or $x = x_2$, the substitution does not apply to the corresponding branches. We can then handle the corresponding cases using lemma 20 directly without resorting to induction, similar to what we did in the case for lambda abstractions.

$t = \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_1$ If $x = x_1$, then $[t' \,/\, x]t = t$ and the result follows by lemma 20. If $x \neq x_1$, then we have $[t' \,/\, x]t = \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_1$. By lemma 17, there is a derivation of the form

$$\dfrac{\dfrac{\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-Fix}] \quad \Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau} \quad \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.[t' \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we therefore have $\Sigma \mid \widehat{\Gamma}, x_1 : \widehat{\tau}_1 \,\&\, \xi_1, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$. We can swap the last two bindings in the context and infer $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ by [T-Fix], for $x \neq x_1$. Lastly, applying rule [T-Sub] results in $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \mu x_1 : \widehat{\tau}_1 \,\&\, \xi_1.t_1 : \widehat{\tau} \,\&\, \xi$.

$t = \text{seq}\ t_1\ t_2$ We have $[t' \,/\, x]\text{seq}\ t_1\ t_2 = \text{seq}\ ([t' \,/\, x]t_1)\ ([t' \,/\, x]t_2)$. By lemma 17, we get a derivation

$$\dfrac{\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1 \qquad \Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_2 : \widehat{\tau}_2 \,\&\, \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{seq}\ ([t' \,/\, x]t_1)\ ([t' \,/\, x]t_2) : \widehat{\tau}_2 \,\&\, \xi_2}\,[\text{T-Seq}] \quad \Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \quad \Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{seq}\ ([t' \,/\, x]t_1)\ ([t' \,/\, x]t_2) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

By induction, we get $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau}_1 \,\&\, \xi_1$ and $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_2 : \widehat{\tau}_2 \,\&\, \xi_2$. Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{seq}\ t_1\ t_2 : \widehat{\tau} \,\&\, \xi$ by [T-Seq] and [T-Sub].

$t = \text{ann}_\ell(t_1)$ We have $[t' \,/\, x]\text{ann}_\ell(t_1) = \text{ann}_\ell([t' \,/\, x]t_1)$. By lemma 18, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi$ and $\Sigma \vdash_{\text{sub}} \ell \sqsubseteq \xi$.
By induction, we get $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \widehat{\tau} \,\&\, \xi$. Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{ann}_\ell(t_1) : \widehat{\tau} \,\&\, \xi$ by [T-Ann].

$t = \Lambda\beta :: \kappa.t_1$ We have $[t' \,/\, x]\Lambda\beta :: \kappa.t_1 = \Lambda\beta :: \kappa.[t' \,/\, x]t_1$. By lemma 17, we get a derivation

$$\dfrac{\dfrac{\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} [t' \,/\, x]t_1 : \widehat{\tau}_1 \,\&\, \xi_1 \qquad \beta \notin \text{fav}(\widehat{\Gamma}) \cup \text{fav}(\xi)}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \Lambda\beta :: \kappa.[t' \,/\, x]t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-AnnApp}] \quad \Sigma \vdash_{\text{sub}} \forall\beta :: \kappa.\widehat{\tau}_1 \leqslant \widehat{\tau} \quad \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \Lambda\beta :: \kappa.[t' \,/\, x]t_1 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

We assume that $\beta \notin \text{dom}(\Sigma)$, because we could simply rename the annotation variable otherwise. We can then extend the typing judgment for t' with this variable, obtaining $\Sigma, \beta :: \kappa \mid \widehat{\Gamma} \vdash_{\text{te}} t' : \widehat{\tau}' \,\&\, \xi'$ by lemma 21.
Applying the induction hypothesis results in $\Sigma, \beta :: \kappa \mid \widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \,\&\, \xi_1$. The side condition is still fulfilled, because $\beta \notin \text{fav}(\widehat{\tau}') \cup \text{fav}(\xi')$, as the typing judgment $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t' : \widehat{\tau}' \,\&\, \xi'$ does not refer to β.

Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\text{te}} \Lambda\beta :: \kappa.t_1 : \widehat{\tau} \mathbin{\&} \xi$ by [T-AnnAbs] and [T-Sub].

$t = t_1 \langle \xi_1 \rangle$ We have $[t' / x]t_1 \langle \xi_1 \rangle = [t' / x]t_1 \langle \xi_2 \rangle$. The ξ_2 subterm is unaffected because the substitution only applies to term variables. By lemma 17, we get a derivation

$$\cfrac{\cfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' / x]t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \mathbin{\&} \xi_1 \qquad \Sigma \vdash_{\text{s}} \xi_2 : \kappa}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' / x]t_1 \langle \xi_2 \rangle : [\xi_2 / \beta]\widehat{\tau}_1 \mathbin{\&} \xi_1}\,[\text{T-AnnApp}] \\ \Sigma \vdash_{\text{sub}} [\xi_2 / \beta]\widehat{\tau}_1 \leqslant \widehat{\tau} \\ \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} [t' / x]t_1 \langle \xi_2 \rangle : \widehat{\tau} \mathbin{\&} \xi}\,[\text{T-Sub}]$$

By induction, we get $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\text{te}} t_1 : \forall\beta :: \kappa.\widehat{\tau}_1 \mathbin{\&} \xi_1$. Thus, we can derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\text{te}} t_1 \langle \xi_2 \rangle : \widehat{\tau} \mathbin{\&} \xi$ by [T-AnnApp] and [T-Sub].

The following lemma provides the noninterference property for an individual reduction step.

Lemma 27. *Let t be a target term such that $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\text{te}} t : \widehat{\tau} \mathbin{\&} \xi$ and $\emptyset \vdash_{\text{sub}} \xi' \not\sqsubseteq \xi$. Let t' be a target term.*

If there is a t_1 with $\emptyset \mid \emptyset \vdash_{\text{te}} t_1 : \widehat{\tau}' \mathbin{\&} \xi'$ such that $[t_1 / x]t \to t'$, then there is a t'' such that $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\text{te}} t'' : \widehat{\tau} \mathbin{\&} \xi$ and for all t_2 with $\emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$ we have $[t_2 / x]t \to [t_2 / x]t''$.

Proof. By induction on $[t_1 / x]t \to t'$. The general idea in all of the cases is to use the assumption $\emptyset \vdash_{\text{sub}} \xi' \not\sqsubseteq \xi$ to rule out certain cases for the term t so that the same reduction rule applies regardless of the substituted subterm.

[E-Abs] We have $[t_1 / x]t = (\lambda y : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3)\ t_4$. Without loss of generality we assume $x \neq y$. Due to the assumption $\emptyset \vdash_{\text{sub}} \xi' \not\sqsubseteq \xi$, we can rule out the cases where $t = x$ and $t = x\ t_4'$. Therefore, there are terms t_3' and t_3' such that $t = (\lambda y : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3')\ t_4'$ and $[t_1 / x]t = (\lambda y : \widehat{\tau}_1 \mathbin{\&} \xi_1.[t_1 / x]t_3')\ [t_1 / x]t_4'$. But then we have for arbitrary t_2 with $\emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$ that $(\lambda y : \widehat{\tau}_1 \mathbin{\&} \xi_1.[t_2 / x]t_3')\ [t_2 / x]t_4' \to [[t_2 / x]t_4' / y][t_2 / x]t_3'$ holds. Because y does not occur free in t_2, we can reorder the substitutions and arrive at $(\lambda y : \widehat{\tau}_1 \mathbin{\&} \xi_1.[t_2 / x]t_3')\ [t_2 / x]t_4' \to [t_2 / x][t_4' / y]t_3'$. By defining $t'' = [t_4' / y]t_3'$, we reach the desired conclusion $[t_2 / x]t \to [t_2 / x]t''$.

[E-AnnAbs] We have $[t_1 / x]t = (\Lambda\beta :: \kappa.t_3)\ \langle \xi_1 \rangle$. Again, we can rule out $t = x$ and $t = x\ \langle \xi_1 \rangle$. Hence, there is a t_3' such that $t = (\Lambda\beta :: \kappa.t_3')\ \langle \xi_1 \rangle$.
But then $[t_2 / x]t \to [\xi_1 / \beta][t_2 / x]t_3'$ is a valid derivation by [E-AnnAbs] for all t_2 such that $\emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$. Since β cannot occur free in t_2, $[\xi_1 / \beta][t_2 / x]t_3' = [t_2 / x][\xi_1 / \beta]t_3'$. Therefore, we choose $t'' = [\xi_1 / \beta]t_3'$.

[E-Fix] We have $[t_1 / x]t = \mu y : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3$. Again, we know $t \neq x$.
If $x = y$, then $t = \mu x : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3$ and x does not occur free in t. But then we always have $\mu x : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3 \to [\mu y : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3 / x]t_3$ by [T-Fix]. Moreover, x also does not occur free in $[\mu y : \widehat{\tau}_1 \mathbin{\&} \xi_1.t_3 / x]t_3$, implying

$[t_2\,/\,x][\mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3\,/\,x]t_3 = [\mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3\,/\,x]t_3$ for all t_2. Hence, choosing $t'' = [\mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3\,/\,x]t_3$ leads to the desired result.

If $x \neq y$, then $t = \mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3'$ for some t_3'. Let t_2 be arbitrary such that $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$. Then $[t_2\,/\,x]t = \mu y : \widehat{\tau}_1 \,\&\, \xi_1.[t_2\,/\,x]t_3'$ and we have $\mu y : \widehat{\tau}_1 \,\&\, \xi_1.[t_2\,/\,x]t_3' \to [\mu y : \widehat{\tau}_1 \,\&\, \xi_1.[t_2\,/\,x]t_3'\,/\,y][t_2\,/\,x]t_3'$ by [T-Fix]. Since y does not occur free in t_2, we can reorder the substitutions as follows: $[\mu y : \widehat{\tau}_1 \,\&\, \xi_1.[t_2\,/\,x]t_3'\,/\,y][t_2\,/\,x]t_3' = [t_2\,/\,x][\mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3'\,/\,y]t_3'$. Thus, choosing $t'' = [\mu y : \widehat{\tau}_1 \,\&\, \xi_1.t_3'\,/\,y]t_3'$ leads to the desired result.

[E-Proj] We have $[t_1\,/\,x]t = \mathrm{proj}_i(t_3, t_4)$, hence t must be of the form $t = \mathrm{proj}_i(e_1, e_2)$ for $\emptyset \vdash_{\mathrm{sub}} \xi' \sqsubseteq \xi$ to hold. Then indeed for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$, we have $[t_2\,/\,x]\mathrm{proj}_i(t_3, t_4) = \mathrm{proj}_i([t_2\,/\,x]e_1, [t_2\,/\,x]e_2)$ and therefore $[t_2\,/\,x]t \to [t_2\,/\,x]e_i$ by [E-Proj]. We choose $t'' = e_i$.

[E-CaseInl] We have $[t_1\,/\,x]t = \textbf{case}\ \mathrm{inl}_\tau(e_1)\ \textbf{of}\ \{\mathrm{inl}(x_1) \to e_2; \mathrm{inr}(x_2) \to e_3\}$. As we can rule out the cases where $t = x$ and $t = \textbf{case}\ x\ \textbf{of}\ \{...\}$, we must have $t = \textbf{case}\ \mathrm{inl}_\tau(e_1')\ \textbf{of}\ \{\mathrm{inl}(x_1) \to e_2'; \mathrm{inr}(x_2) \to e_3'\}$ for some e_1', e_2' and e_3'. We assume $x \neq x_1$ and $x \neq x_2$ as we have seen previously how to handle constructs where name shadowing occurs.

Let t_2 be any target term such that $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ holds. Then, $[t_2\,/\,x]\ \textbf{case}\ \mathrm{inl}_\tau(e_1')\ \textbf{of}\ \{\mathrm{inl}(x_1) \to e_2'; \mathrm{inr}(x_2) \to e_3'\} = \textbf{case}\ \mathrm{inl}_\tau([t_2\,/\,x]e_1')\ \textbf{of}\ \{\mathrm{inl}(x_1) \to [t_2/x]e_2'; \mathrm{inr}(x_2) \to [t_2/x]e_3'\}$ and therefore $\textbf{case}\ \mathrm{inl}_\tau([t_2/x]e_1')\ \textbf{of}\ \{\mathrm{inl}(x_1) \to [t_2/x]e_2'; \mathrm{inr}(x_2) \to [t_2/x]e_3'\} \to [[t_2/x]e_1'/x_1][t_2/x]e_2'$. We choose $t'' = [e_1'\,/\,x_1]e_2'$. We can reorder the substitutions because x_1 does not occur free in t_2, thus $[t_2\,/\,x]t'' = [[t_2\,/\,x]e_1'\,/\,x_1][t_2\,/\,x]e_2'$. Hence, we get the desired $[t_2\,/\,x]t \to [t_2\,/\,x]t''$.

[E-CaseInr] Analogous to previous case.

[E-Context] We show this case exemplarily for $C = \mathrm{proj}_i(\Box)$, but it can be proven in a similar way for other evaluation contexts. Thus, we have $[t_1\,/\,x]t = \mathrm{proj}_i(t_3)$ and $t' = \mathrm{proj}_i(t_3')$ such that $t_3 \to t_3'$.

Due to the assumption $\emptyset \vdash_{\mathrm{sub}} \xi' \not\sqsubseteq \xi$, we can rule out the cases where $t = x$ and $t = \mathrm{proj}_i(x)$. This is because the typing rule [T-Proj] ensures that the effect of the argument of the projection is the same as the effect of the projection itself. Therefore, $t = \mathrm{proj}_i(t_4)$ for some term $t_4 \neq x$, but possibly containing x. This also implies $[t_1\,/\,x]t = \mathrm{proj}_i([t_1\,/\,x]t_4)$.

Then we also have $t_3 = [t_1\,/\,x]t_4$ and thus $[t_1\,/\,x]t_4 \to t_3'$. By lemma 17, there is a derivation for $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{proj}_i(t_4) : \widehat{\tau} \,\&\, \xi$ of the following form

$$\dfrac{\dfrac{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_4 : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{proj}_i(t_4) : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \quad \emptyset \vdash_{\mathrm{sub}} \xi_i \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{proj}_i(t_4) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

As $\emptyset \vdash_{\mathrm{sub}} \xi_i \sqsubseteq \xi$ holds, we also have $\emptyset \vdash_{\mathrm{sub}} \xi' \not\sqsubseteq \xi_i$.

Therefore, all prerequisites for the induction hypothesis are met, and we get a term t_4' such that for all terms t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ we have $[t_2\,/\,x]t_4 \to [t_2\,/\,x]t_4'$. Moreover, $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} t_4' : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i$.

But then we also have for all such t_2

$$\frac{[t_2 \,/\, x]t_4 \rightarrow [t_2 \,/\, x]t_4'}{\mathrm{proj}_i([t_2 \,/\, x]t_4) \rightarrow \mathrm{proj}_i([t_2 \,/\, x]t_4')}\ [\text{E-Context}]$$

and hence by setting $t'' = \mathrm{proj}_i(t_4')$ the desired $[t_2 \,/\, x]t \rightarrow [t_2 \,/\, x]t''$.

[E-LiftApp] We have $[t_1 \,/\, x]t = \mathrm{ann}_\ell(v_1)\ e_1$ for some $v_1 \in \mathbf{Nf'}$. As $t = x$ or $t = x\ e_1$ would violate the assumptions, $t = \mathrm{ann}_\ell(v_2)\ e_2$ for some v_2, e_2 and $[t_1 \,/\, x]v_2 = v_1$, $[t_1 \,/\, x]e_2 = e_1$.
Using lemma 17, there is a derivation for $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ e_2 : \widehat{\tau} \,\&\, \xi$ of the following form.

$$\cfrac{\cfrac{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2) : \widehat{\tau}_1\langle\xi_1\rangle \rightarrow \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_2 \qquad \emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} e_2 : \widehat{\tau}_1 \,\&\, \xi_1}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ e_2 : \widehat{\tau}_2 \,\&\, \xi_2}\ [\text{T-App}] \\ \emptyset \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau} \\ \emptyset \vdash_{\mathrm{sub}} \xi_2 \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ e_2 : \widehat{\tau} \,\&\, \xi}\ [\text{T-Sub}]$$

Since $[t_1 \,/\, x]v_2 \in \mathbf{Nf'}$, $[t_1 \,/\, x]\mathrm{ann}_\ell(v_2) \in \mathbf{Nf}$.
Let t_2 be an arbitrary target term such that $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ holds. By lemma 25, $[t_2 \,/\, x]\mathrm{ann}_\ell(v_2) \in \mathbf{Nf}$. Then, $[t_2 \,/\, x]v_2 \in \mathbf{Nf'}$ by definition. This means we can derive $[t_2/x]\mathrm{ann}_\ell(v_2)\ e_2 \rightarrow [t_2/x]\mathrm{ann}_\ell(v_2\ e_2)$ by [E-LiftApp]. Thus, choosing $t'' = \mathrm{ann}_\ell(v_2\ e_2)$ leads to the desired conclusion.

[E-LiftAnnApp] We have $[t_1 \,/\, x]t = \mathrm{ann}_\ell(v_1)\ \langle\xi_2\rangle$ for some $v_1 \in \mathbf{Nf'}$. As $t = x$ or $t = x\ \langle\xi_2\rangle$ would violate the assumptions, $t = \mathrm{ann}_\ell(v_2)\ \langle\xi_2'\rangle$ for some v_2, ξ_2' and $[t_1 \,/\, x]v_2 = v_1$, $[t_1 \,/\, x]\xi_2' = \xi_2$.
Using lemma 17, there is a derivation for $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ \langle\xi_2'\rangle : \widehat{\tau} \,\&\, \xi$ of the following form.

$$\cfrac{\cfrac{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2) : \forall\beta\ \kappa.\widehat{\tau}_1 \,\&\, \xi_1 \qquad \emptyset \vdash_{\mathrm{s}} \xi_2 : \kappa}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ \langle\xi_2\rangle : [\xi_2 \,/\, \beta]\widehat{\tau}_1 \,\&\, \xi_1}\ [\text{T-AnnApp}] \\ \emptyset \vdash_{\mathrm{sub}} [\xi_2 \,/\, \beta]\widehat{\tau}_1 \leqslant \widehat{\tau} \\ \emptyset \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\mathrm{te}} \mathrm{ann}_\ell(v_2)\ \langle\xi_2\rangle : \widehat{\tau} \,\&\, \xi}\ [\text{T-Sub}]$$

Since $[t_1 \,/\, x]v_2 \in \mathbf{Nf'}$, $[t_1 \,/\, x]\mathrm{ann}_\ell(v_2) \in \mathbf{Nf}$.
Let t_2 be an arbitrary target term such that $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ holds. By lemma 25, $[t_2 \,/\, x]\mathrm{ann}_\ell(v_2) \in \mathbf{Nf}$. Then, $[t_2 \,/\, x]v_2 \in \mathbf{Nf'}$ by definition. This means we can derive $[t_2 \,/\, x]\mathrm{ann}_\ell(v_2)\ \langle\xi_2\rangle \rightarrow [t_2 \,/\, x]\mathrm{ann}_\ell(v_2\ \langle\xi_2\rangle)$ by [E-LiftAnnApp]. Thus, choosing $t'' = \mathrm{ann}_\ell(v_2\ \langle\xi_2\rangle)$ leads to the desired conclusion.

[E-LiftProj] We have $[t_1 \,/\, x]t = \mathrm{proj}_i(\mathrm{ann}_\ell(v_1))$ for some $v_1 \in \mathbf{Nf'}$. As $t = x$ or $t = \mathrm{proj}_i(x)$ would violate the assumptions, $t = \mathrm{proj}_i(\mathrm{ann}_\ell(v_2))$ for some v_2 and $[t_1 \,/\, x]v_2 = v_1$.

Using lemma 17, there is a derivation for $\emptyset \mid x{:}\widehat{\tau}'\&\xi' \vdash_{\text{te}} \text{proj}_i(\text{ann}_\ell(v_2)){:}\widehat{\tau}\&\xi$ of the following form.

$$\dfrac{\dfrac{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{ann}_\ell(v_2) : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_i}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{proj}_i(\text{ann}_\ell(v_2)) : \widehat{\tau}_i \,\&\, \xi_i}\,[\text{T-Proj}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_i \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_i \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{proj}_i(\text{ann}_\ell(v_2)) : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

Since $[t_1 \,/\, x]v_2 \in \mathbf{Nf}'$, $[t_1 \,/\, x]\text{ann}_\ell(v_2) \in \mathbf{Nf}$.
Let t_2 be an arbitrary target term such that $\emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ holds. By lemma 25, $[t_2 \,/\, x]\text{ann}_\ell(v_2) \in \mathbf{Nf}$. Then, $[t_2 \,/\, x]v_2 \in \mathbf{Nf}'$ by definition. This means we can derive $[t_2 \,/\, x]\text{proj}_i(\text{ann}_\ell(v_2)) \to [t_2 \,/\, x]\text{ann}_\ell(\text{proj}_i(v_2))$ by [E-LiftProj]. Thus, choosing $t'' = \text{ann}_\ell(\text{proj}_i(v_2))$ leads to the desired conclusion.

[E-LiftCase] We have $[t_1 \,/\, x]t = \textbf{case}\ \text{ann}_\ell(v_1)\ \textbf{of}\ \{\text{inl}(x_1) \to e_1; \text{inr}(x_2) \to e_2\}$ for some $v_1 \in \mathbf{Nf}'$. As $t = x$ or $t = \textbf{case}\ x\ \textbf{of}\ \{...\}$ would violate the assumptions, $t = \textbf{case}\ \text{ann}_\ell(v_2)\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\}$ for some v_2, e_1', e_2' and $[t_1 \,/\, x]v_2 = v_1$, $[t_1 \,/\, x]e_1' = e_1$, $[t_1 \,/\, x]e_2' = e_2$.
Using lemma 17, there is a derivation of the following form.

$$\dfrac{\dfrac{\begin{array}{c}\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{ann}_\ell(v_2) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi_3 \\ \emptyset \mid x : \widehat{\tau}' \,\&\, \xi', x_1 : \widehat{\tau}_1 \,\&\, \xi_1 \vdash_{\text{te}} e_1' : \widehat{\tau}_3 \,\&\, \xi_3 \\ \emptyset \mid x : \widehat{\tau}' \,\&\, \xi', x_2 : \widehat{\tau}_2 \,\&\, \xi_2 \vdash_{\text{te}} e_2' : \widehat{\tau}_3 \,\&\, \xi_3\end{array}}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \textbf{case}\ \text{ann}_\ell(v_2)\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\} : \widehat{\tau}_3 \,\&\, \xi_3}\,[\text{T-Case}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_3 \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \textbf{case}\ \text{ann}_\ell(v_2)\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\} : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

Since $[t_1 \,/\, x]v_2 \in \mathbf{Nf}'$, $[t_1 \,/\, x]\text{ann}_\ell(v_2) \in \mathbf{Nf}$.
Let t_2 be an arbitrary target term such that $\emptyset \mid \emptyset \vdash_{\text{te}} t_2 : \widehat{\tau}' \,\&\, \xi'$ holds. By lemma 25, $[t_2 \,/\, x]\text{ann}_\ell(v_2) \in \mathbf{Nf}$. Then, $[t_2 \,/\, x]v_2 \in \mathbf{Nf}'$ by definition. This means we can derive $[t_2 \,/\, x]\ \textbf{case}\ (\text{ann}_\ell(v_2))\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\} \to [t_2 \,/\, x]\text{ann}_\ell(\ \textbf{case}\ v_2\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\})$ by [E-LiftCase]. Thus, choosing $t'' = \text{ann}_\ell(\ \textbf{case}\ v_2\ \textbf{of}\ \{\text{inl}(x_1) \to e_1'; \text{inr}(x_2) \to e_2'\})$ leads to the desired conclusion.

[E-LiftSeq] We have $[t_1 \,/\, x]t = \text{seq}\ (\text{ann}_\ell(v_1))\ t_3$ for some v' and t_3. Again, we can rule out the cases $t = x$ and $t = \text{seq}\ x\ t_3$ due to $\emptyset \vdash_{\text{sub}} \xi' \not\sqsubseteq \xi$. Then, $t = \text{seq}\ (\text{ann}_\ell(v_2))\ t_4$ such that $[t_1 \,/\, x]v_2 = v_1$ and $[t_1 \,/\, x]t_4 = t_3$.
Using lemma 17, there is a derivation for $\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{seq}\ (\text{ann}_\ell(v_2))\ t_4 : \widehat{\tau} \,\&\, \xi$ of the following form.

$$\dfrac{\dfrac{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{ann}_\ell(v_2) : \widehat{\tau}_1' \,\&\, \xi_1 \qquad \emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} t_4 : \widehat{\tau}_1 \,\&\, \xi_1}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{seq}\ (\text{ann}_\ell(v_2))\ t_4 : \widehat{\tau}_1 \,\&\, \xi_1}\,[\text{T-Seq}] \quad \emptyset \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau} \quad \emptyset \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi}{\emptyset \mid x : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \text{seq}\ (\text{ann}_\ell(v_2))\ t_4 : \widehat{\tau} \,\&\, \xi}\,[\text{T-Sub}]$$

We note that $[t_1 \mathbin{/} x]\mathrm{ann}_\ell(v_2) = \mathrm{ann}_\ell(v_1) \in \mathbf{Nf}$. By lemma 25, for all terms t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$, $[t_2 \mathbin{/} x]\mathrm{ann}_\ell(v_2) \in \mathbf{Nf}$ and $[t_1 \mathbin{/} x]\mathrm{ann}_\ell(v_2) \simeq [t_2 \mathbin{/} x]\mathrm{ann}_\ell(v_2)$.
Therefore, the following derivation is valid for all such t_2.

$$\frac{[t_2 \mathbin{/} x]v_2 \in \mathbf{Nf}'}{\mathrm{seq}\ (\mathrm{ann}_\ell([t_2 \mathbin{/} x]v_2))\ ([t_2 \mathbin{/} x]t_4) \to \mathrm{ann}_\ell(\mathrm{seq}\ ([t_2 \mathbin{/} x]v_2)\ ([t_2 \mathbin{/} x]t_4))}\ [\text{E-LiftSeq}]$$

By setting $t'' = \mathrm{ann}_\ell(\mathrm{seq}\ v_2\ t_4)$ we get the desired result.

[E-JoinAnn] We have $[t_1 \mathbin{/} x]t = \mathrm{ann}_{\ell_1}(\mathrm{ann}_{\ell_2}(v_1))$ with $v_1 \in \mathbf{Nf}'$. The assumptions let us rule out the cases $t = x$ and $t = \mathrm{ann}_{\ell_1}(x)$. Hence, $t = \mathrm{ann}_{\ell_1}(\mathrm{ann}_{\ell_2}(v_2))$ for some v_2 such that $[t_1 \mathbin{/} x]v_2 = v_1$. By lemma 18, we have $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\mathrm{te}} \mathrm{ann}_{\ell_2}(v_2) : \widehat{\tau} \mathbin{\&} \xi$.
Note that $\mathrm{ann}_{\ell_2}(v_1) \in \mathbf{Nf}$. We can apply lemma 25, therefore $[t_2/x]\mathrm{ann}_{\ell_2}(v_2) \in \mathbf{Nf}$ for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$. This allows us to derive $[t_2 \mathbin{/} x]\mathrm{ann}_{\ell_1}(\mathrm{ann}_{\ell_2}(v_2)) \to [t_2 \mathbin{/} x]\mathrm{ann}_{\ell_1 \sqcup \ell_2}(v_2)$ by [E-JoinAnn] for all such t_2.
Choosing $t'' = \mathrm{ann}_{\ell_1 \sqcup \ell_2}(v_2)$ results in the desired statement.

By theorem 2, we have $\emptyset \mid \emptyset \vdash_{\mathrm{te}} [t_2 \mathbin{/} x]t'' : \widehat{\tau} \mathbin{\&} \xi$ for all t_2 such that $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$. By lemma 26, $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\mathrm{te}} t'' : \widehat{\tau} \mathbin{\&} \xi$.

And finally, we extend this statement to a sequence of reduction steps using the previous lemmas.

Theorem 7 (Noninterference). *Let t be a target term such that $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\mathrm{te}} t : \widehat{\tau} \mathbin{\&} \xi$ and $\emptyset \vdash_{\mathrm{sub}} \xi' \not\sqsubseteq \xi$. Let v be a value.*

If there is a t_1 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_1 : \widehat{\tau}' \mathbin{\&} \xi'$ such that $[t_1 \mathbin{/} x]t \to^* v$, then there is a t' such that for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$ we have $[t_2 \mathbin{/} x]t \to^* [t_2 \mathbin{/} x]t'$ and $[t_1 \mathbin{/} x]t' \simeq [t_2 \mathbin{/} x]t'$.

Proof. By induction on the length n of the reduction sequence $[t_1 \mathbin{/} x]t \to^* v$.

$n = 0$ In this case, $[t_1 \mathbin{/} x]t = v \in \mathbf{Nf}$. By lemma 25, we have for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$, $[t_2 \mathbin{/} x]t \in \mathbf{Nf}$ and $[t_1 \mathbin{/} x]t \simeq [t_2 \mathbin{/} x]t$. We choose $t' = t$ and get $[t_2 \mathbin{/} x]t \to^* [t_2 \mathbin{/} x]t'$ and $[t_1 \mathbin{/} x]t' \simeq [t_2 \mathbin{/} x]t'$.

$n = n' + 1$ We have $[t_1 \mathbin{/} x]t \to t_3$ and $t_3 \to^* v$. By lemma 27, there is a t'' such that $\emptyset \mid x : \widehat{\tau}' \mathbin{\&} \xi' \vdash_{\mathrm{te}} t'' : \widehat{\tau} \mathbin{\&} \xi$ holds and for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$, $[t_2 \mathbin{/} x]t \to [t_2 \mathbin{/} x]t''$ holds. In particular, $[t_1 \mathbin{/} x]t \to [t_1 \mathbin{/} x]t''$ and by lemma 24, $t_3 = [t_1 \mathbin{/} x]t''$.
By induction, we get a t' such that for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$ we have $[t_2 \mathbin{/} x]t'' \to^* [t_2 \mathbin{/} x]t'$ and $[t_1 \mathbin{/} x]t' \simeq [t_2 \mathbin{/} x]t'$.
Combining this with the previous result, we can conclude for all t_2 with $\emptyset \mid \emptyset \vdash_{\mathrm{te}} t_2 : \widehat{\tau}' \mathbin{\&} \xi'$ that $[t_2 \mathbin{/} x]t \to^* [t_2 \mathbin{/} x]t'$ holds.

E Proofs for Section 6

The following lemma characterizes how the free variables in a pattern type relate to the argument and pattern variables of the corresponding judgment.

Lemma 28. *If for an annotated type $\widehat{\tau}$, a dependency term ξ and sort environments Σ, Σ' the judgment $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \rhd \Sigma'$ holds, then*

1. $\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma') = \emptyset$,
2. $\mathrm{fav}(\widehat{\tau}) \cup \mathrm{fav}(\xi) = \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma')$, *and*
3. $\Sigma, \Sigma' \vdash_{\mathrm{s}} \xi : \star$ *and* $\Sigma, \Sigma' \vdash_{\mathrm{wft}} \widehat{\tau}$.

Proof. By induction on the derivation of $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \rhd \Sigma'$.

[P-Unit] In the premise of the [P-Unit] rule we have $\beta \notin \overline{\alpha_i}$. Since $\mathrm{dom}(\Sigma) = \overline{\alpha_i}$ and $\mathrm{dom}(\Sigma') = \{\beta\}$ in this case, we have $\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma') = \emptyset$.
Furthermore, we have $\mathrm{fav}(\widehat{\mathrm{unit}}) = \emptyset$ and $\mathrm{fav}(\beta\ \overline{\alpha_i}) = \{\beta\} \cup \alpha_i = \mathrm{dom}(\Sigma') \cup \mathrm{dom}(\Sigma)$.
Lastly, we have $\Sigma' = \beta :: \overline{\Sigma(\alpha_i) \Rightarrow} \star$. By repeated application of [S-App], we can derive $\Sigma, \Sigma' \vdash_{\mathrm{s}} \beta\ \overline{\alpha_i} : \star$. By [W-Unit] we have $\Sigma, \Sigma' \vdash_{\mathrm{wft}} \widehat{\mathrm{unit}}$.

[P-Sum] In this case, $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$. The premises of the [P-Sum] rule are $\Sigma \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \rhd \Sigma'_1$, $\Sigma \vdash_p \widehat{\tau}_2 \,\&\, \xi_2 \rhd \Sigma'_2$ and $\beta \notin \Sigma, \Sigma'_1, \Sigma'_2$ with $\Sigma'_1 = \overline{\beta_j :: \kappa_{\beta_j}}$ and $\Sigma'_2 = \overline{\gamma_j :: \kappa_{\gamma_k}}$.
We can then apply the induction hypothesis in order to get
1. $\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma'_1) = \emptyset$ and $\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma'_2) = \emptyset$ and
2. $\mathrm{fav}(\widehat{\tau}_1) \cup \mathrm{fav}(\xi_1) = \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma'_1)$ and $\mathrm{fav}(\widehat{\tau}_2) \cup \mathrm{fav}(\xi_2) = \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma'_2)$, and
3. $\Sigma, \Sigma'_1 \vdash_{\mathrm{s}} \xi_1 : \star$ and $\Sigma, \Sigma'_1 \vdash_{\mathrm{wft}} \widehat{\tau}_1$ and $\Sigma, \Sigma'_2 \vdash_{\mathrm{s}} \xi_2 : \star$ and $\Sigma, \Sigma'_2 \vdash_{\mathrm{wft}} \widehat{\tau}_2$.

Since we have $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \Sigma'_1, \Sigma'_2$ and $\beta \notin \Sigma, \Sigma'_1, \Sigma'_2$ it follows that

$$\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma') = \mathrm{dom}(\Sigma) \cap (\{\beta\} \cup \mathrm{dom}(\Sigma'_1) \cup \mathrm{dom}(\Sigma'_2)) = \emptyset$$

and

$$\begin{aligned}
&\mathrm{fav}(\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle) \cup \mathrm{fav}(\beta\ \overline{\alpha_i}) \\
=&\mathrm{fav}(\widehat{\tau}_1) \cup \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\widehat{\tau}_2) \cup \mathrm{fav}(\xi_2) \cup \mathrm{fav}(\beta\ \overline{\alpha_i}) \\
=&\mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma'_1) \cup \mathrm{dom}(\Sigma'_2) \cup \{\beta\} = \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma').
\end{aligned}$$

Similar to the previous case, we can derive $\Sigma, \Sigma' \vdash_{\mathrm{s}} \beta\ \overline{\alpha_i} : \star$. By application of [W-Sum] and result (3) of the induction hypothesis, we get $\Sigma, \Sigma'_1, \Sigma'_2 \vdash_{\mathrm{wft}} \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$. The latter derivation is possible because all parts of the combined environment have disjoint domains and therefore do not shadow existing bindings (see lemma 9).

[P-Prod] Analogous to the [P-Sum] case.

[P-Arr] In this case, $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$. The premises of the [P-Arr] rule are $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \rhd \Sigma'_1$, $\Sigma, \Sigma'_1 \vdash_p \widehat{\tau}_1 \,\&\, \xi_2 \rhd \Sigma'_2$ and $\beta \notin \Sigma, \Sigma'_1, \Sigma'_2$ with $\Sigma'_1 = \overline{\beta_j :: \kappa_{\beta_j}}$ and $\Sigma'_2 = \overline{\gamma_j :: \kappa_{\gamma_k}}$.
We can then apply the induction hypothesis in order to get
1. $\emptyset \cap \mathrm{dom}(\Sigma'_1) = \emptyset$ and $\mathrm{dom}(\Sigma, \Sigma'_1) \cap \mathrm{dom}(\Sigma'_2) = \emptyset$,
2. $\mathrm{fav}(\widehat{\tau}_1) \cup \mathrm{fav}(\xi_1) = \emptyset \cup \mathrm{dom}(\Sigma'_1)$ and $\mathrm{fav}(\widehat{\tau}_2) \cup \mathrm{fav}(\xi_2) = \mathrm{dom}(\Sigma, \Sigma'_1) \cup \mathrm{dom}(\Sigma'_2)$, and
3. $\Sigma'_1 \vdash_{\mathrm{s}} \xi_1 : \star$ and $\Sigma'_1 \vdash_{\mathrm{wft}} \widehat{\tau}_1$ and $\Sigma, \Sigma'_1, \Sigma'_2 \vdash_{\mathrm{s}} \xi_2 : \star$ and $\Sigma, \Sigma'_1, \Sigma'_2 \vdash_{\mathrm{wft}} \widehat{\tau}_2$.

Since we have $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \Sigma'_2$ and $\beta \notin \Sigma, \Sigma'_1, \Sigma'_2$ it follows that

$$\mathrm{dom}(\Sigma) \cap \mathrm{dom}(\Sigma') = \mathrm{dom}(\Sigma) \cap (\{\beta\} \cup \mathrm{dom}(\Sigma'_2)) = \emptyset$$

and

$$\begin{aligned}
&\mathrm{fav}(\forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle) \cup \mathrm{fav}(\beta\ \overline{\alpha_i}) \\
=&((\mathrm{fav}(\widehat{\tau}_1) \cup \mathrm{fav}(\xi_1) \cup \mathrm{fav}(\widehat{\tau}_2) \cup \mathrm{fav}(\xi_2)) \setminus \overline{\beta_j}) \cup \mathrm{fav}(\beta\ \overline{\alpha_i}) \\
=&((\mathrm{dom}(\Sigma'_1) \cup \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma'_2)) \setminus \mathrm{dom}(\Sigma'_1)) \cup \{\beta\} \cup \mathrm{dom}(\Sigma) \\
=&\,\mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma'_2) \cup \{\beta\} = \mathrm{dom}(\Sigma) \cup \mathrm{dom}(\Sigma').
\end{aligned}$$

Similar to the previous case, we can derive $\Sigma, \Sigma' \vdash_{\mathrm{s}} \beta\ \overline{\alpha_i} : \star$.
We can show $\Sigma, \Sigma' \vdash_{\mathrm{wft}} \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ by repeatedly applying [W-Forall] and then showing that $\Sigma, \Sigma', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{wft}} \widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ holds.
We apply rule [W-Arr] whose premises are given by

1. $\Sigma, \Sigma', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{s}} \xi_1 : \star$ follows by context extension from $\Sigma'_1 \vdash_{\mathrm{s}} \xi_1 : \star$,
2. analogously, $\Sigma, \Sigma', \overline{\beta_j :: \beta_{\kappa_j}} \vdash_{\mathrm{wft}} \widehat{\tau}_1$ follows from $\Sigma'_1 \vdash_{\mathrm{wft}} \widehat{\tau}_1$
3. $\Sigma, \Sigma', \overline{\beta_j :: \beta_{\kappa_j}} \vdash_{\mathrm{s}} \xi_2 : \star$ follows from $\Sigma, \Sigma'_1, \Sigma'_2 \vdash_{\mathrm{s}} \xi_2 : \star$ by exploiting that the domains of Σ'_1 and Σ'_2 are disjoint and that β does not occur free in ξ_2 (which are both known by induction), and
4. analogously $\Sigma, \Sigma', \overline{\beta_j :: \beta_{\kappa_j}} \vdash_{\mathrm{wft}} \widehat{\tau}_1$ follows from $\Sigma, \Sigma'_1, \Sigma'_2 \vdash_{\mathrm{wft}} \widehat{\tau}_1$.

In particular, this lemma lets us conclude that for patterns occurring in argument positions, i.e. where the sort environment Σ is empty, the free variables of the pattern type are exactly the pattern variables of that type.

Moreover, pattern types are unique up to alpha-equivalence, as the following lemma demonstrates.

Lemma 29. *Suppose $\widehat{\tau}$ and $\widehat{\tau}'$ are annotated types and ξ and ξ' are dependency terms such that $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ and $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ and $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \triangleright \Sigma''$ for some Σ, Σ' and Σ''. Then Σ' and Σ'' are equal up to renaming, $\widehat{\tau} \equiv_\alpha [\Sigma' \,/\, \Sigma'']\widehat{\tau}'$ and $\xi \equiv_\alpha [\Sigma' \,/\, \Sigma'']\xi'$, i.e. they are alpha-equivalent up to the names of the pattern variables.*[3]

Proof. Proof by induction on the derivation tree of $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$.

[P-Unit] We have $\widehat{\tau} = \widehat{\mathrm{unit}}$ and from the assumption $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ we can infer that the only rule that possibly applies to $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \triangleright \Sigma'$ is also [P-Unit]. Hence, $\widehat{\tau}' = \widehat{\mathrm{unit}}$ as well.
We have $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$ and $\Sigma'' = \beta' :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$. Therefore, Σ' and Σ'' are equal up to renaming.
From the definition of [P-Unit] we know $\xi = \beta\ \overline{\alpha_i}$ and $\xi' = \beta'\ \overline{\alpha_i}$. Since by lemma 28, Σ' and Σ as well as Σ'' and Σ are disjoint, we have $[\Sigma' \,/\, \Sigma'']\xi' = [\beta \,/\, \beta']\xi' = \xi$ and therefore in particular $\xi \equiv_\alpha [\Sigma' \,/\, \Sigma'']\xi'$.

[3] The notation $[\Sigma'/\Sigma'']$ is used to denote a substitution that replaces every variable in Σ'' with the corresponding variable from Σ'

[P-SUM] We have $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ for some $\widehat{\tau}_1$, ξ_1, $\widehat{\tau}_2$ and ξ_2. Furthermore, we can infer from $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ that the the only rule that possibly applies to $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \rhd \Sigma''$ is [P-SUM]. Therefore, $\widehat{\tau}' = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle$ for some $\widehat{\tau}'_1$, ξ'_1, $\widehat{\tau}'_2$ and ξ'_2.
The premises of $\Sigma \vdash_p \widehat{\tau}\&\xi\rhd\Sigma'$ are $\Sigma \vdash_p \widehat{\tau}_1\&\xi_1\rhd\Sigma'_1$ and $\Sigma \vdash_p \widehat{\tau}_2\&\xi_2\rhd\Sigma'_2$, and the premises of $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \rhd \Sigma'$ are $\Sigma \vdash_p \widehat{\tau}'_1 \,\&\, \xi'_1 \rhd \Sigma''_1$ and $\Sigma \vdash_p \widehat{\tau}'_2 \,\&\, \xi'_2 \rhd \Sigma''_2$.
By induction, we know that Σ'_1 and Σ''_1 as well as Σ'_2 and Σ''_2 are equal up to renaming. Moreover, $\widehat{\tau}_1 \equiv_\alpha [\Sigma'_1 / \Sigma''_1]\widehat{\tau}'_1$, $\widehat{\tau}_2 \equiv_\alpha [\Sigma'_2 / \Sigma''_2]\widehat{\tau}'_2$, $\xi_1 \equiv_\alpha [\Sigma'_1 / \Sigma''_1]\xi'_1$ and $\xi_2 \equiv_\alpha [\Sigma'_2 \,/\, \Sigma''_2]\xi'_2$.
Suppose that $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$. Since $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow}\ \star, \Sigma'_1, \Sigma'_2$ and $\Sigma'' = \beta' :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \Sigma''_1, \Sigma''_2$, they are equal up to renaming.
Hence, we must also have $\widehat{\tau} \equiv_\alpha [\Sigma' \,/\, \Sigma'']\widehat{\tau}'$. Analogously to the previous case, we have $\xi = [\Sigma' \,/\, \Sigma'']\xi'$.

[P-PROD] Analogous to the previous case.

[P-ARR] We have $\widehat{\tau}_1 = \forall\overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ for some $\widehat{\tau}_1$, ξ_1, $\widehat{\tau}_2$ and ξ_2. Furthermore, we can infer from $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ that the only rule that applies to $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \rhd \Sigma''$ is [P-ARR]. Thus, $\widehat{\tau}' = \forall\overline{\beta'_{j'} :: \kappa_{\beta'_{j'}}}.\widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}'_2\langle\xi'_2\rangle$.
The premises of $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \rhd \Sigma'$ are $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \rhd \overline{\beta_j :: \kappa_{\beta_j}}$ and $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_p \widehat{\tau}_2 \,\&\, \xi_2 \rhd \Sigma'_2$, and the premises of $\Sigma \vdash_p \widehat{\tau}' \,\&\, \xi' \rhd \Sigma'$ are $\emptyset \vdash_p \widehat{\tau}'_1 \,\&\, \xi'_1 \rhd \overline{\beta'_{j'} :: \kappa'_{\beta'_{j'}}}$ and $\Sigma, \overline{\beta'_{j'} :: \kappa'_{\beta'_{j'}}} \vdash_p \widehat{\tau}'_2 \,\&\, \xi'_2 \rhd \Sigma''_2$.
By induction, we know that $\overline{\beta_j :: \kappa_{\beta_j}}$ is equal to $\overline{\beta'_{j'} :: \kappa'_{\beta'_{j'}}}$ up to renaming. Therefore, the environments have the same size and $\kappa_{\beta_j} = \kappa'_{\beta'_j}$ for all j. Moreover, $\widehat{\tau}_1 \equiv_\alpha [\overline{\beta_j} \,/\, \overline{\beta'_j}]\widehat{\tau}'_1$ and $\xi_1 \equiv_\alpha [\overline{\beta_j} \,/\, \overline{\beta'_j}]\xi'_1$.
Hence, we can perform renaming in the second premise and get $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_p [\overline{\beta_j} \,/\, \overline{\beta'_j}]\widehat{\tau}'_2 \,\&\, [\overline{\beta_j} \,/\, \overline{\beta'_j}]\xi'_2 \rhd \Sigma''_2$.
Now we can apply induction here as well, hence Σ'_2 and Σ''_2 are equal up to renaming and $\widehat{\tau}_2 \equiv_\alpha [\Sigma'_2 \,/\, \Sigma''_2, \overline{\beta_j} \,/\, \overline{\beta'_j}]\widehat{\tau}'_2$ and $\xi_2 \equiv_\alpha [\Sigma'_2 \,/\, \Sigma''_2, \overline{\beta_j} \,/\, \overline{\beta'_j}]\xi'_2$.
Suppose that $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$. Since $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow}\ \star, \Sigma'_2$ and $\Sigma'' = \beta' :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \Sigma''_2$, they are equal up to renaming.
Since the $\overline{\beta_j}$ are bound in $\widehat{\tau}$ and $\widehat{\tau}'$, we have indeed $\widehat{\tau} \equiv_\alpha [\Sigma' \,/\, \Sigma'']\widehat{\tau}'$ and similar to the previous cases, $\xi = [\Sigma' \,/\, \Sigma'']\xi'$.

The following lemma establishes that a pattern type is always conservative.

Lemma 30. *Let $\widehat{\tau}$ be an annotated type, ξ a dependency term and Σ, Σ' sort environments such that $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \rhd \Sigma'$ holds. Then $\widehat{\tau}$ is conservative.*

Proof. By induction on the derivation of $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \rhd \Sigma'$.

[P-UNIT] We have $\widehat{\tau} = \widehat{\mathrm{unit}}$ which is conservative by definition.

[P-SUM] We have $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$. By induction, both $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative, and therefore $\widehat{\tau}$ is conservative.

[P-PROD] Analogous to the previous case.

[P-ARR] We have $\widehat{\tau} = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ which is conservative because $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ holds by assumption of [P-ARR] and $\widehat{\tau}_2$ is conservative by induction.

Moreover, substitutions preserve conservativeness.

Lemma 31. *Let $\widehat{\tau}$ be a conservative type and θ a substitution on annotation variables. Then $\theta\widehat{\tau}$ is also conservative.*

Proof. By induction on $\widehat{\tau}$. Since $\widehat{\tau}$ is conservative, one of the following cases holds

$\widehat{\tau} = \widehat{\text{unit}}$ Trivial, since $\theta\widehat{\text{unit}} = \widehat{\text{unit}}$.

$\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ By induction, $\theta\widehat{\tau}_1$ and $\theta\widehat{\tau}_2$ are conservative. Therefore, $\theta(\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle) = \theta\widehat{\tau}_1\langle\theta\xi_1\rangle + \theta\widehat{\tau}_2\langle\theta\xi_2\rangle$ is also conservative.

$\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle$ Analogous to the previous case.

$\widehat{\tau} = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ We have $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\widehat{\tau}_2$ is conservative. By lemma 28, $\text{fav}(\widehat{\tau}_1) \cup \text{fav}(\xi_1) = \overline{\beta_j}$. Define $\theta' := \theta \restriction_{\text{dom}(\theta)\setminus\overline{\beta_j}}$. Then, $\theta\widehat{\tau} = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \theta'\widehat{\tau}_2\langle\theta'\xi_2\rangle$. By induction, $\theta'\widehat{\tau}_2$ is conservative. Hence $\theta\widehat{\tau}$, is also conservative.

The following lemma establishes basic correctness properties of the type completion relation. In particular, the completed type is always a pattern type.

Lemma 32. *Suppose there are Σ, τ, $\widehat{\tau}$, ξ and Σ' such that $\Sigma \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ holds, then*

1. *$\tau = \lfloor\widehat{\tau}\rfloor$, and*
2. *$\widehat{\tau} \,\&\, \xi$ is a pattern type, i.e. $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ holds*

Proof. By induction on the derivation tree of $\Sigma \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$.

[C-UNIT] By definition of the rule [C-UNIT], $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \text{unit}$, $\widehat{\tau} = \widehat{\text{unit}}$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$.
By definition, $\lfloor\widehat{\text{unit}}\rfloor = \text{unit}$ and by rule [P-UNIT], $\Sigma \vdash_p \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ holds.

[C-SUM] By definition of the rule [C-SUM], $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \tau_1 + \tau_2$, $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\beta_j :: \kappa_{\beta_j}}, \overline{\gamma_k :: \kappa_{\gamma_k}}$.
The premises are $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_c \tau_1 : \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_c \tau_2 : \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$.
By induction, we have $\lfloor\widehat{\tau}_1\rfloor = \tau_1$ and $\lfloor\widehat{\tau}_2\rfloor = \tau_1$, and therefore $\lfloor\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle\rfloor = \lfloor\widehat{\tau}_1\rfloor + \lfloor\widehat{\tau}_2\rfloor = \tau_1 + \tau_2$. Moreover, we get $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$ from which we can conclude $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \beta\ \overline{\alpha_i} \triangleright \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\beta_j :: \kappa_{\beta_j}}, \overline{\gamma_k :: \kappa_{\gamma_k}}$ by applying rule [P-SUM].

[C-PROD] Analogous to [C-SUM].

[C-ARR] In this case we have $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \tau_1 \to \tau_2$, $\widehat{\tau} = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\gamma_k :: \kappa_{\gamma_k}}$.
The premises are $\emptyset \vdash_c \tau_1 : \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_c \tau_2 : \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$.

By induction, we have $\lfloor \widehat{\tau}_1 \rfloor = \tau_1$ and $\lfloor \widehat{\tau}_2 \rfloor = \tau_1$, and therefore

$$\lfloor \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle \xi_1 \rangle \to \widehat{\tau}_2\langle \xi_2 \rangle \rfloor = \lfloor \widehat{\tau}_1\langle \xi_1 \rangle \to \widehat{\tau}_2\langle \xi_2 \rangle \rfloor = \lfloor \widehat{\tau}_1 \rfloor \to \lfloor \widehat{\tau}_2 \rfloor = \tau_1 \to \tau_2.$$

Moreover, we get $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_p \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$ from which we can conclude by [P-Arr]

$$\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle \xi_1 \rangle \to \widehat{\tau}_2\langle \xi_2 \rangle \,\&\, \beta\ \overline{\alpha_i} \triangleright \beta :: \overline{\kappa_{\alpha_i}} \Rightarrow \star, \overline{\gamma_k :: \kappa_{\gamma_k}}.$$

The following lemma shows that these terms indeed correspond to the least elements of the lattices.

Lemma 33. *For all sorts κ and all sort environments Σ, least annotations terms $\bot_\kappa$ are (1) well-sorted, i.e. $\Sigma \vdash_{\mathrm{s}} \bot_\kappa : \kappa$ and (2) least w.r.t. the subsumption relation $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$.*

Proof. 1. We show by induction on κ that $\Sigma \vdash_{\mathrm{s}} \bot_\kappa : \kappa$ holds for all Σ.

$\kappa = \star$ — By definition $\Sigma \vdash_{\mathrm{s}} \bot : \star$.

$\kappa = \kappa_1 \Rightarrow \kappa_2$ — By the induction hypothesis, $\Sigma, \beta :: \kappa_1 \vdash_{\mathrm{s}} \bot_{\kappa_2} : \kappa_2$. Therefore we can conclude $\Sigma \vdash_{\mathrm{s}} \lambda\beta :: \kappa_1.\bot_{\kappa_2} : \kappa_1 \Rightarrow \kappa_2$.

2. We show by induction on κ that $[\![\bot_\kappa]\!]_\rho = \bot \in V_\kappa$, i.e. that it is the bottom value of lattice V_κ for all environments ρ.
Suppose ρ is an arbitrary environment.

$\kappa = \star$ — By definition, $[\![\bot_\kappa]\!]_\rho = [\![\bot]\!]_\rho = \bot \in V_\star$.

$\kappa = \kappa_1 \Rightarrow \kappa_2$ — $[\![\bot_{\kappa_1 \Rightarrow \kappa_2}]\!]_\rho = [\![\lambda\beta :: \kappa_1.\bot_{\kappa_2}]\!]_\rho = \lambda v \in V_{\kappa_1}.\, [\![\bot_{\kappa_2}]\!]_{\rho[\beta \mapsto v]} = \lambda v \in V_{\kappa_1}.\bot$ which is exactly the bottom value of the pointwise extension lattice $V_{\kappa_1} \to V_{\kappa_2}$.

Consequentially, for all environments ρ and annotation terms ξ of sort κ we have $[\![\bot_\kappa]\!]_\rho = \bot \sqsubseteq [\![\xi]\!]_\rho$. Since this holds in particular for all environments compatible with an arbitrary Σ, we can conclude $\Sigma \vdash_{\mathrm{sub}} \bot_\kappa \sqsubseteq \xi$.

We proceed by showing that least type completions are well-formed, conservative types of the correct shape.

Lemma 34. *For all types τ of the underlying type system, (1) $\bot_\tau$ is conservative, (2) $\Sigma \vdash_{\mathrm{wft}} \bot_\tau$ for all sort environments Σ, i.e. $\bot_\tau$ is well-formed in any environment, and (3) $\lfloor \bot_\tau \rfloor = \tau$.*

Proof. Recall the definition of $\bot_\tau$:

$$\bot_\tau = [\overline{\bot_{\kappa_i} \,/\, \beta_i}]\widehat{\tau} \text{ for } \emptyset \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \overline{\beta_i :: \kappa_i}.$$

We define the substitution $\theta = [\overline{\bot_{\kappa_i} \,/\, \beta_i}] : \mathrm{dom}(\overline{\beta_i :: \kappa_i}) \to \mathbf{AnnTm}$. By lemma 33, the terms in the image of θ are always well-sorted.

By lemma 32, we have $\emptyset \vdash_p \widehat{\tau} \,\&\, \xi \triangleright \overline{\beta_i :: \kappa_i}$ and $\lfloor \widehat{\tau} \rfloor = \tau$. Since θ only applies to nested annotations, but does not change the shape of the type, we can conclude $\lfloor \theta\widehat{\tau} \rfloor = \tau$. Note that $\mathrm{fav}(\theta\beta) = \emptyset$ for all β and $\mathrm{fav}(\widehat{\tau}) = \overline{\beta_i}$ by lemma 28.

By lemma 30, $\widehat{\tau}$ is conservative and $\overline{\beta_i :: \kappa_i} \vdash_{\mathrm{wft}} \widehat{\tau}$. By lemma 11, $\emptyset \vdash_{\mathrm{wft}} \theta\widehat{\tau}$ and by lemma 31, $\theta\widehat{\tau}$ is conservative. By lemma 9 $\Sigma \vdash_{\mathrm{wft}} \theta\widehat{\tau}$ holds for any Σ. Since $\bot_\tau = \theta\widehat{\tau}$, this completes the proof.

We demonstrate that the given definition of least type completions indeed leads to least types with respect to the subtyping relation. For this, we need a slightly more general statement.

Lemma 35. *Suppose there are Σ, τ, $\widehat{\tau}$, ξ and Σ' such that $\Sigma \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ holds. Furthermore, let $\theta : \mathrm{dom}(\Sigma') \to$ **AnnTm** denote a substitution defined by $\theta(\beta) = \bot_{\Sigma'(\beta)}$. Then, for all conservative $\widehat{\tau}'$ and sort environments Σ'' such that $\lfloor \widehat{\tau}' \rfloor = \tau$ and $\Sigma'' \vdash_{\mathrm{wft}} \widehat{\tau}'$, we have (1) $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau} \leqslant \widehat{\tau}'$ and (2) $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi \sqsubseteq \bot$.*

Proof. By induction on the derivation tree of $\Sigma \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$.

[C-Unit] By definition of the rule [C-Unit], $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \mathrm{unit}$, $\widehat{\tau} = \widehat{\mathrm{unit}}$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$.
By definition $\theta\xi = (\lambda \overline{x_i :: \kappa_{\alpha_i}}.\bot_\star)\ \overline{\alpha_i}$. We have $\Sigma \vdash_{\mathrm{sub}} (\lambda \overline{x_i :: \kappa_{\alpha_i}}.\bot_\star)\ \overline{\alpha_i} \sqsubseteq \bot_\star$. Hence, we can apply lemma 33 and get $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi \sqsubseteq \bot$.
Since $\widehat{\mathrm{unit}}$ is the only conservative type $\widehat{\tau}'$ such that $\lfloor \widehat{\tau}' \rfloor = \mathrm{unit}$, we also have $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau} \leqslant \widehat{\tau}'$ by [Sub-Refl].

[C-Sum] By definition of the rule [C-Sum], $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \tau_1 + \tau_2$, $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\beta_j :: \kappa_{\beta_j}}, \overline{\gamma_k :: \kappa_{\gamma_k}}$.
The premises are $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_c \tau_1 : \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_c \tau_2 : \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$.
By definition of $\lfloor\cdot\rfloor$ and the fact that $\widehat{\tau}'$ is conservative, we know that $\widehat{\tau}' = \widehat{\tau}_1'\langle\xi_1'\rangle + \widehat{\tau}_2'\langle\xi_2'\rangle$ for some $\widehat{\tau}_1', \widehat{\tau}_2', \xi_1', \xi_2'$.
Note that the substitutions occurring in the induction hypothesis are simply restrictions of the substitution θ.
By induction, we have $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau}_1 \leqslant \widehat{\tau}_1'$ and $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi_1 \sqsubseteq \bot$ as well as $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau}_2 \leqslant \widehat{\tau}_2'$ and $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi_2 \sqsubseteq \bot$.
From the transitivity of the subsumption relation on dependency terms we can deduce $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi_1 \sqsubseteq \xi_1'$ and $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi_2 \sqsubseteq \xi_2'$, as $\bot$ is the least element of this order.
Subtyping rule [S-Sum] lets us conclude $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau} \leqslant \widehat{\tau}'$.
Analogously to the first case, we can also prove $\Sigma'' \vdash_{\mathrm{sub}} \theta\xi \sqsubseteq \bot$.

[C-Prod] Analogous to [C-Sum].

[C-Arr] In this case we have $\Sigma = \overline{\alpha_i :: \kappa_{\alpha_i}}$, $\tau = \tau_1 \to \tau_2$, $\widehat{\tau} = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$, $\xi = \beta\ \overline{\alpha_i}$ and $\Sigma' = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\gamma_k :: \kappa_{\gamma_k}}$.
The premises are $\emptyset \vdash_c \tau_1 : \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_c \tau_2 : \widehat{\tau}_2 \,\&\, \xi_2 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$.
By definition of $\lfloor\cdot\rfloor$ and the fact that $\widehat{\tau}'$ is conservative, we know that $\widehat{\tau}' = \forall \overline{\beta_j' :: \kappa_{\beta_j'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle$ for some $\widehat{\tau}_1', \widehat{\tau}_2', \xi_1', \xi_2'$. Since the number and sorts of the variables that are quantified over are determined by the definition

of conservative types and the shape of $\widehat{\tau}_1'$, we can assume without loss of generality that $\beta_j = \beta_j'$ for all j.
Due to $\widehat{\tau}'$ being conservative, we also know $\emptyset \vdash_p \widehat{\tau}_1' \& \xi_1' \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$. By lemma 32 applied to the first premise we have $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$. Since pattern types are unique up to alpha-equivalence (see lemma 29), we can again assume without loss of generality that $\widehat{\tau}_1' = \widehat{\tau}_1$ and $\xi_1' = \xi_1$. Then we have $\overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_1$ by [SUB-REFL] and $\overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \xi_1' \sqsubseteq \xi_1$ from the reflexivity of the subsumption relation. We can extend the contexts with arbitrary variables not occurring free in $\widehat{\tau}_1$ and ξ_1 (using lemmas 8 and 14), therefore $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_1$ and $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \xi_1' \sqsubseteq \xi_1$ also hold.
By induction, we have $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \theta\widehat{\tau}_2 \leqslant \widehat{\tau}_2'$ and $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \theta\xi_2 \sqsubseteq \bot$.
From the transitivity of the subsumption relation on dependency terms we can deduce $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \theta\xi_2 \sqsubseteq \xi_2'$, as $\bot$ is the least element of this order.
Subtyping rule [S-ARR] lets us conclude $\Sigma'', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \widehat{\tau}_1\langle\xi_1\rangle \to \theta\widehat{\tau}_2\langle\theta\xi_2\rangle \leqslant \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle$. The substitution θ only applies to the return type, because all of the free variables of $\widehat{\tau}_1$ and ξ_1 are in bound in $\widehat{\tau}$ (see also lemma 28).
Repeated application of [SUB-FORALL] gives us the desired $\Sigma'' \vdash_{\mathrm{sub}} \theta\widehat{\tau} \leqslant \widehat{\tau}'$.
Analogously to the previous cases, we can also prove $\Sigma'' \vdash_{\mathrm{sub}} \xi \sqsubseteq \bot$.

The previous lemma established that substituting the pattern variables of a pattern type in an arbitrary environment with the least annotations of the right sort always yields the least type under that environment. We now relate this statement to the definition of $\bot_\tau$.

Lemma 36. *For all types τ, the least type completions $\bot_\tau$ is least w.r.t. the subtyping relation $\leqslant$, i.e. for all conservative types $\widehat{\tau}$ with $\lfloor\widehat{\tau}\rfloor = \tau$ and sort environments Σ such that $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}$, we have $\Sigma \vdash_{\mathrm{sub}} \bot_\tau \leqslant \widehat{\tau}$.*

Proof. Let Σ be an arbitrary sort environment and $\widehat{\tau}'$ be a conservative type such that $\lfloor\widehat{\tau}'\rfloor = \tau$. Furthermore, there are ξ and $\widehat{\tau}$ such that $\emptyset \vdash_c \tau : \widehat{\tau} \,\&\, \xi \triangleright \Sigma'$ holds. By lemma 35, we have $\Sigma \vdash_{\mathrm{sub}} \theta\widehat{\tau} \leqslant \widehat{\tau}$ where θ is the substitution as defined in the lemma. We note that this exactly the same substitution that is used in the definition of $\bot_\tau$, thereby completing the proof.

Example 2. Consider the type ((unit $\times$ unit $\to$ unit + unit) $\to$ unit) $\to$ unit. The least completion is obtained by instantiating all its pattern variables with least annotations of the correct sort, resulting in the following type (with further reduction of dependency terms).

$$
\begin{array}{l}
\forall \beta_5\ \beta_4'. \\
\quad (\forall \beta_4\ \beta_1'\ \beta_2'\ \beta_3'. \\
\qquad (\forall \beta_1\ \beta_2\ \beta_3.(\mathrm{unit}\langle\beta_1\rangle \times \mathrm{unit}\langle\beta_2\rangle)\langle\beta_3\rangle \\
\qquad \to t_1 + t_2\ t_3\langle\cdot\rangle)\ \beta_4 \\
\qquad \to \mathrm{unit}\langle\beta_4'\ \beta_4\ \beta_1'\ \beta_2'\ \beta_3'\rangle\langle\cdot\rangle)\ \beta_5 \\
\qquad \to \mathrm{unit}\langle\bot\rangle \,\&\, \bot\langle\cdot\rangle
\end{array}
$$

where $ti = \mathrm{unit}\langle \mathit{betai}'\ \beta_1\ \beta_2\ \beta_3\rangle$ for $i = 1, 2, 3$.

We now move on to the proofs for soundness and completeness of the algorithm.

Since the reconstruction algorithm applies canonicalization after every step, the following two lemmas are needed to show that this still leads to a valid derivation. It does not matter what $[\![\xi]\!]_\Sigma$ evaluates to, as long as it is semantically equivalent to ξ.

Lemma 37. *If we have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t}{:}\widehat{\tau}\&\xi$, then we also have $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t}{:}\widehat{\tau}\&[\![\xi]\!]_\Sigma$.*

Proof. A fundamental property of canonicalization is that $[\![\xi]\!]_\Sigma$ is semantically equivalent to ξ under the environment Σ, that is to say $\Sigma \vdash_{\mathrm{sub}} \xi \sqsubseteq [\![\xi]\!]_\Sigma$ and $\Sigma \vdash_{\mathrm{sub}} [\![\xi]\!]_\Sigma \sqsubseteq \xi$. The conclusion follows from a single application of [T-SUB].

Lemma 38. *If we have $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}$, then $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau} \leqslant [\![\widehat{\tau}]\!]_\Sigma$ and $\Sigma \vdash_{\mathrm{sub}} [\![\widehat{\tau}]\!]_\Sigma \leqslant \widehat{\tau}$.*

Proof. By induction on the derivation tree of $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}$ using the fact that canonical forms preserve semantics.

Next, we establish the correctness of the matching algorithm $\mathcal{M}$ that is used in the type reconstruction algorithm.

Lemma 39. *Let $\widehat{\tau}'$ be a pattern type such that $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}' \mathbin{\&} \beta\ \overline{\alpha_i} \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\widehat{\tau}$ be a conservative type such that $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ and $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\mathrm{wft}} \widehat{\tau}$. Then there is a substitution $\theta = \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \widehat{\tau}'; \widehat{\tau})$ such that $\theta\widehat{\tau}' = \widehat{\tau}$ and $\Sigma \vdash_{\mathrm{s}} \theta\beta_j : \kappa_{\beta_j}$ for all $\beta_j \in \mathrm{dom}(\theta) = \overline{\beta_j} \setminus \{\beta\}$.*

Proof. By induction on the derivation of $\Sigma \vdash_p \widehat{\tau}' \mathbin{\&} \xi' \triangleright \overline{\beta_i :: \kappa_{\beta_i}}$.

[P-UNIT] We have $\widehat{\tau}' = \widehat{\mathrm{unit}}$ and $\overline{\beta_j :: \kappa_{\beta_j}} = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$. Since $\widehat{\tau}$ is conservative and $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$, we also have $\widehat{\tau} = \widehat{\mathrm{unit}}$. Therefore, $\theta = \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \widehat{\tau}'; \widehat{\tau}) = [\,]$. Clearly, $\theta\widehat{\tau}' = \theta\widehat{\mathrm{unit}} = \widehat{\mathrm{unit}} = \widehat{\tau}$ and $\emptyset = \mathrm{dom}(\theta) = \{\beta\} \setminus \{\beta\}$.

[P-SUM] We have $\widehat{\tau}' = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle$ for some $\widehat{\tau}'_1$, ξ'_1, $\widehat{\tau}'_2$ and ξ'_2. The premises of the rule are $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}'_1\&\xi'_1 \triangleright \overline{\gamma_k :: \kappa_{\gamma_k}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_p \widehat{\tau}'_2\&\xi'_2 \triangleright \overline{\gamma'_{k'} :: \kappa_{\gamma'_{k'}}}$ with $\overline{\beta_j :: \kappa_{\beta_j}} = \beta :: \overline{\kappa_{\alpha_i} \Rightarrow} \star, \overline{\gamma_k :: \kappa_{\gamma_k}}, \overline{\gamma'_{k'} :: \kappa_{\gamma'_{k'}}}$. By the definition of the pattern rules, $\xi'_1 = \beta'\ \overline{\alpha_i}$ and $\xi'_2 = \beta''\ \overline{\alpha_i}$ for some $\beta' :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$ and $\beta'' :: \overline{\kappa_{\alpha_i} \Rightarrow} \star$. Because $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ and $\widehat{\tau}$ is conservative, $\widehat{\tau} = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$ for some $\widehat{\tau}_1$, ξ_1, $\widehat{\tau}_2$ and ξ_2. By definition, both $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative as well. From $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\mathrm{wft}} \widehat{\tau}$ we can infer $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\mathrm{wft}} \widehat{\tau}_1$ and $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\mathrm{wft}} \widehat{\tau}_2$ as they are the premises of [W-SUM] which is the only rule that applies.
By induction, we get $\theta' = \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \widehat{\tau}'_1; \widehat{\tau}_1)$ and $\theta'' = \mathcal{M}(\alpha_i :: \kappa_{\alpha_i}; \widehat{\tau}'_2; \widehat{\tau}_2)$. It follows

$$\theta = \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \widehat{\tau}'; \widehat{\tau}) = [\beta' \mapsto \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}.\xi_1, \beta'' \mapsto \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}.\xi_2] \circ \theta' \circ \theta''.$$

We have $\mathrm{dom}(\theta) = \{\beta', \beta''\} \cup \mathrm{dom}(\theta') \cup \mathrm{dom}(\theta'') = \{\beta', \beta''\} \cup (\overline{\gamma_k} \setminus \{\beta'\}) \cup (\overline{\gamma'_{k'}} \setminus \{\beta''\}) = \overline{\beta_j} \setminus \{\beta\}$. Moreover, $\theta(\widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}'_2\langle\xi'_2\rangle) = \theta\widehat{\tau}'_1\langle\theta\xi'_1\rangle + \theta\widehat{\tau}'_2\langle\theta\xi'_2\rangle = \theta'\widehat{\tau}'_1\langle\xi_1\rangle + \theta''\widehat{\tau}'_2\langle\xi_2\rangle = \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle = \widehat{\tau}$ where the second step is justified by lemma 28.

Lastly, we need to show $\Sigma \vdash_s \theta\beta' : \overline{\kappa_{\alpha_i}} \Rightarrow \star$ and $\Sigma \vdash_s \theta\beta'' : \overline{\kappa_{\alpha_i}} \Rightarrow \star$. These statements can be derived by repeated application of [S-Abs].

$$\frac{\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_s \xi_1 : \star}{\Sigma \vdash_s \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}.\xi_1 : \overline{\kappa_{\alpha_i}} \Rightarrow \star}\ \overline{[\text{S-Abs}]_i}$$

The assumption follows as a premise of $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\text{wft}} \widehat{\tau}$. This holds analogously for $\theta\beta''$ and the remaining judgments follow by induction. Hence, $\Sigma \vdash_s \theta\beta_j : \kappa_{\beta_j}$ for all $\beta_j \in \text{dom}(\theta)$.

[P-Prod] Analogous to previous case.

[P-Arr] We have $\widehat{\tau}' = \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle$ for some $\widehat{\tau}_1'$, ξ_1', $\widehat{\tau}_2'$ and ξ_2'. The premises of the rule are $\emptyset \vdash_p \widehat{\tau}_1' \,\&\, \xi_1' \triangleright \overline{\beta_k' :: \kappa_{\beta_k'}}$ and $\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}} \vdash_p \widehat{\tau}_2' \,\&\, \xi_2' \triangleright \overline{\gamma_l :: \kappa_{\gamma_l}}$ where $\overline{\beta_j :: \kappa_{\beta_k}} = \beta :: \overline{\kappa_{\alpha_i}} \Rightarrow \star, \overline{\gamma_l :: \kappa_{\gamma_l}}$.

Since we have $\lfloor\widehat{\tau}\rfloor = \lfloor\widehat{\tau}'\rfloor$ and $\widehat{\tau}$ is conservative, $\widehat{\tau} = \forall\overline{\beta_l'' :: \kappa_{\beta_l''}}.\widehat{\tau}_1\langle\xi_1\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ for some $\widehat{\tau}_1$, $\widehat{\tau}_2$, ξ_1 and ξ_2 with $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \xi_1 \triangleright \overline{\beta_l'' :: \kappa_{\beta_l''}}$. Because pattern types are unique up to renaming (see lemma 29), we can assume without loss of generality that $\overline{\beta_l'' :: \kappa_{\beta_l''}} = \overline{\beta_k' :: \kappa_{\beta_k'}}$ and therefore $\widehat{\tau}_1 = \widehat{\tau}_1'$ and $\xi_1 = \xi_1'$. By the definition of the pattern rules, $\xi_2 = \gamma'\ \overline{\alpha_i}\ \overline{\beta_k'}$ for some $\gamma' :: \overline{\kappa_{\alpha_i}} \Rightarrow \overline{\kappa_{\beta_k'}} \Rightarrow \star$. By assumption, $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}} \vdash_{\text{wft}} \widehat{\tau}$. But then we must also have $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}} \vdash_{\text{wft}} \widehat{\tau}_2$.

We can now apply the induction hypothesis. We get a substitution

$$\theta' = \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}; \widehat{\tau}_2'; \widehat{\tau}_2)$$

such that $\theta'\widehat{\tau}_2' = \widehat{\tau}_2$ and $\Sigma \vdash_s \theta'\gamma_l : \kappa_{\gamma_l}$ for all $\gamma_l \in \text{dom}(\theta') = \overline{\gamma_l} \setminus \{\gamma'\}$. Unfolding the definition of $\mathcal{M}$ yields

$$\begin{aligned}
\theta =&\mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \widehat{\tau}'; \widehat{\tau})\\
=&\mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}; \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle; \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2\langle\xi_2\rangle)\\
&\vdots\\
=&\mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}; \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2'\langle\xi_2'\rangle; \widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2\langle\xi_2\rangle)\\
=&[\gamma' \mapsto \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}.\xi_2] \circ \mathcal{M}(\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}; \widehat{\tau}_2'; \widehat{\tau}_2)\\
=&[\gamma' \mapsto \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}.\xi_2] \circ \theta'.
\end{aligned}$$

Clearly, $\text{dom}(\theta) = \overline{\gamma_l} = \overline{\beta_j} \setminus \{\beta\}$. From the well-formedness of $\widehat{\tau}$ we can infer $\Sigma, \overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}} \vdash_s \xi_2 : \star$. Therefore, $\Sigma \vdash_s \lambda\overline{\alpha_i :: \kappa_{\alpha_i}}, \overline{\beta_k' :: \kappa_{\beta_k'}}.\xi_2 : \overline{\kappa_{\alpha_i}} \Rightarrow \overline{\kappa_{\beta_k'}} \Rightarrow \star$ by repeated application of [S-Abs]. Also, $\theta\widehat{\tau}' = \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\theta\widehat{\tau}_1'\langle\theta\xi_1'\rangle \to \theta\widehat{\tau}_2'\langle\theta\xi_2'\rangle = \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \theta\widehat{\tau}_2'\langle\theta\xi_2'\rangle = \forall\overline{\beta_k' :: \kappa_{\beta_k'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_2\langle\xi_2\rangle = \widehat{\tau}$. The second step is justified by the fact that the free variables of $\widehat{\tau}_1'$ and ξ_1' are disjoint from $\text{dom}(\theta)$ (see lemma 28). The third step follows from the induction hypothesis.

Similarly, we show some useful properties of the join $\widehat{\tau}_1 \sqcup \widehat{\tau}_2$ of two annotated types.

Lemma 40. *Let $\widehat{\tau}_1$ and $\widehat{\tau}_2$ be conservative types and Σ a sort environment such that $\lfloor \widehat{\tau}_1 \rfloor = \lfloor \widehat{\tau}_2 \rfloor = \tau$ for some τ, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_1$ and $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_2$.*

Then $\lfloor \widehat{\tau}_1 \sqcup \widehat{\tau}_2 \rfloor = \tau$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_1 \sqcup \widehat{\tau}_2$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_1 \sqcup \widehat{\tau}_2$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_1 \sqcup \widehat{\tau}_2$ and $\widehat{\tau}_1 \sqcup \widehat{\tau}_2$ is conservative.

Proof. By induction on τ (i.e. the underlying structure of both $\widehat{\tau}_1$ and $\widehat{\tau}_2$).

$\tau = \mathrm{unit}$ Then $\widehat{\tau}_1 = \widehat{\tau}_2 = \widehat{\mathrm{unit}}$ because $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are also conservative. By definition, $\widehat{\mathrm{unit}} \sqcup \widehat{\mathrm{unit}} = \widehat{\mathrm{unit}}$. Clearly, $\lfloor \widehat{\mathrm{unit}} \rfloor = \mathrm{unit}$ by definition, $\Sigma \vdash_{\mathrm{sub}} \widehat{\mathrm{unit}} \leqslant \widehat{\mathrm{unit}}$ by [Sub-Refl], $\Sigma \vdash_{\mathrm{wft}} \widehat{\mathrm{unit}}$ by [W-Unit] and $\widehat{\mathrm{unit}}$ conservative by definition.

$\tau = \tau' + \tau''$ Then $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ and $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ such that $\lfloor \widehat{\tau}'_1 \rfloor = \tau' = \lfloor \widehat{\tau}'_2 \rfloor$ and $\lfloor \widehat{\tau}''_1 \rfloor = \tau'' = \lfloor \widehat{\tau}''_2 \rfloor$, because $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative. Moreover, we know $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_1$ and $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_2$, and the only rule applying here is [W-Sum]. Therefore, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}'_1$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}''_1$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}'_2$ and $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}''_2$.
By induction, we get $\lfloor \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 \rfloor = \tau'$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_2 \leqslant \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2$ and $\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2$ conservative. Analogously, $\lfloor \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2 \rfloor = \tau''$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}''_2 \leqslant \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2$ and $\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2$ conservative.
We have $\widehat{\tau}_1 \sqcup \widehat{\tau}_2 = (\widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle) \sqcup (\widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle) = (\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle$. Clearly, $\lfloor (\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle \rfloor = \lfloor \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 \rfloor + \lfloor \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2 \rfloor = \tau' + \tau'' = \tau$.
We can derive

$$\frac{\begin{array}{cc}\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 & \Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_2 \leqslant \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2 \\ \Sigma \vdash_{\mathrm{sub}} \xi'_1 \sqsubseteq \xi'_1 \sqcup \xi'_2 & \Sigma \vdash_{\mathrm{sub}} \xi''_1 \sqsubseteq \xi''_1 \sqcup \xi''_2\end{array}}{\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle \leqslant (\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle}\ [\textsc{Sub-Sum}]$$

and $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle \leqslant (\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle$ follows analogously. The resulting type $(\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle$ is conservative, as its parts are conservative by induction.
Lastly,

$$\frac{\begin{array}{cc}\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 & \Sigma \vdash_{\mathrm{s}} \xi'_1 \sqcup \xi'_2 : \star \\ \Sigma \vdash_{\mathrm{wft}} \widehat{\tau}''_1 \sqcup \widehat{\tau}''_2 & \Sigma \vdash_{\mathrm{s}} \xi''_1 \sqcup \xi''_2 : \star\end{array}}{\Sigma \vdash_{\mathrm{wft}} (\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2)\langle\xi'_1 \sqcup \xi'_2\rangle + (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle}$$

where the well-sortedness of the annotations follows from the well-formedness of $\widehat{\tau}_1$ and $\widehat{\tau}_2$ and [S-Join].

$\tau = \tau' \times \tau''$ Analogous to the previous case.

$\tau = \tau' \to \tau''$ Then $\widehat{\tau}_1 = \forall\overline{\beta_i :: \kappa_{\beta_i}}.\widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$ and $\widehat{\tau}_2 = \forall\overline{\beta'_j :: \kappa_{\beta'_j}}.\widehat{\tau}'_2\langle\xi'_2\rangle \to \widehat{\tau}''_2\langle\xi''_2\rangle$ since both are conservative. Since $\lfloor \widehat{\tau}_1 \rfloor = \tau = \lfloor \widehat{\tau}_2 \rfloor$, we also know $\lfloor \widehat{\tau}'_1 \rfloor = \tau' = \lfloor \widehat{\tau}'_2 \rfloor$ and $\lfloor \widehat{\tau}''_1 \rfloor = \tau'' = \lfloor \widehat{\tau}''_2 \rfloor$.
By definition of conservativeness, $\emptyset \vdash_p \widehat{\tau}'_1 \mathbin{\&} \xi'_1 \triangleright \overline{\beta_i :: \kappa_{\beta_i}}$ and $\emptyset \vdash_p \widehat{\tau}'_2 \mathbin{\&} \xi'_2 \triangleright \overline{\beta'_j :: \kappa_{\beta'_j}}$. By lemma 29, patterns are unique up to renaming. Hence, we can assume without loss of generality that $\overline{\beta_i :: \kappa_{\beta_i}} = \overline{\beta'_j :: \kappa_{\beta'_j}}$, and therefore $\widehat{\tau}'_1 = \widehat{\tau}'_2$ and $\xi'_1 = \xi'_2$.
Then, $\widehat{\tau}_1 \sqcup \widehat{\tau}_2 = \forall\overline{\beta_i :: \kappa_{\beta_i}}.\widehat{\tau}'_1\langle\xi'_1\rangle \to (\widehat{\tau}''_1 \sqcup \widehat{\tau}''_2)\langle\xi''_1 \sqcup \xi''_2\rangle$.

From $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1$ and $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$ we can infer $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{wft}} \widehat{\tau}_1''$ and $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{wft}} \widehat{\tau}_2''$.
The induction hypothesis yields $\lfloor \widehat{\tau}_1'' \sqcup \widehat{\tau}_2'' \rfloor = \tau''$, $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{sub}} \widehat{\tau}_1'' \leqslant \widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$, $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{sub}} \widehat{\tau}_2'' \leqslant \widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$, $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{wft}} \widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$ and the conservativeness of $\widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$.
Clearly, $\lfloor \widehat{\tau}_1 \sqcup \widehat{\tau}_2 \rfloor = \lfloor \forall \overline{\beta_i :: \kappa_{\beta_i}}. \widehat{\tau}_1' \langle \xi_1' \rangle \to (\widehat{\tau}_1'' \sqcup \widehat{\tau}_2'')\langle \xi_1'' \sqcup \xi_2'' \rangle \rfloor = \lfloor \widehat{\tau}_1' \rfloor \to \lfloor \widehat{\tau}_1'' \sqcup \widehat{\tau}_2'' \rfloor = \tau' \to \tau''$.
Using [Sub-Arr], we can derive $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{sub}} \widehat{\tau}_1'\langle \xi_1' \rangle \to \widehat{\tau}_1''\langle \xi_1'' \rangle \leqslant \widehat{\tau}_1'\langle \xi_1' \rangle \to (\widehat{\tau}_1'' \sqcup \widehat{\tau}_2'')\langle \xi_1'' \sqcup \xi_2'' \rangle$ and $\Sigma, \overline{\beta_i :: \kappa_{\beta_i}} \vdash_{\text{sub}} \widehat{\tau}_1'\langle \xi_1' \rangle \to \widehat{\tau}_2''\langle \xi_2'' \rangle \leqslant \widehat{\tau}_1'\langle \xi_1' \rangle \to (\widehat{\tau}_1'' \sqcup \widehat{\tau}_2'')\langle \xi_1'' \sqcup \xi_2'' \rangle$. The contravariant position can be treated invariantly, because we could assume $\widehat{\tau}_1' = \widehat{\tau}_2'$.
Repeated application of [Sub-Forall] results in the desired $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_1 \sqcup \widehat{\tau}_2$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_1 \sqcup \widehat{\tau}_2$.
As $\widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$ is conservative and $\emptyset \vdash_p \widehat{\tau}_1' \& \xi_1' \rhd \overline{\beta_i :: \kappa_{\beta_i}}$, $\widehat{\tau}_1 \sqcup \widehat{\tau}_2$ is also conservative.
Similarly, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1 \sqcup \widehat{\tau}_2$ follows from $\Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_1'$ and $\Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_1'' \sqcup \widehat{\tau}_2''$.

Based on this knowledge, we can now show that algorithm $\mathcal{R}$ is sound w.r.t. to the declarative type system.

Theorem 5. *Let t be a source term, Σ a sort environment and $\widehat{\Gamma}$ an annotated type environment well-formed under Σ such that $\mathcal{R}(\widehat{\Gamma}; \Sigma; t) = \widehat{t} : \widehat{\tau} \,\&\, \xi$ for some $\widehat{t}$, $\widehat{\tau}$ and ξ.*
Then, $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t} : \widehat{\tau} \,\&\, \xi$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}$, $\Sigma \vdash_{\text{s}} \xi : \star$ and $\widehat{\tau}$ is conservative.

Proof. By induction on t. All of the following cases assume an implicit application of the foregoing lemmas 37 and 38, as we only show that the types and effects before canonicalization are derivable in the annotated type system.

$t = x$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; x) = x : \widehat{\Gamma}(x)$. Suppose $\widehat{\Gamma}(x) = \widehat{\tau} \,\&\, \xi$. By assumption, $\widehat{\Gamma}$ is well-formed under Σ, therefore $\widehat{\tau}$ is conservative, $\Sigma \vdash_{\text{wft}} \widehat{\tau}$ and $\Sigma \vdash_{\text{s}} \xi : \star$.
$t = ()$ $\mathcal{R}(\widehat{\Gamma}; \Sigma; ()) = () : \widehat{\text{unit}} : \bot$. Trivially, $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} : \widehat{\text{unit}} \,\&\, \bot$ by [T-Unit], $\Sigma \vdash_{\text{wft}} \widehat{\text{unit}}$ by [W-Unit], $\Sigma \vdash_{\text{s}} \bot : \star$ by [S-Lat] and $\widehat{\text{unit}}$ is conservative by definition.
$t = \text{ann}_\ell(t')$ In this case $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{ann}_\ell(t')) = \text{ann}_\ell(\widehat{t'}) : \widehat{\tau}' \,\&\, \llbracket \xi' \sqcup \ell \rrbracket_\Sigma$ where $\widehat{t'} : \widehat{\tau}' \,\&\, \xi' = \mathcal{R}(\widehat{\Gamma}; \Sigma; t')$.
By induction, $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}' \& \xi'$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}'$, $\Sigma \vdash_{\text{s}} \xi' : \star$ and $\widehat{\tau}'$ is conservative.
We can then derive

$$\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}' \,\&\, \xi' \qquad \dfrac{}{\Sigma \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau}'}\,[\text{Sub-Refl}] \qquad \Sigma \vdash_{\text{sub}} \xi' \sqsubseteq \xi' \sqcup \ell}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau}' \,\&\, \xi' \sqcup \ell}\,[\text{T-Sub}]$$

and subsequently

$$\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} t : \widehat{\tau} \,\&\, \xi' \sqcup \ell \qquad \Sigma \vdash_{\text{sub}} \ell \sqsubseteq \xi' \sqcup \ell}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{ann}_\ell(\widehat{t'}) : \widehat{\tau} \,\&\, \xi' \sqcup \ell}\,[\text{T-Ann}]$$

Moreover, $\Sigma \vdash_{\text{s}} \xi' \sqcup \ell : \star$ by [S-Join].

$t = \text{seq } t_1\ t_2$ In this case $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{seq } t_1\ t_2) = \text{seq } \widehat{t_1}\ \widehat{t_2} : \widehat{\tau}_2 \,\&\, [\![\xi_1 \sqcup \xi_2]\!]_\Sigma$ where $\widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_1)$ and $\widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_2)$.

We can derive

$$\dfrac{\dfrac{\text{by induction}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1} \qquad \Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_1 \qquad \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_1 \sqcup \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1 \sqcup \xi_2}\ [\text{T-Sub}]$$

and

$$\dfrac{\dfrac{\text{by induction}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2} \qquad \Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_2 \qquad \Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi_1 \sqcup \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_1 \sqcup \xi_2}\ [\text{T-Sub}]$$

concluding with

$$\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1 \sqcup \xi_2 \qquad \Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_1 \sqcup \xi_2}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{seq } \widehat{t_1}\ \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_1 \sqcup \xi_2}\ [\text{T-Seq}]$$

Moreover, the conservativeness of $\widehat{\tau}_2$ and $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$ follow by induction and $\Sigma \vdash_{\text{s}} \xi_1 \sqcup \xi_2 : \star$ follows by induction and rule [S-Join].

$t = (t_1, t_2)$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1, t_2) = (\widehat{t_1}, \widehat{t_2}) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot$ where $\widehat{t_1} : \widehat{\tau}_1 \& \xi_1 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_1)$ and $\widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_2)$.

We can derive

$$\dfrac{\dfrac{\text{by induction}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1} \qquad \dfrac{\text{by induction}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} (\widehat{t_1}, \widehat{t_2}) : \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\ [\text{T-Pair}]$$

By induction, we also have $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$, $\Sigma \vdash_{\text{s}} \xi_1 : \star$ and $\Sigma \vdash_{\text{s}} \xi_2 : \star$. This allows us to conclude $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$. By [S-Lat], $\Sigma \vdash_{\text{s}} \bot : \star$. Lastly, since $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative by induction, so is $\widehat{\tau}_1\langle\xi_1\rangle + \widehat{\tau}_2\langle\xi_2\rangle$.

$t = \text{inl}_{\tau_2}(t_1)$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{inl}_{\tau_2}(t_1)) = \text{inl}_{\tau_2}(\widehat{t_1}) : \widehat{\tau}_1\langle\xi_1\rangle + \bot_{\tau_2}\langle\bot\rangle \,\&\, \bot$ where $\widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_1)$.

By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1$, $\Sigma \vdash_{\text{s}} \xi_1 : \star$ and $\widehat{\tau}_1$ is conservative. By lemma 34, $\bot_{\tau_2}$ is conservative, $\Sigma \vdash_{\text{wft}} \bot_{\tau_2}$ and $\lfloor \bot_{\tau_2} \rfloor = \tau_2$. We can therefore apply [T-Inl] and get $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{inl}_{\tau_2}(\widehat{t_1}) : \widehat{\tau}_1\langle\xi_1\rangle + \bot_{\tau_2}\langle\bot\rangle \,\&\, \bot$. By [W-Sum], $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle + \bot_{\tau_2}\langle\bot\rangle$ and by [S-Lat], $\Sigma \vdash_{\text{s}} \bot : \star$. Lastly, as both $\widehat{\tau}_1$ and $\bot_{\tau_2}$ are conservative, so is $\widehat{\tau}_1\langle\xi_1\rangle + \bot_{\tau_2}\langle\bot\rangle$.

$t = \text{inl}_{\tau_1}(t_2)$ Analogous to the previous case.

$t = \text{proj}_i(t')$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{proj}_i(t')) = \text{proj}_i(\widehat{t'}) : \widehat{\tau}_i \,\&\, [\![\xi \sqcup \xi_i]\!]_\Sigma$ where $\widehat{t'} : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi = \mathcal{R}(\widehat{\Gamma}; \Sigma; t')$ and $i = 1$ or $i = 2$.

By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \xi$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle$, $\Sigma \vdash_{\text{s}} \xi : \star$ and $\widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle$ is conservative. This implies that $\Sigma \vdash_{\text{wft}} \widehat{\tau}_i$ and $\Sigma \vdash_{\text{s}} \xi_i : \star$ hold and that $\widehat{\tau}_i$ is conservative.

We can derive the following subtyping judgment

$$\dfrac{\begin{array}{cc} \Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_1 & \Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_2 \\ \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_1 \sqcup \xi & \Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi_2 \sqcup \xi \end{array}}{\Sigma \vdash_{\text{sub}} \widehat{\tau}_1\langle\xi_1\rangle \times \widehat{\tau}_2\langle\xi_2\rangle \leqslant \widehat{\tau}_1\langle\xi_1 \sqcup \xi\rangle \times \widehat{\tau}_2\langle\xi_2 \sqcup \xi\rangle}\ [\textsc{Sub-Prod}]$$

Together with $\Sigma \vdash_{\text{sub}} \xi \sqsubseteq \xi_i \sqcup \xi$ we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}_1\langle\xi_1 \sqcup \xi\rangle \times \widehat{\tau}_2\langle\xi_2 \sqcup \xi\rangle \,\&\, \xi_i \sqcup \xi$ using [T-Sub]. This has the right form to apply [T-Proj], leaving us with $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \text{proj}_i(\widehat{t'}) : \widehat{\tau}_i \,\&\, \xi_i \sqcup \xi$.

$t = \textbf{case}\ t'\ \textbf{of}\ \{\text{inl}(x) \to t_1; \text{inr}(y) \to t_2\}$ In this case, $\mathcal{R}(\widehat{\Gamma}; \Sigma; t) = \textbf{case}\ \widehat{t'}\ \textbf{of}\ \{\text{inl}(x) \to \widehat{t_1}; \text{inr}(y) \to \widehat{t_2}\} : [\![\widehat{\tau}_1 \sqcup \widehat{\tau}_2]\!]_\Sigma \,\&\, [\![\xi_1 \sqcup \xi_2 \sqcup \xi_2]\!]_\Sigma$ where $\widehat{t_1} : \widehat{\tau}\langle\xi\rangle + \widehat{\tau}'\langle\xi'\rangle \,\&\, \xi_1 = \mathcal{R}(\widehat{\Gamma}; \Sigma; t_1)$, $\widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2 = \mathcal{R}(\widehat{\Gamma}, x : \widehat{\tau} \,\&\, \xi; \Sigma; t_2)$ and $\widehat{t_3} : \widehat{\tau}_3 \,\&\, \xi_3 = \mathcal{R}(\widehat{\Gamma}, y : \widehat{\tau}' \,\&\, \xi'; \Sigma; t_3)$.

By induction, $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}\langle\xi\rangle + \widehat{\tau}'\langle\xi'\rangle \,\&\, \xi_1$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}\langle\xi\rangle + \widehat{\tau}'\langle\xi'\rangle$, $\Sigma \vdash_{\text{s}} \xi_1 : \star$ and $\widehat{\tau}\langle\xi\rangle + \widehat{\tau}'\langle\xi'\rangle$ is conservative. Hence, both $\widehat{\Gamma}, x : \widehat{\tau} \,\&\, \xi$ and $\widehat{\Gamma}, x : \widehat{\tau}' \,\&\, \xi'$ are well-formed under Σ and we can apply the induction hypothesis to the remaining recursive calls. This results in $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau} \,\&\, \xi \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$, $\Sigma \vdash_{\text{s}} \xi_2 : \star$ and $\widehat{\tau}_2$ conservative, and similarly for the third call.

Using [T-Sub] and lemma 40, we can derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}\langle\xi\rangle + \widehat{\tau}'\langle\xi'\rangle \,\&\, \xi_1 \sqcup \xi_2 \sqcup \xi_3$, $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau} \,\&\, \xi \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \sqcup \widehat{\tau}_3 \,\&\, \xi_1 \sqcup \xi_2 \sqcup \xi_3$ and $\Sigma \mid \widehat{\Gamma}, y : \widehat{\tau}' \,\&\, \xi' \vdash_{\text{te}} \widehat{t_3} : \widehat{\tau}_2 \sqcup \widehat{\tau}_3 \,\&\, \xi_1 \sqcup \xi_2 \sqcup \xi_3$. This makes it possible to apply [T-Case], leaving us with the desired

$$\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \textbf{case}\ \widehat{t_1}\ \textbf{of}\ \{\text{inl}(x) \to \widehat{t_2}; \text{inr}(y) \to \widehat{t_3}\} : \widehat{\tau}_2 \sqcup \widehat{\tau}_2 \,\&\, \xi_1 \sqcup \xi_2 \sqcup \xi_3$$

Also by lemma 40, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2 \sqcup \widehat{\tau}_3$ and $\widehat{\tau}_2 \sqcup \widehat{\tau}_3$ conservative. By [S-Join], $\Sigma \vdash_{\text{s}} \xi_1 \sqcup \xi_2 \sqcup \xi_3 : \star$.

$t = \lambda x : \tau_1.t'$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; \lambda x : \tau_1.t') = \Lambda\overline{\beta_i :: \kappa_i}.\lambda x : \widehat{\tau}_1 \& \beta.\widehat{t'} : \forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot$ where $\widehat{\tau}_1 \,\&\, \beta \rhd \overline{\beta_i :: \kappa_i} = \mathcal{C}([\,]; \tau_1)$ and $\widehat{t'} : \widehat{\tau}_2 \,\&\, \xi_2 = \mathcal{R}(\widehat{\Gamma}, x : \widehat{\tau}_1 \,\&\, \beta; \Sigma, \overline{\beta_i :: \kappa_i}; t')$.

By lemma 32 we get $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \beta \rhd \overline{\beta_i :: \kappa_i}$. Applying lemma 28 to the latter judgment gives us $\overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_1$ and $\overline{\beta_i :: \kappa_i} \vdash_{\text{s}} \beta : \star$. We can now extend the context in the previous two judgments with Σ since the $\overline{\beta_i}$ are fresh. For the same reason, $\widehat{\Gamma}$ is well-formed under $\Sigma, \overline{\beta_i :: \kappa_i}$. By lemma 30, $\widehat{\tau}_1$ is conservative. We can conclude that the extended environment $\widehat{\Gamma}, x : \widehat{\tau}_1 \,\&\, \beta$ is well-formed under $\Sigma, \overline{\beta_i :: \kappa_i}$ as well.

We can derive

$$\dfrac{\dfrac{\dfrac{\text{by induction}}{\Sigma, \overline{\beta_i :: \kappa_i} \mid \widehat{\Gamma}, x : \widehat{\tau}_1 \,\&\, \beta \vdash_{\text{te}} \widehat{t'} : \widehat{\tau}_2 \,\&\, \xi_2}}{\Sigma, \overline{\beta_i :: \kappa_i} \mid \widehat{\Gamma} \vdash_{\text{te}} \lambda x : \widehat{\tau}_1 \,\&\, \beta.\widehat{t'} : \widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\ [\text{T-Abs}]}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \Lambda\overline{\beta_i :: \kappa_i}.\lambda x : \widehat{\tau}_1 \,\&\, \beta.\widehat{t'} : \forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot}\ \overline{[\textsc{T-AnnAbs}]}_i$$

where $\overline{[\textsc{T-AnnAbs}]}_i$ denotes an application of rule [T-AnnAbs] for every i. We have omitted the side-conditions of [T-AnnAbs] in the tree above.

However, since the $\overline{\beta_i}$ are fresh, we have $\beta_i \notin \text{fav}(\widehat{\Gamma}) \cup \text{fav}(\bot) = \text{fav}(\widehat{\Gamma})$ for all i.
Similarly, we can derive

$$\dfrac{\dfrac{\dfrac{\text{by lemma 28}}{\Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_1} \quad \dfrac{\text{by induction}}{\Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_2} \quad \Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{s}} \beta : \star \quad \Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{s}} \xi_2 : \star}{\Sigma, \overline{\beta_i :: \kappa_i} \vdash_{\text{wft}} \widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle}\,[\text{W-Arr}]}{\Sigma \vdash_{\text{wft}} \forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle}\,\overline{[\text{W-Forall}]_i}$$

$\Sigma \vdash_{\text{s}} \bot : \star$ follows trivially from [S-Lat]. By induction, $\widehat{\tau}_2$ is conservative, and since we established earlier that $\emptyset \vdash_p \widehat{\tau}_1 \,\&\, \beta \rhd \overline{\beta_i :: \kappa_i}$ holds, the resulting type $\forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle$ is conservative.

$t = t_1\ t_2$ In this case $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1\ t_2) = \widehat{t_1}\ \langle\overline{\theta\ \beta_i}\rangle\ \widehat{t_2} : \llbracket\theta\ \widehat{\tau}'\rrbracket_\Sigma \,\&\, \llbracket\xi_1 \sqcup \theta\ \xi'\rrbracket_\Sigma$ where $\widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1 = \mathcal{R}(\Gamma; \Sigma; t_1)$, $\widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2 = \mathcal{R}(\Gamma; \Sigma; t_2)$, $\widehat{\tau}_2'\langle\beta\rangle \to \widehat{\tau}'\langle\xi'\rangle \rhd \overline{\beta_i} = \mathcal{I}(\widehat{\tau}_1)$ and $\theta = [\beta \mapsto \xi_2] \circ \mathcal{M}([\,]; \widehat{\tau}_2'; \widehat{\tau}_2)$.
By induction, we have $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1$ and $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2$, $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1$ and $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$, $\Sigma \vdash_{\text{s}} \xi_1 : \star$ and $\Sigma \vdash_{\text{s}} \xi_2 : \star$ and both $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative.
Since $\mathcal{I}(\widehat{\tau}_1) = \widehat{\tau}_2'\langle\beta\rangle \to \widehat{\tau}'\langle\xi'\rangle \rhd \overline{\beta_i}$, we know that $\widehat{\tau}_1 = \forall\overline{\beta_i' :: \kappa_{\beta_i'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_1''\langle\xi_2'\rangle$ for some $\overline{\beta_i'}$. Additionally, $\widehat{\tau}_1$ is conservative. This implies $\emptyset \vdash_p \widehat{\tau}_1' \,\&\, \xi_1' \rhd \overline{\beta_i' :: \kappa_{\beta_i'}}$. By the definitions of the pattern rules we can infer that $\xi_1' = \beta'$ for some $\beta' \in \overline{\beta_i'}$.
Furthermore, due to the way $\mathcal{I}$ is defined, $[\overline{\beta_i/\beta_i'}](\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_1''\langle\xi_2'\rangle) = \widehat{\tau}_2'\langle\beta\rangle \to \widehat{\tau}'\langle\xi'\rangle$.
Since by induction $\Sigma \vdash_{\text{wft}} \widehat{\tau}_2$ and $\Sigma \vdash_{\text{s}} \xi_2 : \star$, we can apply lemma 39 in order to get $\Sigma \vdash_{\text{s}} \theta\beta_i : \kappa_{\beta_i'}$ for all i.
We can derive

$$\dfrac{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1} : \forall\overline{\beta_i' :: \kappa_{\beta_i'}}.\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_1''\langle\xi_2'\rangle \,\&\, \xi_1 \qquad \overline{\Sigma \vdash_{\text{s}} \theta\beta_i : \kappa_{\beta_i'}}}{\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1}\ \overline{\langle\theta\beta_i\rangle} : [\overline{\theta\beta_i\ /\ \beta_i'}]\ (\widehat{\tau}_1'\langle\xi_1'\rangle \to \widehat{\tau}_1''\langle\xi_2'\rangle) \,\&\, \xi_1}\,\overline{[\text{T-AnnApp}]_i}$$

We define $\theta' = [\overline{\theta\beta_i\ /\ \beta_i'}]$. Note that $\theta' = \theta \circ [\overline{\beta_i/\beta_i'}]$. It follows, by using lemma 39, that $\theta'\widehat{\tau}_1' = \theta([\overline{\beta_i/\beta_i'}]\widehat{\tau}_1') = \theta\widehat{\tau}_2' = \widehat{\tau}_2$. Similarly, $\theta'\xi_1' = \theta([\overline{\beta_i/\beta_i'}]\xi_1') = \theta\beta = \xi_2$, $\theta'\widehat{\tau}_1'' = \theta([\overline{\beta_i/\beta_i'}]\widehat{\tau}_1'') = \theta\widehat{\tau}'$ and $\theta'\xi_2' = \theta([\overline{\beta_i/\beta_i'}]\xi_2') = \theta\xi'$.
This allows us to rewrite the above derivation as $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1}\ \overline{\langle\theta\beta_i\rangle} : \widehat{\tau}_2\langle\xi_2\rangle \to \theta\widehat{\tau}_1'\langle\theta\xi'\rangle \,\&\, \xi_1$. By applying [T-Sub] and [Sub-Arr], we can further derive $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1}\ \overline{\langle\theta\beta_i\rangle} : \widehat{\tau}_2\langle\xi_2\rangle \to \theta\widehat{\tau}_1'\langle\xi_1 \sqcup \theta\xi'\rangle \,\&\, \xi_1 \sqcup \theta\xi'$.
This has the right shape for applying [T-App] which lets us conclude $\Sigma \mid \widehat{\Gamma} \vdash_{\text{te}} \widehat{t_1}\ \overline{\langle\theta\beta_i\rangle}\ \widehat{t_2} : \theta\widehat{\tau}' \,\&\, \xi_1 \sqcup \theta\xi'$.
Because all terms in the image of θ are of the right sort and we know $\Sigma \vdash_{\text{wft}} \widehat{\tau}_1$, we can conclude $\Sigma \vdash_{\text{wft}} \theta\widehat{\tau}'$ and $\Sigma \vdash_{\text{s}} \theta\xi' : \star$ from its premises. Applying [S-Join] results in $\Sigma \vdash_{\text{s}} \xi_1 \sqcup \theta\xi' : \star$.
Since $\widehat{\tau}_1$ is conservative, so is $\widehat{\tau}_1''$ and therefore also $\theta'\widehat{\tau}_1'' = \theta\widehat{\tau}'$.

$t = \mu x : \tau.t'$ In this case $\mathcal{R}(\widehat{\Gamma}; \Sigma; \mu x : \tau.t') = \mu x : \widehat{\tau}_n \mathbin{\&} \xi_n.\widehat{t'} : \widehat{\tau}_n \mathbin{\&} \xi_n$ for some $\widehat{t'}$, $\widehat{\tau}_n$ and ξ_n such that $\widehat{t'} : \widehat{\tau}_n \mathbin{\&} \xi_n = \mathcal{R}(\widehat{\Gamma}, x : \widehat{\tau}_n \mathbin{\&} \xi_n; \Sigma; t')$. These exist because by assumption, $\mathcal{R}$ has produced a result, and that only happens if the fixpoint iteration terminated after a finite number of iterations. We denote the number of iterations by n.

We have $\xi_0 = \bot$. Hence, $\Sigma \vdash_{\mathrm{s}} \xi_0 : \star$ follows trivially from [S-LAT]. $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_0$ and the fact that $\widehat{\tau}_0$ is conservative for $\widehat{\tau}_0 = \bot_\tau$ follows from lemma 34.

For every iteration i we can apply the induction hypothesis in order to get $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_{i+1}$ and $\Sigma \vdash_{\mathrm{s}} \xi_{i+1} : \star$ and the conservativeness of $\widehat{\tau}_{i+1}$ from the facts established about the previous iteration.

By repeating this step n times, we can eventually derive $\Sigma \mid \widehat{\Gamma}, x : \widehat{\tau}_n \mathbin{\&} \xi_n \vdash_{\mathrm{te}} \widehat{t'} : \widehat{\tau}_n \mathbin{\&} \xi_n$, $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}_n$ and $\Sigma \vdash_{\mathrm{s}} \xi_n : \star$. Furthermore, $\widehat{\tau}_n$ is conservative.

We can conclude $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \mu x : \widehat{\tau}_n \mathbin{\&} \xi_n.\widehat{t'} : \widehat{\tau}_n \mathbin{\&} \xi_n$ by [T-FIX].

A straightforward way to decide semantic equivalence is to enumerate all possible environments and compare the denotations of the two terms in all of these (possibly after some semantics preserving normalization). This works if the dependency lattice $\mathcal{L}$ is finite:

Lemma 41. *If the dependency lattice $\mathcal{L}$ is finite, then the semantic equality of two dependency terms ξ_1 and ξ_2 under an environment Σ is decidable.*

Proof. First, we show by induction on sorts κ that the universes V_κ are finite.

$\kappa = \star$ In this case we have $V_\kappa = V_\star = \mathcal{L}$ which by assumption is finite.

$\kappa = \kappa_1 \Rightarrow \kappa_2$

By induction, V_{κ_1} and V_{κ_2} are finite. Therefore, the set $V_{\kappa_2}^{V_{\kappa_1}}$ of functions from V_{κ_1} to V_{κ_2} is also finite (with cardinality $|V_{\kappa_2}^{V_{\kappa_1}}| = |V_{\kappa_2}|^{|V_{\kappa_1}|}$). Since $V_{\kappa_1 \Rightarrow \kappa_2} \subseteq V_{\kappa_2}^{V_{\kappa_1}}$, it must also be finite.

For any environment ρ that is compatible with Σ we know $\mathrm{dom}(\rho) = \mathrm{dom}(\Sigma)$ and for all $\beta \in \mathrm{dom}(\rho)$ we have $\rho(\beta) \in V_{\Sigma(\beta)}$. $\mathrm{dom}(\Sigma)$ is finite because sort environments are finite. Moreover, the image $\mathrm{im}(\rho) \subseteq \bigcup\{V_\kappa \mid \kappa \in \mathrm{im}(\Sigma)\} \subseteq \bigcup\{V_\kappa \mid \kappa \in \mathbf{AnnSort}\}$ is finite because the image of Σ can only consist of finitely many sorts and a finite union of finite sets is again finite, even though there are infinitely many sorts.

Hence, there can only be a finite number of environments ρ that are compatible with Σ.

The denotation function $[\![-]\!]_\rho$ is structurally recursive on terms and can therefore be computed in finite time.

Thus, equality of two dependency terms ξ_1 and ξ_2 can be decided by enumerating all environments ρ, computing the denotations $[\![\xi_1]\!]_\rho$ and $[\![\xi_2]\!]_\rho$ and then comparing the results.

Definition 3. *Let τ be an underlying type and let Σ be a sort environment. We define*

1. *the set of conservative annotated type and effect pairs with corresponding underlying type* τ *well-formed under* Σ,

$$\mathbf{TyEff}_\tau(\Sigma) = \left\{\widehat{\tau} \,\&\, \xi \in \widehat{\mathbf{Ty}} \times \mathbf{AnnTm} \mid \Sigma \vdash_{\mathrm{wft}} \widehat{\tau} \wedge \Sigma \vdash_{\mathrm{s}} \xi : \star \wedge \widehat{\tau} \textit{ conservative} \wedge \lfloor \widehat{\tau} \rfloor = \tau \right\}$$

2. *the equivalence relation* $\equiv_{\tau;\Sigma} \subseteq \mathbf{TyEff}_\tau(\Sigma) \times \mathbf{TyEff}_\tau(\Sigma)$ *given by*

$$\begin{aligned} &\widehat{\tau}_1 \,\&\, \xi_1 \equiv_{\tau;\Sigma} \widehat{\tau}_2 \,\&\, \xi_2 \\ \iff &\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2 \wedge \Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_1 \wedge \Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2 \wedge \Sigma \vdash_{\mathrm{sub}} \xi_2 \sqsubseteq \xi_1, \end{aligned}$$

3. $\mathbf{TyEff}_\tau(\Sigma)/\equiv_{\tau;\Sigma}$ *to be the quotient of the set of well-formed annotated type and effect pairs by the equivalence relation* $\equiv_{\tau;\Sigma}$,
4. $[\widehat{\tau} \,\&\, \xi]_\Sigma$ *to denote the equivalence class of* $\widehat{\tau} \,\&\, \xi \in \mathbf{TyEff}_{\lfloor\widehat{\tau}\rfloor}(\Sigma)$.

When the underlying type τ *is obvious from the context, we write* $\equiv_\Sigma$ *instead of* $\equiv_{\tau;\Sigma}$.

Lemma 42. $\equiv_\Sigma$ *is an equivalence relation.*

Proof. We need to show that $\equiv_\Sigma$ is reflexive, symmetric and transitive.

Reflexivity For all $\widehat{\tau} \,\&\, \xi \in \mathbf{TyEff}_\tau(\Sigma)$ we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau} \leqslant \widehat{\tau}$ by [SUB-REFL] and $\Sigma \vdash_{\mathrm{sub}} \xi \sqsubseteq \xi$ by the reflexivity of subsumption. Therefore $\widehat{\tau} \,\&\, \xi \equiv_\Sigma \widehat{\tau}\ \xi$.

Symmetry Given $\widehat{\tau}_1 \,\&\, \xi_1$ and $\widehat{\tau}_2 \,\&\, \xi_2$ such that $\widehat{\tau}_1 \,\&\, \xi_1 \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2$, then we also have $\widehat{\tau}_2 \,\&\, \xi_2 \equiv_\Sigma \widehat{\tau}_1 \,\&\, \xi_1$ by the symmetry of $\wedge$.

Transitivity Given $\widehat{\tau}_1 \,\&\, \xi_1$, $\widehat{\tau}_2 \,\&\, \xi_2$ and $\widehat{\tau}_3 \,\&\, \xi_3$ such that $\widehat{\tau}_1 \,\&\, \xi_1 \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2$ and $\widehat{\tau}_2 \,\&\, \xi_2 \equiv_\Sigma \widehat{\tau}_3 \,\&\, \xi_3$. Then we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_1$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_3$ and $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_2$. By [SUB-TRANS], we can infer $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$ and $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_1$. By the transitivity of subsumption, we have $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_3$ and $\Sigma \vdash_{\mathrm{sub}} \xi_3 \sqsubseteq \xi_1$. Hence $\widehat{\tau}_1 \,\&\, \xi_1 \equiv_\Sigma \widehat{\tau}_3 \,\&\, \xi_3$.

An important ingredient in the completeness proof is the fact that the set of equivalence classes defined above is finite.

Lemma 43. $\mathbf{TyEff}_\tau(\Sigma)/\equiv_{\tau;\Sigma}$ *is finite.*

Proof. Suppose not. Then there is an infinite number of type and effect pairs $(\widehat{\tau}_i \,\&\, \xi_i)_{i\in\mathbb{N}}$ such that none of them is equivalent to any other pair. Since the type and effect pairs are all of the same shape, they can only differ in the (potentially nested) annotations. Hence, there must be an infinite number of dependency terms $(\xi'_i)_{i\in\mathbb{N}}$ occurring in the same position that are not semantically equivalent to any other (because there are only finitely many positions where an annotation can occur in a type). For every pair of such terms ξ'_i, ξ'_j, there is an environment $\rho_{i,j}$ such that $[\![\xi'_i]\!]_{\rho_{i,j}} \neq [\![\xi'_j]\!]_{\rho_{i,j}}$.

In the proof of lemma 41 we established that there can only be a finite number of environments compatible with any given sort environment. Therefore, there is an environment ρ and an infinite subset $S \subseteq \mathbb{N}$ such that for all $i, j \in S$ with $i \neq j$ we have $[\![\xi'_i]\!]_\rho \neq [\![\xi'_j]\!]_\rho$. However, by assumption, the underlying lattice is finite. Contradiction.

Next, we define an order relation $\sqsubseteq_{\tau;\Sigma}$ on equivalence classes $\mathbf{TyEff}_\tau(\Sigma)/\equiv_{\tau;\Sigma}$ as follows:

$$[\widehat{\tau}_1 \,\&\, \xi_1]_\Sigma \sqsubseteq_{\tau;\Sigma} [\widehat{\tau}_2 \,\&\, \xi_2]_\Sigma \iff \Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2 \wedge \Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_2.$$

Lemma 44. *$\sqsubseteq_\Sigma$ is well-defined and a partial order.*

Proof. We first show that the definition of $\sqsubseteq_{\tau;\Sigma}$ is independent of the chosen representative. Let $\mathcal{E}_1, \mathcal{E}_2 \in \mathbf{TyEff}_\tau(\Sigma)/\equiv_{\tau;\Sigma}$ denote two arbitrary equivalence classes and let $\widehat{\tau}_1 \,\&\, \xi_1, \widehat{\tau}'_1 \,\&\, \xi'_1 \in \mathcal{E}_1$ and $\widehat{\tau}_2 \,\&\, \xi_2, \widehat{\tau}'_2 \,\&\, \xi'_2 \in \mathcal{E}_2$ be arbitrary representatives of theses classes. We have $\widehat{\tau}_1 \,\&\, \xi_1 \equiv_{\tau;\Sigma} \widehat{\tau}'_1 \,\&\, \xi'_1$ and $\widehat{\tau}_2 \,\&\, \xi_2 \equiv_{\tau;\Sigma} \widehat{\tau}'_2 \,\&\, \xi'_2$.

Suppose that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ and $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_2$ hold. By definition 3, we additionally have (among others) $\Sigma \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}_1 \wedge \Sigma \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi_1$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}'_2 \wedge \Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi'_2$. By [Sub-Trans], $\Sigma \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$ and by the transitivity of subsumption we have $\Sigma \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi'_2$. Hence, $\sqsubseteq_\Sigma$ is well-defined.

In order to show that $\sqsubseteq_\Sigma$ is a partial order, we show that the relation is reflexive, transitive and anti-symmetric.

Reflexivity By [Sub-Refl] and the reflexivity of subsumption, $\Sigma \vdash_{\text{sub}} \widehat{\tau} \leqslant \widehat{\tau}$ and $\Sigma \vdash_{\text{sub}} \xi \sqsubseteq \xi$ for all $\widehat{\tau} \,\&\, \xi \in \mathbf{TyEff}_\tau(\Sigma)$. Hence, $[\widehat{\tau} \,\&\, \xi]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau} \,\&\, \xi]_\Sigma$.

Transitivity Given $\widehat{\tau}_1 \,\&\, \xi_1$, $\widehat{\tau}_2 \,\&\, \xi_2$ and $\widehat{\tau}_3 \,\&\, \xi_3$ such that $[\widehat{\tau}_1 \,\&\, \xi_1]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_2 \,\&\, \xi_2]_\Sigma$ and $[\widehat{\tau}_2 \& \xi_2]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_3 \& \xi_3]_\Sigma$. Then we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_3$ as well as $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_2$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi_2$. By [Sub-Trans], we can infer $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_3$. By the transitivity of subsumption, we have $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_3$. Hence $[\widehat{\tau}_1 \,\&\, \xi_1]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_3 \,\&\, \xi_3]_\Sigma$.

Antisymmetry Suppose that $[\widehat{\tau}_1 \& \xi_1]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_2 \& \xi_2]_\Sigma$ and $[\widehat{\tau}_2 \& \xi_2]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_1 \& \xi_1]_\Sigma$. Then, by definition $\widehat{\tau}_1 \,\&\, \xi_1 \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2$. Therefore, $[\widehat{\tau}_1 \,\&\, \xi_1]_\Sigma = [\widehat{\tau}_2 \,\&\, \xi_2]_\Sigma$.

And we define a pointwise join operation on type and effect pairs as follows:

$$(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) := \widehat{\tau}_1 \sqcup \widehat{\tau}_2 \,\&\, \xi_1 \sqcup \xi_2.$$

It is easily verified to be associative, commutative and idempotent.

Lemma 45. *The join operation $\sqcup$ on type and effect pairs is*

1. *associative, i.e. for all $\widehat{\tau}_1 \,\&\, \xi_1, \widehat{\tau}_2 \,\&\, \xi_2, \widehat{\tau}_3 \,\&\, \xi_3 \in \mathbf{TyEff}_\tau(\Sigma)$ we have $((\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2)) \sqcup (\widehat{\tau}_3 \,\&\, \xi_3) \equiv_\Sigma (\widehat{\tau}_1 \,\&\, \xi_1) \sqcup ((\widehat{\tau}_2 \,\&\, \xi_2) \sqcup (\widehat{\tau}_3 \,\&\, \xi_3))$,*
2. *commutative, i.e. for all $\widehat{\tau}_1 \,\&\, \xi_1, \widehat{\tau}_2 \,\&\, \xi_2 \in \mathbf{TyEff}_\tau(\Sigma)$ we have $(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) \equiv_\Sigma (\widehat{\tau}_2 \,\&\, \xi_2) \sqcup (\widehat{\tau}_1 \,\&\, \xi_1)$, and*
3. *idempotent, i.e. for all $\widehat{\tau}_1 \,\&\, \xi_1 \in \mathbf{TyEff}_\tau(\Sigma)$ we have $(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_1 \,\&\, \xi_1) \equiv_\Sigma \widehat{\tau}_1 \,\&\, \xi_1$.*

Proof. All three properties follow from the associativity, commutativity and idempotence of the underlying lattice. We show an exemplary proof for commutativity, the other two can be done similarly.

Let $\widehat{\tau}_1 \& \xi_1, \widehat{\tau}_2 \& \xi_2 \in \mathbf{TyEff}_\tau(\Sigma)$ be arbitrary. We show $(\widehat{\tau}_1 \& \xi_1) \sqcup (\widehat{\tau}_2 \& \xi_2) \equiv_\Sigma (\widehat{\tau}_2 \& \xi_2) \sqcup (\widehat{\tau}_1 \& \xi_1)$ by induction on $\widehat{\tau}_1$. In all cases, we have $\Sigma \vdash \xi_1 \sqcup \xi_2 \equiv \xi_2 \sqcup \xi_1$ by the commutativity of the underlying lattice.

$\widehat{\tau}_1 = \widehat{\mathrm{unit}}$ In this case, $\widehat{\tau}_2 = \widehat{\mathrm{unit}}$ and therefore $\widehat{\tau}_1 \sqcup \widehat{\tau}_2 = \widehat{\mathrm{unit}} = \widehat{\tau}_2 \sqcup \widehat{\tau}_1$ as well. The equivalence follows from reflexivity.

$\widehat{\tau}_1 = \forall\beta :: \kappa.\widehat{\tau}'_1$ Then $\widehat{\tau}_2 = \forall\beta' :: \kappa'.\widehat{\tau}'_2$ for some $\widehat{\tau}'_2$. We can assume $\beta' = \beta$ due to alpha-equivalence. By induction, we have $(\widehat{\tau}'_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}'_2 \,\&\, \xi_2) \equiv_{\Sigma,\beta::\kappa} (\widehat{\tau}'_2 \,\&\, \xi_2) \sqcup (\widehat{\tau}'_1 \,\&\, \xi_1)$.
We have $(\widehat{\tau}_1 \& \xi_1) \sqcup (\widehat{\tau}_2 \& \xi_2) = \forall\beta::\kappa.\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 \& \xi_1 \sqcup \xi_2 \equiv_\Sigma \forall\beta::\kappa.\widehat{\tau}'_2 \sqcup \widehat{\tau}'_1 \& \xi_2 \sqcup \xi_1 = (\widehat{\tau}_2 \,\&\, \xi_2) \sqcup (\widehat{\tau}_1 \,\&\, \xi_1)$ by [Sub-Forall].

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ Then $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ for some $\widehat{\tau}'_2$, $\widehat{\tau}''_2$, ξ'_2 and ξ''_2. By induction, we have $(\widehat{\tau}'_1 \,\&\, \xi'_1) \sqcup (\widehat{\tau}'_2 \,\&\, \xi'_2) \equiv_\Sigma (\widehat{\tau}'_2 \,\&\, \xi'_2) \sqcup (\widehat{\tau}'_1 \,\&\, \xi'_1)$ and $(\widehat{\tau}''_1 \,\&\, \xi''_1) \sqcup (\widehat{\tau}''_2 \,\&\, \xi''_2) \equiv_\Sigma (\widehat{\tau}''_2 \,\&\, \xi''_2) \sqcup (\widehat{\tau}''_1 \,\&\, \xi''_1)$. Therefore, we must also have $(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) \equiv_\Sigma (\widehat{\tau}_2 \,\&\, \xi_2) \sqcup (\widehat{\tau}_1 \,\&\, \xi_1)$ by [Sub-Sum].

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}''_1\langle\xi''_1\rangle$ Analogously to the previous case.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$ Similarly to the previous case, but noting that the contravariant positions of the function arrows must be the same.

The following lemma shows that the join operation interacts with the subtyping and subsumption relation in a way we would expect from a lattice.

Lemma 46. *For all* $\widehat{\tau}_1 \,\&\, \xi_1, \widehat{\tau}_2 \,\&\, \xi_2 \in \mathbf{TyEff}_\tau(\Sigma)$, *we have*

$$\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2 \wedge \Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2 \iff (\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2.$$

Proof. Suppose that $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2 \wedge \Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$ holds. We show $(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2$ by induction on $\widehat{\tau}_1$.

$\widehat{\tau}_1 = \widehat{\mathrm{unit}}$ Then $\widehat{\tau}_2 = \widehat{\mathrm{unit}}$ as well. Because we have $\Sigma \vdash \xi_1 \sqcup \xi_2 \equiv \xi_2$ due to the lattice properties, it follows that $(\widehat{\mathrm{unit}} \,\&\, \xi_1) \sqcup (\widehat{\mathrm{unit}} \,\&\, \xi_2) \equiv_\Sigma \widehat{\mathrm{unit}} \,\&\, \xi_2$ must also hold.

$\widehat{\tau}_1 = \forall\beta :: \kappa.\widehat{\tau}'_1$ The $\widehat{\tau}_2 = \forall\beta::\kappa.\widehat{\tau}'_2$ for some $\widehat{\tau}'_2$. By lemma 13, we have $\Sigma, \beta::\kappa \vdash_{\mathrm{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$. By induction, we have $(\widehat{\tau}'_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}'_2 \,\&\, \xi_2) \equiv_\Sigma \widehat{\tau}'_2 \,\&\, \xi_2$. By applying [Sub-Forall], we get $\forall\beta :: \kappa.\widehat{\tau}'_1 \sqcup \widehat{\tau}'_2 \,\&\, \xi_1 \sqcup \xi_2 \equiv_\Sigma \forall\beta :: \kappa.\widehat{\tau}'_2 \,\&\, \xi_2$.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ Then $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ for some $\widehat{\tau}'_2$, $\widehat{\tau}''_2$, ξ'_2 and ξ''_2. By lemma 13, we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$, $\Sigma \vdash_{\mathrm{sub}} \xi'_1 \sqsubseteq \xi'_2$, $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$ and $\Sigma \vdash_{\mathrm{sub}} \xi''_1 \sqsubseteq \xi''_2$.
By induction, we have $(\widehat{\tau}'_1 \,\&\, \xi'_1) \sqcup (\widehat{\tau}'_2 \,\&\, \xi'_2) \equiv_\Sigma (\widehat{\tau}'_2 \,\&\, \xi'_2)$ and $(\widehat{\tau}''_1 \,\&\, \xi''_1) \sqcup (\widehat{\tau}''_2 \,\&\, \xi''_2) \equiv_\Sigma \widehat{\tau}''_2 \,\&\, \xi''_2$. Therefore, we must also have $(\widehat{\tau}_1 \,\&\, \xi_1) \sqcup (\widehat{\tau}_2 \,\&\, \xi_2) \equiv_\Sigma \widehat{\tau}_2 \,\&\, \xi_2$ by [Sub-Sum].

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}''_1\langle\xi''_1\rangle$ Analogously to the previous case.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$ Similarly to the previous case, but noting that the contravariant positions of the function arrows must be the same.

We also need this lemma relating the substitutions obtained when matching a subtype of another type with the same pattern type.

Lemma 47. *If we have* $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$, $\theta_1 = \mathcal{M}(\Sigma'; \widehat{\tau}; \widehat{\tau}_1)$ *and* $\theta_2 = \mathcal{M}(\Sigma'; \widehat{\tau}; \widehat{\tau}_2)$, *then* $\Sigma \vdash_{\mathrm{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$ *for all* $\beta \in \mathrm{dom}(\theta_1) = \mathrm{dom}(\theta_2)$.

Proof. By induction on the derivation of $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$.

[Sub-Refl] Trivial, since $\widehat{\tau}_1 = \widehat{\tau}_2$ and therefore also $\theta_1 = \theta_2$.

[Sub-Trans] We have $\widehat{\tau}'$ such that $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}'$ and $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}' \leqslant \widehat{\tau}_2$. By induction, we get $\theta_1 = \mathcal{M}(\Sigma'; \widehat{\tau}; \widehat{\tau}_1)$, $\theta' = \mathcal{M}(\Sigma'; \widehat{\tau}; \widehat{\tau}')$, $\theta_2 = \mathcal{M}(\Sigma'; \widehat{\tau}; \widehat{\tau}_2)$ such that $\Sigma \vdash_{\text{sub}} \theta_1(\beta) \sqsubseteq \theta'(\beta)$ and $\Sigma \vdash_{\text{sub}} \theta'(\beta) \leqslant \theta_2(\beta)$ for all β. We apply [Sub-Trans] once for every variable and get $\Sigma \vdash_{\text{sub}} \theta_1(\beta) \leqslant \theta_2(\beta)$ for all β.

[Sub-Forall] We have $\widehat{\tau}_1 = \forall \beta :: \kappa.\widehat{\tau}'_1$ and $\widehat{\tau}_2 = \forall \beta :: \kappa.\widehat{\tau}'_2$ such that $\Sigma, \Sigma', \beta :: \kappa \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$. Therefore, $\theta_1 = \mathcal{M}(\Sigma', \beta :: \kappa; \widehat{\tau}; \widehat{\tau}'_1)$ and $\theta_2 = \mathcal{M}(\Sigma', \beta :: \kappa; \widehat{\tau}; \widehat{\tau}'_2)$.
By induction, $\Sigma \vdash_{\text{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$ for all $\beta \in \text{dom}(\theta_1) = \text{dom}(\theta_2)$.

[Sub-Sum] We have $\widehat{\tau} = \widehat{\tau}'\langle\beta\rangle + \widehat{\tau}''\langle\beta'\rangle$, $\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ and $\widehat{\tau}_2 \quad \cdot + \cdot$ $(\widehat{\tau}'_2\langle\xi'_2\rangle)$ $(\widehat{\tau}''_2\langle\xi''_2\rangle)$ such that $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$, $\Sigma, \Sigma' \vdash_{\text{sub}} \xi'_1 \sqsubseteq \xi'_2$, $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$ and $\Sigma, \Sigma' \vdash_{\text{sub}} \xi''_1 \sqsubseteq \xi''_2$.
Suppose that $\Sigma' = \overline{\beta_i :: \kappa_{\beta_i}}$. We get

$$\theta_1 = [\beta \mapsto \lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi'_1, \beta' \mapsto \lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi''_1] \circ \mathcal{M}(\Sigma'; \widehat{\tau}'; \widehat{\tau}'_1) \circ \mathcal{M}(\Sigma; \widehat{\tau}''; \widehat{\tau}''_1)$$

and

$$\theta_2 = [\beta \mapsto \lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi'_2, \beta' \mapsto \lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi''_2] \circ \mathcal{M}(\Sigma'; \widehat{\tau}'; \widehat{\tau}'_2) \circ \mathcal{M}(\Sigma; \widehat{\tau}''; \widehat{\tau}''_2).$$

For β, we have to show $\Sigma \vdash_{\text{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$, i.e. for all compatible environments ρ we have $[\![\lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi'_1]\!]_\rho \sqsubseteq [\![\lambda\overline{\beta_i :: \kappa_{\beta_i}}.\xi'_2]\!]_\rho$. We need to compare the resulting functions pointwise. But by assumption, we already have $[\![\xi'_1]\!]_{\rho'} \sqsubseteq [\![\xi'_2]\!]_{\rho'}$ for all environments compatible with Σ, Σ', and therefore in particular for all possible arguments that could be passed to these functions. The same holds for β' using a similar reasoning and the remaining subsumption statements follow by induction.

[Sub-Prod] Analogously to the previous case.

[Sub-Arr] Analogously to the previous cases, but only recursing on the covariant part of the function arrow, since the contravariant positions are necessarily equal.

The following lemma shows that the subtyping relation between conservative types is preserved by certain kinds of substitutions.

Lemma 48. *Let Σ and Σ' be sort environments with disjoint domains and let $\widehat{\tau}_1$ and $\widehat{\tau}_2$ be conservative types. If we have $\Sigma, \Sigma' \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ then we also have $\Sigma \vdash_{\text{sub}} \theta_1\widehat{\tau}_1 \leqslant \theta_2\widehat{\tau}_2$ for all substitutions $\theta_1, \theta_2 : \text{dom}(\Sigma') \to \textbf{AnnTm}$ such that $\Sigma \vdash_{\text{sub}} \theta_1(\beta) \sqsubseteq \theta_2(\beta)$, $\Sigma \vdash_{\text{s}} \theta_1(\beta) : \Sigma'(\beta)$ and $\Sigma \vdash_{\text{s}} \theta_2(\beta) : \Sigma'(\beta)$ for all $\beta \in \text{dom}(\Sigma')$.*

Proof. By induction on the conservative type $\widehat{\tau}_1$. By lemma 12, we know that $\widehat{\tau}_2$ has the same shape as $\widehat{\tau}_1$.

$\widehat{\tau}_1 = \widehat{\text{unit}}$ In this case, $\widehat{\tau}_2 = \widehat{\text{unit}}$ as well. Since $\theta_1\widehat{\text{unit}} = \widehat{\text{unit}} = \theta_2\widehat{\text{unit}}$, $\Sigma \vdash_{\text{sub}} \theta_1\widehat{\tau}_1 \leqslant \theta_2\widehat{\tau}_2$ holds by assumption.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle$ We must have $\widehat{\tau}_2 = \widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle$ for some $\widehat{\tau}'_2$, $\widehat{\tau}''_2$, ξ'_2 and ξ''_2. By lemma 13, $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \widehat{\tau}'_1 \leqslant \widehat{\tau}'_2$, $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$, $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \xi'_1 \sqsubseteq \xi'_2$ and $\Sigma, \Sigma' \vdash_{\mathrm{sub}} \xi''_1 \sqsubseteq \xi''_2$.

Then, by induction $\Sigma \vdash_{\mathrm{sub}} \theta_1\widehat{\tau}'_1 \leqslant \theta_2\widehat{\tau}'_2$, $\Sigma \vdash_{\mathrm{sub}} \theta_1\widehat{\tau}''_1 \leqslant \theta_2\widehat{\tau}''_2$ and by lemma 6, $\Sigma \vdash_{\mathrm{sub}} \theta_1\xi'_1 \sqsubseteq \theta_1\xi'_2$ and $\Sigma \vdash_{\mathrm{sub}} \theta_1\xi''_1 \sqsubseteq \theta_2\xi''_2$. Hence, we can derive $\Sigma \vdash_{\mathrm{sub}} \theta_1(\widehat{\tau}'_1\langle\xi'_1\rangle + \widehat{\tau}''_1\langle\xi''_1\rangle) \leqslant \theta_2(\widehat{\tau}'_2\langle\xi'_2\rangle + \widehat{\tau}''_2\langle\xi''_2\rangle)$.

$\widehat{\tau}_1 = \widehat{\tau}'_1\langle\xi'_1\rangle \times \widehat{\tau}''_1\langle\xi''_1\rangle$ This case can be proven analogously to the previous case.

$\widehat{\tau}_1 = \forall\overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}'_1\langle\xi'_1\rangle \to \widehat{\tau}''_1\langle\xi''_1\rangle$ This case needs special care in handling the contravariant argument position of the function type. We have $\widehat{\tau}_2 = \forall\overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}'_2\langle\xi'_2\rangle \to \widehat{\tau}''_2\langle\xi''_2\rangle$. As $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative, we have $\emptyset \vdash_p \widehat{\tau}'_1 \,\&\, \xi'_1 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$ and $\emptyset \vdash_p \widehat{\tau}'_2 \,\&\, \xi'_2 \triangleright \overline{\beta_j :: \kappa_{\beta_j}}$.

We assume that the variables $\overline{\beta_j}$ are disjoint from those in Σ and Σ', because we could always obtain an equivalent type through renaming. By lemma 28, $\mathrm{fav}(\widehat{\tau}'_1) \cup \mathrm{fav}(\xi'_1) \subseteq \overline{\beta_j}$ and $\mathrm{fav}(\widehat{\tau}'_2) \cup \mathrm{fav}(\xi'_2) \subseteq \overline{\beta_j}$. Therefore, none of the free variables of $\widehat{\tau}'_1$ and ξ'_1 are free in $\widehat{\tau}_1$, and the same holds for $\widehat{\tau}_2$.

As the substitutions do not apply to the free variables of the argument positions,

$$\theta_1\widehat{\tau}_1 = \forall\overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}'_1\langle\xi'_1\rangle \to \theta_1\widehat{\tau}''_1\langle\theta_1\xi''_1\rangle \text{ and } \theta_2\widehat{\tau}_2 = \forall\overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}'_2\langle\xi'_2\rangle \to \theta_2\widehat{\tau}''_2\langle\theta_2\xi''_2\rangle.$$

We can assume $\widehat{\tau}'_1 = \widehat{\tau}'_2$ and $\xi'_1 = \xi'_2$ due to lemma 29.

Through repeated application of lemma 13, we can infer $\Sigma, \Sigma', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$ and $\Sigma, \Sigma', \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \xi''_1 \sqsubseteq \xi''_2$. The contexts in the judgments above can be reordered to $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}}, \Sigma' \vdash_{\mathrm{sub}} \widehat{\tau}''_1 \leqslant \widehat{\tau}''_2$ and $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}}, \Sigma' \vdash_{\mathrm{sub}} \xi''_1 \sqsubseteq \xi''_2$.

By induction, respectively lemma 6, we get $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \theta_1\widehat{\tau}''_1 \leqslant \theta_2\widehat{\tau}''_2$ and $\Sigma, \overline{\beta_j :: \kappa_j} \vdash_{\mathrm{sub}} \theta_1\xi''_1 \sqsubseteq \theta_2\xi''_2$. The judgments $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \widehat{\tau}'_2 \leqslant \widehat{\tau}'_1$ as well as $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\mathrm{sub}} \xi'_2 \sqsubseteq \xi'_1$ follow by removing the variables in Σ' from the context, as they are not free in the respective types and effects (using lemma 14).

Lastly, we can derive $\Sigma \vdash_{\mathrm{sub}} \theta_1\widehat{\tau}_1 \leqslant \theta_2\widehat{\tau}_2$ by [Sub-Arr] and [Sub-Forall].

Note that, in contrast to lemma 15, the proof of the foregoing lemma crucially relies on the fact that conservative types are invariant in the argument types of function arrows.

We now prove the following monotonicity lemma stating that increasing a type or effect in the environment (w.r.t. the subtyping or subsumption relations) passed to $\mathcal{R}$ also leads to greater or equal type and effect being computed.

Lemma 49. *Let Σ be a sort environment, let $\widehat{\Gamma_1}$, $\widehat{\Gamma_2}$ be type and effect environments with identical domains well-formed under Σ and let t be a source term such that we have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t) = \widehat{t}_1 : \widehat{\tau}_1 \,\&\, \xi_1$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t) = \widehat{t}_2 : \widehat{\tau}_2 \,\&\, \xi_2$.*

If for all $x \in \mathrm{dom}(\widehat{\Gamma_1})$ with $\widehat{\Gamma_1}(x) = \widehat{\tau} \,\&\, \xi$ and $\widehat{\Gamma_2}(x) = \widehat{\tau}' \,\&\, \xi'$ we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau} \leqslant \widehat{\tau}'$ and $\Sigma \vdash_{\mathrm{sub}} \xi \sqsubseteq \xi'$, then also $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ and $\Sigma \vdash_{\mathrm{sub}} \xi_1 \sqsubseteq \xi_2$.

Proof. By induction on t.

$t = x$ We have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; x) = x : \widehat{\Gamma_1}(x)$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; x) = x : \widehat{\Gamma_2}(x)$. The result follows by assumption.

$t = ()$ The assigned type and effect pair is always $\widehat{\text{unit}} \mathbin{\&} \bot$, therefore the result is trivially true by reflexivity.

$t = \text{ann}_\ell(t')$ We have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; \text{ann}_\ell(t')) = \text{ann}_\ell(\widehat{t_1'}) : \widehat{\tau}_1' \mathbin{\&} [\![\xi_1' \sqcup \ell]\!]_\Sigma$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; \text{ann}_\ell(t')) = \text{ann}_\ell(\widehat{t_2'}) : \widehat{\tau}_2' \mathbin{\&} [\![\xi_2' \sqcup \ell]\!]_\Sigma$.
By induction, $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t') = \widehat{t_1'} : \widehat{\tau}_1' \mathbin{\&} \xi_1'$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t') = \widehat{t_2'} : \widehat{\tau}_2' \mathbin{\&} \xi_2'$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$ and $\Sigma \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_2'$. Using the lattice properties, we can derive $\Sigma \vdash_{\text{sub}} \xi_1' \sqcup \ell \sqsubseteq \xi_2' \sqcup \ell$.

$t = \text{seq } t' \ t''$ We have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; \text{seq } t' \ t'') = \text{seq } \widehat{t_1'} \ \widehat{t_1''} : \widehat{\tau}_1'' \mathbin{\&} [\![\xi_1' \sqcup \xi_1'']\!]_\Sigma$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; \text{seq } t' \ t'') = \text{seq } \widehat{t_2'} \ \widehat{t_2''} : \widehat{\tau}_2'' \mathbin{\&} [\![\xi_2' \sqcup \xi_2'']\!]_\Sigma$.
By induction, $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1'' \leqslant \widehat{\tau}_2''$, $\Sigma \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_2'$ and $\Sigma \vdash_{\text{sub}} \xi_1'' \sqsubseteq \xi_2''$. Using the lattice properties, we can derive $\Sigma \vdash_{\text{sub}} \xi_1' \sqcup \xi_1'' \sqsubseteq \xi_2' \sqcup \xi_2''$.

$t = (t_1, t_2)$ By applying [Sub-Prod] to the induction hypotheses for both recursive calls.

$t = \text{inl}_{\tau_2}(t_1)$ By applying [Sub-Sum] to the induction hypothesis for the recursive call and lemmas 33 and 35.

$t = \text{inl}_{\tau_1}(t_2)$ Analogous to the previous case.

$t = \text{proj}_i(t')$ By applying lemma 13 to the induction hypothesis.

$t = \textbf{case } t_1 \textbf{ of } \{\text{inl}(x) \to t_2; \text{inr}(x) \to t_3\}$ By induction, we get

$$\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t_1) = \widehat{t_1} : \widehat{\tau}_1^l\langle\xi_1^l\rangle + \widehat{\tau}_1^r\langle\xi_1^r\rangle \mathbin{\&} \xi_1 \text{ and } \mathcal{R}(\widehat{\Gamma_2}; \Sigma; t_1) = \widehat{t_4} : \widehat{\tau}_4^l\langle\xi_4^l\rangle + \widehat{\tau}_4^r\langle\xi_4^r\rangle \mathbin{\&} \xi_4$$

such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^l\langle\xi_1^l\rangle + \widehat{\tau}_1^r\langle\xi_1^r\rangle \leqslant \widehat{\tau}_4^l\langle\xi_4^l\rangle + \widehat{\tau}_4^r\langle\xi_4^r\rangle$ and $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_4$. By lemma 13, we get $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^l \leqslant \widehat{\tau}_4^l$ and $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^r \leqslant \widehat{\tau}_4^r$ as well as $\Sigma \vdash_{\text{sub}} \xi_1^l \sqsubseteq \xi_4^l$ and $\Sigma \vdash_{\text{sub}} \xi_1^r \sqsubseteq \xi_4^r$.
This allows us to apply the induction hypothesis to the recursive call for both branches of the case-expression, and we get

$$\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^l \mathbin{\&} \xi_1^l; \Sigma; t_2) = \widehat{t_2} : \widehat{\tau}_2 \mathbin{\&} \xi_2 \text{ and } \mathcal{R}(\widehat{\Gamma_2}, x : \widehat{\tau}_4^l \mathbin{\&} \xi_4^l; \Sigma; t_2) = \widehat{t_5} : \widehat{\tau}_5 \mathbin{\&} \xi_5$$

such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \leqslant \widehat{\tau}_5$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqsubseteq \xi_5$. Similarly, for the other branch we have

$$\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^r \mathbin{\&} \xi_1^r; \Sigma; t_3) = \widehat{t_3} : \widehat{\tau}_3 \mathbin{\&} \xi_3 \text{ and } \mathcal{R}(\widehat{\Gamma_2}, x : \widehat{\tau}_4^r \mathbin{\&} \xi_4^r; \Sigma; t_3) = \widehat{t_6} : \widehat{\tau}_6 \mathbin{\&} \xi_6$$

such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_3 \leqslant \widehat{\tau}_6$ and $\Sigma \vdash_{\text{sub}} \xi_3 \sqsubseteq \xi_6$.
Using lemma 45 and lemma 46, we can derive

$$\begin{aligned}
&((\widehat{\tau}_2 \mathbin{\&} \xi_2) \sqcup (\widehat{\tau}_3 \mathbin{\&} \xi_3)) \sqcup ((\widehat{\tau}_5 \mathbin{\&} \xi_5) \sqcup (\widehat{\tau}_6 \mathbin{\&} \xi_6)) \\
\equiv_\Sigma &((\widehat{\tau}_2 \mathbin{\&} \xi_2) \sqcup (\widehat{\tau}_5 \mathbin{\&} \xi_5)) \sqcup ((\widehat{\tau}_3 \mathbin{\&} \xi_3) \sqcup (\widehat{\tau}_6 \mathbin{\&} \xi_6)) \\
\equiv_\Sigma &(\widehat{\tau}_5 \mathbin{\&} \xi_5) \sqcup (\widehat{\tau}_6 \mathbin{\&} \xi_6)
\end{aligned}$$

and therefore $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2 \sqcup \widehat{\tau}_3 \leqslant \widehat{\tau}_5 \sqcup \widehat{\tau}_6$ and $\Sigma \vdash_{\text{sub}} \xi_2 \sqcup \xi_3 \sqsubseteq \xi_5 \sqcup \xi_6$. Using the lattice properties, we also have $\Sigma \vdash_{\text{sub}} \xi_1 \sqcup \xi_2 \sqcup \xi_3 \sqsubseteq \xi_4 \sqcup \xi_5 \sqcup \xi_6$.
Since $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t) = \textbf{case } \widehat{t_1} \textbf{ of } \{\text{inl}(x) \to \widehat{t_2}; \text{inr}(x) \to \widehat{t_3}\} : [\![\widehat{\tau}_2 \sqcup \widehat{\tau}_3]\!]_\Sigma \mathbin{\&} [\![\xi_1 \sqcup \xi_2 \sqcup \xi_3]\!]_\Sigma$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t) = \textbf{case } \widehat{t_4} \textbf{ of } \{\text{inl}(x) \to \widehat{t_5}; \text{inr}(x) \to \widehat{t_6}\} : [\![\widehat{\tau}_5 \sqcup \widehat{\tau}_6]\!]_\Sigma \mathbin{\&} [\![\xi_4 \sqcup \xi_5 \sqcup \xi_6]\!]_\Sigma$, this completes the proof.

$t = \lambda x : \tau_1.t'$ We note that the argument position can be treated invariantly due to the fact that pattern completions are unique. The desired result then follows by applying [Sub-Arr] using the induction hypothesis for the covariant position, followed by repeated applications of [Sub-Forall].

$t = t'\ t''$ By induction, we get $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t') = \widehat{t_1'} : \widehat{\tau}_1' \,\&\, \xi_1'$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t') = \widehat{t_2'} : \widehat{\tau}_2' \& \xi_2'$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$ and $\Sigma \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_2'$ as well as $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t'') = \widehat{t_1''} : \widehat{\tau}_1'' \,\&\, \xi_1''$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t'') = \widehat{t_2''} : \widehat{\tau}_2'' \,\&\, \xi_2''$ such that $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1'' \leqslant \widehat{\tau}_2''$ and $\Sigma \vdash_{\text{sub}} \xi_1'' \sqsubseteq \xi_2''$.

We know that $\widehat{\tau}_1'$ and $\widehat{\tau}_2'$ are conservative and function types, therefore they are of the form $\widehat{\tau}_1' = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_1'^{a}\langle \xi_1'^{a}\rangle \to \widehat{\tau}_1'^{r}\langle \xi_1'^{r}\rangle$ and $\widehat{\tau}_2' = \forall \overline{\beta_j :: \kappa_{\beta_j}}.\widehat{\tau}_2'^{a}\langle \xi_2'^{a}\rangle \to \widehat{\tau}_2'^{r}\langle \xi_2'^{r}\rangle$.

We can deconstruct the derivation of $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1' \leqslant \widehat{\tau}_2'$ by repeated application of lemma 13, once for every quantifier and finally for the arrow type itself. We end up with $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\text{sub}} \widehat{\tau}_1'^{r} \leqslant \widehat{\tau}_2'^{r}$ and $\Sigma, \overline{\beta_j :: \kappa_{\beta_j}} \vdash_{\text{sub}} \xi_1'^{r} \sqsubseteq \xi_2'^{r}$.

Without loss of generality we can assume that the $\overline{\beta_j}$ are fresh and therefore

$$\widehat{\tau}_1'^{a}\langle \xi_1'^{a}\rangle \to \widehat{\tau}_1'^{r}\langle \xi_1'^{r}\rangle \triangleright \overline{\beta_j :: \kappa_{\beta_j}} = \mathcal{I}(\widehat{\tau}_1') \text{ and } \widehat{\tau}_2'^{a}\langle \xi_2'^{a}\rangle \to \widehat{\tau}_2'^{r}\langle \xi_2'^{r}\rangle \triangleright \overline{\beta_j :: \kappa_{\beta_j}} = \mathcal{I}(\widehat{\tau}_2').$$

Since $\widehat{\tau}_1'^{a} \,\&\, \xi_1'^{a}$ and $\widehat{\tau}_2'^{a} \,\&\, \xi_2'^{a}$ are patterns of the same underlying type, we can assume that $\widehat{\tau}_1'^{a} \& \xi_1'^{a} = \widehat{\tau}_2'^{a} \& \xi_2'^{a}$ because pattern types are unique up to renaming. We know that $\xi_1'^{a}$ and $\xi_2'^{a}$ must be single variables because they are part of a pattern, so suppose $\xi_1'^{a} = \xi_2'^{a} = \beta_1$.

Then we get one substitution for each call to $\mathcal{R}$ given by

$$\theta_1 = [\beta_1 \mapsto \xi_1''] \circ \mathcal{M}([]; \widehat{\tau}_1'^{a}; \widehat{\tau}_1'')) \text{ and } \theta_2 = [\beta_1 \mapsto \xi_2''] \circ \mathcal{M}([]; \widehat{\tau}_2'^{a}; \widehat{\tau}_2'')).$$

By lemma 47, $\Sigma \vdash_{\text{sub}} \theta_1(\gamma) \sqsubseteq \theta_2(\gamma)$ holds for all variables γ. That allows us to apply lemma 6 and lemma 48 in order to derive $\Sigma \vdash_{\text{sub}} \theta_1 \widehat{\tau}_1'^{r} \leqslant \theta_2 \widehat{\tau}_2'^{r}$ and $\Sigma \vdash_{\text{sub}} \theta_1 \xi_1'^{r} \sqsubseteq \theta_2 \xi_2'^{r}$. Together with $\Sigma \vdash_{\text{sub}} \xi_1' \sqsubseteq \xi_2'$ deduced in the beginning, we also have $\Sigma \vdash_{\text{sub}} \xi_1' \sqcup \xi_1'^{r} \sqsubseteq \xi_2' \sqcup \xi_2'^{r}$.

As we have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; t'\ t'') = \widehat{t_1'}\ \widehat{t_1''} : [\![\theta_1 \widehat{\tau}_1''^{r}]\!]_\Sigma \& [\![\xi_1' \sqcup \xi_1'^{r}]\!]_\Sigma$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; t'\ t'') = \widehat{t_2'}\ \widehat{t_2''} : [\![\theta_2 \widehat{\tau}_2''^{r}]\!]_\Sigma \,\&\, [\![\xi_2' \sqcup \xi_2'^{r}]\!]_\Sigma$, this completes the proof.

$t = \mu x : \tau.t'$ By assumption, we have $\mathcal{R}(\widehat{\Gamma_1}; \Sigma; \mu x : \tau.t') = \mu x : \widehat{\tau}_1 \& \xi_1.\widehat{t_1'} : \widehat{\tau}_1 \& \xi_1$ and $\mathcal{R}(\widehat{\Gamma_2}; \Sigma; \mu x : \tau.t') = \mu x : \widehat{\tau}_2 \,\&\, \xi_2.\widehat{t_2'} : \widehat{\tau}_2 \,\&\, \xi_2$. Therefore, both fixpoint iterations converged and there are sequences $(\widehat{\tau}_1^{(i)} \,\&\, \xi_1^{(i)})_{i \leqslant m}$ and $(\widehat{\tau}_2^{(i)} \,\&\, \xi_2^{(i)})_{i \leqslant n}$ such that $\widehat{\tau}_1^{(0)} \& \xi_1^{(0)} = \widehat{\tau}_2^{(0)} \& \xi_2^{(0)} = \bot_\tau \& \bot$, $t_1'^{(i+1)} : \widehat{\tau}_1^{(i+1)} \& \xi_1^{(i+1)} = \mathcal{R}(\widehat{\Gamma_1}, x{:}\widehat{\tau}_1^{(i)} \& \xi_1^{(i)}; \Sigma; t')$ and $t_2'^{(i+1)} : \widehat{\tau}_2^{(i+1)} \& \xi_2^{(i+1)} = \mathcal{R}(\widehat{\Gamma_2}, x{:}\widehat{\tau}_2^{(i)} \& \xi_2^{(i)}; \Sigma; t')$ for some sequences $(t_1'^{(i)})_{1 \leqslant i \leqslant m}$ and $(t_2'^{(i)})_{1 \leqslant i \leqslant n}$.

We proceed by showing that both sequences of type and effect pairs are monotonically increasing.

Claim: For all $i \leqslant m$ we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i)} \leqslant \widehat{\tau}_1^{(i+1)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i)} \sqsubseteq \xi_1^{(i+1)}$.

Proof: By induction on i.

$i = 0$ We have $\widehat{\tau}_1^{(0)} = \bot_\tau$ and $\xi_1^{(0)} = \bot_\star$. By lemma 35, $\Sigma \vdash_{\text{sub}} \bot_\tau \leqslant \widehat{\tau}_1^{(1)}$ and by lemma 33 $\Sigma \vdash_{\text{sub}} \bot_\star \sqsubseteq \xi_1^{(1)}$.

$i = i' + 1$ By induction, we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i')} \leqslant \widehat{\tau}_1^{(i'+1)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i')} \sqsubseteq \xi_1^{(i'+1)}$. Since we have $\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^{(i')} \,\&\, \xi_1^{(i')}; \Sigma; t') = \widehat{t_1'}^{(i'+1)} : \widehat{\tau}_1^{(i'+1)} \,\&\, \xi_1^{(i'+1)}$ and $\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^{(i'+1)} \,\&\, \xi_1^{(i'+1)}; \Sigma; t') = \widehat{t_1'}^{(i'+2)} : \widehat{\tau}_1^{(i'+2)} \,\&\, \xi_1^{(i'+2)}$, we can apply the outer induction hypothesis, resulting in $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i'+1)} \leqslant \widehat{\tau}_1^{(i'+2)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i'+1)} \sqsubseteq \xi_1^{(i'+2)}$.

■

Claim: For all $i \leqslant n$ we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2^{(i)} \leqslant \widehat{\tau}_2^{(i+1)}$ and $\Sigma \vdash_{\text{sub}} \xi_2^{(i)} \sqsubseteq \xi_2^{(i+1)}$.
Proof: Analogously to that of the previous claim. ■
We can now relate both sequences.
Claim: For all $i \leqslant \min\{m, n\}$ we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i)} \leqslant \widehat{\tau}_2^{(i)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i)} \sqsubseteq \xi_2^{(i)}$.
Proof: By induction on i.
$i = 0$ $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(0)} \leqslant \widehat{\tau}_2^{(0)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(0)} \sqsubseteq \xi_2^{(0)}$ hold by [Sub-Refl] and the reflexivity of subsumption.
$i = i' + 1$ By induction, we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i')} \leqslant \widehat{\tau}_2^{(i')}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i')} \sqsubseteq \xi_2^{(i')}$. Moreover, we can apply the outer induction hypothesis to $\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^{(i')} \,\&\, \xi_1^{(i')}; \Sigma; t') = \widehat{t_1'}^{(i'+1)} : \widehat{\tau}_1^{(i'+1)} \,\&\, \xi_1^{(i'+1)}$ and $\mathcal{R}(\widehat{\Gamma_2}, x : \widehat{\tau}_2^{(i')} \,\&\, \xi_2^{(i')}; \Sigma; t') = \widehat{t_2'}^{(i'+1)} : \widehat{\tau}_2^{(i'+1)} \,\&\, \xi_2^{(i'+1)}$. This results in $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i'+1)} \leqslant \widehat{\tau}_2^{(i'+1)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i'+1)} \sqsubseteq \xi_2^{(i'+1)}$.

■

We distinguish three cases.

- If $m = n$, from the previous claim we get $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(m)} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(m)} \sqsubseteq \xi_2^{(n)}$.
- If $m < n$, we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(m)} \leqslant \widehat{\tau}_2^{(m)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(m)} \sqsubseteq \xi_2^{(m)}$ as well as $\Sigma \vdash_{\text{sub}} \widehat{\tau}_2^{(i)} \leqslant \widehat{\tau}_2^{(i+1)}$ and $\Sigma \vdash_{\text{sub}} \xi_2^{(i)} \sqsubseteq \xi_2^{(i+1)}$ for all $m \leqslant i < n$. By transitivity, we get $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(m)} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(m)} \sqsubseteq \xi_2^{(n)}$.
- If $m > n$, we can prove the following claim by induction.
  Claim: For all $n \leqslant i < m$ we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i)} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i)} \sqsubseteq \xi_2^{(n)}$.
  Proof: By induction on i.
  $i = n$ This follows from a previous claim.
  $i = i' + 1$ By induction, we have $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i')} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i')} \sqsubseteq \xi_2^{(n)}$. Moreover, we can apply the outer induction hypothesis to $\mathcal{R}(\widehat{\Gamma_1}, x : \widehat{\tau}_1^{(i')} \,\&\, \xi_1^{(i')}; \Sigma; t') = \widehat{t_1'}^{(i'+1)} : \widehat{\tau}_1^{(i'+1)} \,\&\, \xi_1^{(i'+1)}$ and $\mathcal{R}(\widehat{\Gamma_2}, x : \widehat{\tau}_2^{(n)} \,\&\, \xi_2^{(n)}; \Sigma; t') = \widehat{t_2'}^{(n)} : \widehat{\tau}_2^{(n)} \,\&\, \xi_2^{(n)}$. The latter holds because $\widehat{\tau}_2^{(n)} \,\&\, \xi_2^{(n)}$ is a fixpoint. This results in $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(i'+1)} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(i'+1)} \sqsubseteq \xi_2^{(n)}$.

  ■

  In particular, we can conclude $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1^{(m)} \leqslant \widehat{\tau}_2^{(n)}$ and $\Sigma \vdash_{\text{sub}} \xi_1^{(m)} \sqsubseteq \xi_2^{(n)}$.

Since $\widehat{\tau}_1 \mathbin{\&} \xi_1 = \widehat{\tau}_1^{(m)} \mathbin{\&} \xi_1^{(m)}$ and $\widehat{\tau}_2 \mathbin{\&} \xi_2 = \widehat{\tau}_2^{(m)} \mathbin{\&} \xi_2^{(m)}$, we have shown the desired result $\Sigma \vdash_{\text{sub}} \widehat{\tau}_1 \leqslant \widehat{\tau}_2$ and $\Sigma \vdash_{\text{sub}} \xi_1 \sqsubseteq \xi_2$.

We have now covered the preliminaries required for proving the completeness of our type reconstruction algorithm.

Theorem 6 (Completeness). *Given a source term t, a sort environment Σ, an annotated type environment $\widehat{\Gamma}$ well-formed under Σ, and an underlying type τ such that $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t : \tau$, then there are $\widehat{t}$, $\widehat{\tau}$ and ξ such that $\mathcal{R}(\widehat{\Gamma}; \Sigma; t) = \widehat{t} : \widehat{\tau} \mathbin{\&} \xi$ and $\lfloor \widehat{\tau} \rfloor = \tau$, $\lfloor \widehat{t} \rfloor = t$.*

Proof. By induction on t.

$t = x$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; x) = x : \widehat{\Gamma}(x)$. Suppose $\widehat{\Gamma}(x) = \widehat{\tau} \mathbin{\&} \xi$. By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} x : \tau$. As [U-Var] is the only matching rule, we have $\lfloor \widehat{\Gamma} \rfloor (x) = \tau$ and therefore $\lfloor \widehat{\tau} \rfloor = \tau$. Also, $\lfloor x \rfloor = x$.

$t = ()$ We have $\mathcal{R}(\widehat{\Gamma}; \Sigma; ()) = () : \widehat{\text{unit}} : \bot$ and clearly $\lfloor () \rfloor = ()$. By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} () : \tau$. As the only rule that applies is [U-Unit], we have $\tau = \text{unit}$ and therefore $\lfloor \widehat{\text{unit}} \rfloor = \text{unit} = \tau$.

$t = \text{ann}_\ell(t')$ By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} \text{ann}_\ell(t') : \tau$. This must be by [U-Ann], and therefore $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t' : \tau$. By induction, $\mathcal{R}(\widehat{\Gamma}; \Sigma; t') = \widehat{t'} : \widehat{\tau}' \mathbin{\&} \xi'$ such that $\lfloor \widehat{t'} \rfloor = t'$ and $\lfloor \widehat{\tau}' \rfloor = \tau$.
Then we also have $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{ann}_\ell(t')) = \text{ann}_\ell(\widehat{t'}) : \widehat{\tau}' \mathbin{\&} \llbracket \xi \sqcup \ell \rrbracket_\Sigma$ with $\lfloor \text{ann}_\ell(\widehat{t'}) \rfloor = \text{ann}_\ell(\lfloor \widehat{t'} \rfloor) = \text{ann}_\ell(t')$ and $\lfloor \widehat{\tau}' \rfloor = \tau$.

$t = \text{seq}\ t_1\ t_2$ By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} \text{seq}\ t_1\ t_2 : \tau$. The only rule that applies is [U-Seq], and therefore $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t_1 : \tau_1$ for some τ_1 and $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t_2 : \tau$.
By induction, we have $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1) = \widehat{t_1} : \widehat{\tau}_1 \mathbin{\&} \xi_1$ and $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_2) = \widehat{t_2} : \widehat{\tau}_2 \mathbin{\&} \xi_2$ such that $\lfloor \widehat{t_1} \rfloor = t_1$, $\lfloor \widehat{t_2} \rfloor = t_2$, $\lfloor \widehat{\tau}_1 \rfloor = \tau_1$ and $\lfloor \widehat{\tau}_2 \rfloor = \tau$.
Then we also have $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{seq}\ t_1\ t_2) = \text{seq}\ \widehat{t_1}\ \widehat{t_2} : \widehat{\tau}_2 \mathbin{\&} \llbracket \xi_1 \sqcup \xi_2 \rrbracket_\Sigma$ with $\lfloor \text{seq}\ \widehat{t_1}\ \widehat{t_2} \rfloor = \text{seq}\ \lfloor \widehat{t_1} \rfloor\ \lfloor \widehat{t_2} \rfloor = \text{seq}\ t_1\ t_2$ and $\lfloor \widehat{\tau}_2 \rfloor = \tau$.

$t = (t_1, t_2)$ By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} (t_1, t_2) : \tau$. The only matching rule is [U-Pair], therefore we have $\tau = \tau_1 \times \tau_2$ for some τ_1, τ_2, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t_1 : \tau_1$ and $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t_2 : \tau_2$.
By induction we get $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1) = \widehat{t_1} : \widehat{\tau}_1 \mathbin{\&} \xi_1$ and $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_2) = \widehat{t_2} : \widehat{\tau}_2 \mathbin{\&} \xi_2$ such that $\lfloor \widehat{t_1} \rfloor = t_1$, $\lfloor \widehat{\tau}_1 \rfloor = \tau_1$, $\lfloor \widehat{t_2} \rfloor = t_2$ and $\lfloor \widehat{\tau}_2 \rfloor = \tau_2$.
Then, $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1, t_2) = (\widehat{t_1}, \widehat{t_2}) : \widehat{\tau}_1 \langle \xi_1 \rangle \times \widehat{\tau}_2 \langle \xi_2 \rangle \mathbin{\&} \bot$ and we have $\lfloor (\widehat{t_1}, \widehat{t_2}) \rfloor = (\lfloor \widehat{t_1} \rfloor, \lfloor \widehat{t_2} \rfloor) = (t_1, t_2)$ and $\lfloor \widehat{\tau}_1 \langle \xi_1 \rangle \times \widehat{\tau}_2 \langle \xi_2 \rangle \rfloor = \lfloor \widehat{\tau}_1 \rfloor \times \lfloor \widehat{\tau}_2 \rfloor = \tau_1 \times \tau_2 = \tau$.

$t = \text{inl}_{\tau_2}(t_1)$ By assumption, $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} \text{inl}_{\tau_2}(t_1) : \tau$. The only matching rule is [U-Inl], therefore we have $\tau = \tau_1 + \tau_2$ for some τ_1 such that $\lfloor \widehat{\Gamma} \rfloor \vdash_{\text{t}} t_1 : \tau_1$.
By induction we get $\mathcal{R}(\widehat{\Gamma}; \Sigma; t_1) = \widehat{t_1} : \widehat{\tau}_1 \mathbin{\&} \xi_1$ such that $\lfloor \widehat{t_1} \rfloor = t_1$ and $\lfloor \widehat{\tau}_1 \rfloor = \tau_1$.
Then, $\mathcal{R}(\widehat{\Gamma}; \Sigma; \text{inl}_{\tau_2}(t_1)) = \text{inl}_{\tau_2}(\widehat{t_1}) : \widehat{\tau}_1 \langle \xi_1 \rangle + \bot_{\tau_2} \langle \bot \rangle \mathbin{\&} \bot$ and we have $\lfloor \text{inl}_{\tau_2}(\widehat{t_1}) \rfloor = \text{inl}_{\tau_2}(\lfloor \widehat{t_1} \rfloor) = \text{inl}_{\tau_2}(t_1)$ and $\lfloor \widehat{\tau}_1 \langle \xi_1 \rangle + \bot_{\tau_2} \langle \bot \rangle \rfloor = \lfloor \widehat{\tau}_1 \rfloor + \lfloor \bot_{\tau_2} \rfloor = \tau_1 + \tau_2 = \tau$. In the second to last step, $\lfloor \bot_{\tau_2} \rfloor = \tau_2$ follows from Lemma 34.

$t = \text{inl}_{\tau_1}(t_2)$ Analogously to the previous case.

$t = \mathrm{proj}_i(t')$ By assumption, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} \mathrm{proj}_i(t') : \tau$. The only matching rule is [U-PROJ], therefore we have $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} t' : \tau_1 \times \tau_2$ for some τ_1 and τ_2 such that $\tau = \tau_i$.
By induction, $\mathcal{R}(\widehat{\Gamma}; \Sigma; t') = \widehat{t'} : \widehat{\tau}' \,\&\, \xi'$ such that $\lfloor\widehat{t'}\rfloor = t'$ and $\lfloor\widehat{\tau}'\rfloor = \tau_1 \times \tau_2$.
By theorem 5, $\widehat{\tau}'$ is conservative. Therefore, $\widehat{\tau}' = \widehat{\tau}_1'\langle\xi_1'\rangle \times \widehat{\tau}_2'\langle\xi_2'\rangle$ such that $\lfloor\widehat{\tau}_1'\rfloor = \tau_1$ and $\lfloor\widehat{\tau}_2'\rfloor = \tau_2$.
Then, $\mathcal{R}(\widehat{\Gamma}; \Sigma; \mathrm{proj_i}(t')) = \mathrm{proj}_i(\widehat{t'}) : \widehat{\tau}_i' \,\&\, \llbracket \xi \sqcup \xi_i \rrbracket_\Sigma$ and $\lfloor \mathrm{proj}_i(\widehat{t'})\rfloor = \mathrm{proj}_i(\lfloor\widehat{t'}\rfloor) = \mathrm{proj}_i(t')$, $\lfloor\widehat{\tau}_i'\rfloor = \tau_i = \tau$.

$t = \textbf{case}\ t'\ \textbf{of}\ \{\mathrm{inl}(x) \to t_1; \mathrm{inr}(x) \to t_2\}$ By assumption, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}}$ $\textbf{case}\ t'\ \textbf{of}\ \{\mathrm{inl}(x) \to t_1; \mathrm{inr}(x) \to t_2\} : \tau$. Only [U-CASE] applies, hence $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} t' : \tau_1 + \tau_2$ for some τ_1, τ_2 and $\lfloor\widehat{\Gamma}\rfloor, x : \tau_1 \vdash_{\mathrm{t}} t_1 : \tau$ and $\lfloor\widehat{\Gamma}\rfloor, x : \tau_2 \vdash_{\mathrm{t}} t_2 : \tau$.
By induction, $\mathcal{R}(\widehat{\Gamma}; \Sigma; t') = \widehat{t'} : \widehat{\tau}' \,\&\, \xi'$ such that $\lfloor\widehat{t'}\rfloor = t'$ and $\lfloor\widehat{\tau}'\rfloor = \tau_1 + \tau_2$.
By theorem 5, $\widehat{\tau}'$ is conservative and $\Sigma \vdash_{\mathrm{wft}} \widehat{\tau}'$. Therefore, $\widehat{\tau}' = \widehat{\tau}_1'\langle\xi_1'\rangle + \widehat{\tau}_2'\langle\xi_2'\rangle$ such that $\lfloor\widehat{\tau}_1'\rfloor = \tau_1$ and $\lfloor\widehat{\tau}_2'\rfloor = \tau_2$.
Note that only [W-SUM] applies to the well-formedness judgment. But then $\widehat{\tau}_1'$ and $\widehat{\tau}_2'$ are also conservative and $\Sigma \vdash_{\mathrm{s}} \xi_1' : \star$ and $\Sigma \vdash_{\mathrm{s}} \xi_2' : \star$ hold. Hence, $\widehat{\Gamma}, x : \widehat{\tau}_1' \,\&\, \xi_1'$ and $\widehat{\Gamma}, x : \widehat{\tau}_2' \,\&\, \xi_2'$ are both well-formed under Σ. Moreover, $\lfloor\widehat{\Gamma}, x : \widehat{\tau}_1' \,\&\, \xi_1'\rfloor = \lfloor\widehat{\Gamma}\rfloor, x : \tau_1$ and $\lfloor\widehat{\Gamma}, x : \widehat{\tau}_2' \,\&\, \xi_2'\rfloor = \lfloor\widehat{\Gamma}\rfloor, x : \tau_2$.
Now we can apply the induction hypothesis to both case branches, resulting in $\mathcal{R}(\widehat{\Gamma}, x : \widehat{\tau}_1' \,\&\, \xi_1'; \Sigma; t_1) = \widehat{t_1} : \widehat{\tau}_1 \,\&\, \xi_1$ and $\mathcal{R}(\widehat{\Gamma}, x : \widehat{\tau}_2' \,\&\, \xi_2'; \Sigma; t_2) = \widehat{t_2} : \widehat{\tau}_2 \,\&\, \xi_2$ such that $\lfloor\widehat{t_1}\rfloor = t_1$, $\lfloor\widehat{\tau}_1\rfloor = \tau$, $\lfloor\widehat{t_2}\rfloor = t_2$ and $\lfloor\widehat{\tau}_2\rfloor = \tau$.
Then, $\mathcal{R}(\widehat{\Gamma}; \Sigma;\ \textbf{case}\ t'\ \textbf{of}\ \{\mathrm{inl}(x) \to t_1; \mathrm{inr}(x) \to t_2\}) = \textbf{case}\ \widehat{t'}\ \textbf{of}\ \{\mathrm{inl}(x) \to \widehat{t_1}; \mathrm{inr}(x) \to \widehat{t_2}\} : \llbracket \widehat{\tau}_1 \sqcup \widehat{\tau}_2 \rrbracket_\Sigma \,\&\, \llbracket \xi' \sqcup \xi_1 \sqcup \xi_2 \rrbracket$.. Now, clearly

$$\begin{aligned}
&\lfloor\ \textbf{case}\ \widehat{t'}\ \textbf{of}\ \{\mathrm{inl}(x) \to \widehat{t_1}; \mathrm{inr}(x) \to \widehat{t_2}\}\rfloor \\
=&\textbf{case}\ \lfloor\widehat{t'}\rfloor\ \textbf{of}\ \{\mathrm{inl}(x) \to \lfloor\widehat{t_1}\rfloor; \mathrm{inr}(x) \to \lfloor\widehat{t_2}\rfloor\} \\
=&\textbf{case}\ t'\ \textbf{of}\ \{\mathrm{inl}(x) \to t_1; \mathrm{inr}(x) \to t_2\}.
\end{aligned}$$

By lemma 40, $\lfloor\llbracket \widehat{\tau}_1 \sqcup \widehat{\tau}_2 \rrbracket_\Sigma\rfloor = \lfloor\widehat{\tau}_1 \sqcup \widehat{\tau}_2\rfloor = \tau$.

$t = \lambda x : \tau_1.t'$ By assumption, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} \lambda x : \tau_1.t' : \tau$. The only rule that applies is [U-ABS], therefore $\tau = \tau_1 \to \tau_2$ for some τ_2 and the premise is $\lfloor\widehat{\Gamma}\rfloor, x : \tau_1 \vdash_{\mathrm{t}} t' : \tau_2$.
We have $\widehat{\tau}_1 \,\&\, \beta \triangleright \overline{\beta_i :: \kappa_i} = \mathcal{C}([]; \tau_1)$. We define $\widehat{\Gamma'} = \widehat{\Gamma}, x : \widehat{\tau}_1 \,\&\, \beta$. By lemma 32 we have $\lfloor\widehat{\tau}_1\rfloor = \tau_1$. Then $\lfloor\widehat{\Gamma'}\rfloor = \lfloor\widehat{\Gamma}\rfloor, x : \lfloor\widehat{\tau}_1\rfloor = \lfloor\widehat{\Gamma}\rfloor, x : \tau_1$. By a reasoning similar to the corresponding soundness proof, we know $\widehat{\Gamma'}$ is well-formed under $\Sigma, \overline{\beta_i :: \kappa_i}$.
We can now apply the induction hypothesis, so we have $\mathcal{R}(\widehat{\Gamma'}; \Sigma, \overline{\beta_i :: \kappa_i}; t') = \widehat{t'} : \widehat{\tau}_2 \,\&\, \xi_2$ such that $\lfloor\widehat{t'}\rfloor = t'$ and $\lfloor\widehat{\tau}_2\rfloor = \tau_2$.
Then, $\mathcal{R}(\widehat{\Gamma}; \Sigma; \lambda x : \tau_1.t') = \Lambda\overline{\beta_i :: \kappa_i}.\lambda x : \widehat{\tau}_1 \,\&\, \beta.\widehat{t'} : \forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle \,\&\, \bot$ with $\lfloor\Lambda\overline{\beta_i :: \kappa_i}.\lambda x : \widehat{\tau}_1 \,\&\, \beta.\widehat{t'}\rfloor = \lfloor\lambda x : \widehat{\tau}_1 \,\&\, \beta.\widehat{t'}\rfloor = \lambda x : \lfloor\widehat{\tau}_1\rfloor.\lfloor\widehat{t'}\rfloor = \lambda x : \tau_1.t'$ and $\lfloor\forall\overline{\beta_i :: \kappa_i}.\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle\rfloor = \lfloor\widehat{\tau}_1\langle\beta\rangle \to \widehat{\tau}_2\langle\xi_2\rangle\rfloor = \lfloor\widehat{\tau}_1\rfloor \to \lfloor\widehat{\tau}_2\rfloor = \tau_1 \to \tau_2$.

$t = t_1\ t_2$ By assumption, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} t_1\ t_2 : \tau$. Only [U-APP] applies, therefore $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} t_1 : \tau_2 \to \tau$ and $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} t_2 : \tau_2$ for some τ_2.

By induction, $\mathcal{R}(\lfloor\widehat{\Gamma}\rfloor;\Sigma;t_1) = \widehat{t_1}:\widehat{\tau}_1 \,\&\, \xi_1$ and $\mathcal{R}(\lfloor\widehat{\Gamma}\rfloor;\Sigma;t_2) = \widehat{t_2}:\widehat{\tau}_2 \,\&\, \xi_2$ such that $\lfloor\widehat{t_1}\rfloor = t_1$, $\lfloor\widehat{\tau}_1\rfloor = \tau_2 \to \tau$, $\lfloor\widehat{t_2}\rfloor = t_2$ and $\lfloor\widehat{\tau}_2\rfloor = \tau_2$. By theorem 5, $\widehat{\tau}_1$ and $\widehat{\tau}_2$ are conservative.

As $\widehat{\tau}_1$ is a function type and $\mathcal{I}$ is structurally recursive, the call always succeeds, yielding $\widehat{\tau}'_2\langle\beta\rangle \to \widehat{\tau}\langle\xi\rangle \rhd \overline{\beta_i} = \mathcal{I}(\widehat{\tau}_1)$. The call to $\mathcal{M}$ match also succeeds by lemma 39, resulting in $\theta = [\beta \mapsto \xi_2] \circ \mathcal{M}([\,];\widehat{\tau}'_2;\widehat{\tau}_2)$.

Since $\lfloor\widehat{\tau}_1\rfloor = \tau_2 \to \tau$, we must also have $\lfloor\widehat{\tau}'_2\langle\xi'_2\rangle \to \widehat{\tau}\langle\xi\rangle\rfloor = \tau_2 \to \tau$ by definition of $\mathcal{I}$. That implies $\lfloor\widehat{\tau}\rfloor = \tau$.

Then, $\mathcal{R}(\widehat{\Gamma};\Sigma;t_1\ t_2) = \widehat{t_1}\ \langle\overline{\theta\ \beta_i}\rangle\ \widehat{t_2}{:}\llbracket\theta\widehat{\tau}\rrbracket_\Sigma \& \llbracket\xi_1 \sqcup \theta\ \xi\rrbracket_\Sigma$. We have $\lfloor\widehat{t_1}\ \langle\overline{\theta\ \beta_i}\rangle\ \widehat{t_2}\rfloor = \lfloor\widehat{t_1}\rfloor\ \lfloor\widehat{t_2}\rfloor = t_1\ t_2$ and $\lfloor\llbracket\theta\widehat{\tau}\rrbracket_\Sigma\rfloor = \lfloor\theta\widehat{\tau}\rfloor = \lfloor\widehat{\tau}\rfloor = \tau$. The second step is justified by the fact that substitutions do not change the structure of types.

$t = \mu x:\tau.t'$ By assumption, $\lfloor\widehat{\Gamma}\rfloor \vdash_{\mathrm{t}} \mu x : \tau.t' : \tau$. The only matching rule is [U-Fix], therefore $\lfloor\widehat{\Gamma}\rfloor, x:\tau \vdash_{\mathrm{t}} t':\tau$ must hold as its premise.

We define the sequences $(\widehat{t'}_i)_{i\in\mathbb{N}^+}$ and $(\widehat{\tau}_i \,\&\, \xi_i)_{i\in\mathbb{N}}$ by

$$\widehat{\tau}_0 \,\&\, \xi_0 := \bot_\tau \,\&\, \bot_\star$$
$$\widehat{t'}_{i+1} : \widehat{\tau}_{i+1} \,\&\, \xi_{i+1} := \mathcal{R}(\widehat{\Gamma}, x:\widehat{\tau}_i \,\&\, \xi_i;\Sigma;t')$$

The well-definedness of this definition follows from the induction hypothesis. Moreover, $\lfloor\widehat{\tau}_i\rfloor = \tau$ follows from lemma 34 for $i = 0$ and from the induction hypothesis for $i > 0$.

Claim: For all $i \in \mathbb{N}$ we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_i \leqslant \widehat{\tau}_{i+1}$ and $\Sigma \vdash_{\mathrm{sub}} \xi_i \sqsubseteq \xi_{i+1}$.

Proof: By induction on i.

$i = 0$ We have $\widehat{\tau}_0 = \bot_\tau$ and $\xi_0 = \bot_\star$. By lemma 35, $\Sigma \vdash_{\mathrm{sub}} \bot_\tau \leqslant \widehat{\tau}_1$ and by lemma 33 $\Sigma \vdash_{\mathrm{sub}} \bot_\star \sqsubseteq \xi_1$.

$i = i' + 1$ By induction, we have $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_{i'} \leqslant \widehat{\tau}_{i'+1}$ and $\Sigma \vdash_{\mathrm{sub}} \xi_{i'} \sqsubseteq \xi_{i'+1}$. Since we have $\mathcal{R}(\widehat{\Gamma}, x:\widehat{\tau}_{i'} \,\&\, \xi_{i'};\Sigma;t') = \widehat{t'}_{i'+1} : \widehat{\tau}_{i'+1} \,\&\, \xi_{i'+1}$ and $\mathcal{R}(\widehat{\Gamma}, x{:}\widehat{\tau}_{i'+1} \& \xi_{i'+1};\Sigma;t') = \widehat{t'}_{i'+2}{:}\widehat{\tau}_{i'+2} \& \xi_{i'+2}$, we can apply lemma 49, resulting in $\Sigma \vdash_{\mathrm{sub}} \widehat{\tau}_{i'+1} \leqslant \widehat{\tau}_{i'+2}$ and $\Sigma \vdash_{\mathrm{sub}} \xi_{i'+1} \sqsubseteq \xi_{i'+2}$.

∎

This implies $[\widehat{\tau}_i \,\&\, \xi_i]_\Sigma \sqsubseteq_\Sigma [\widehat{\tau}_{i+1} \,\&\, \xi_{i+1}]_\Sigma$ for all i by definition. By lemma 43, there is only a finite number of such equivalence classes. Hence, there is an index j such that $[\widehat{\tau}_i \,\&\, \xi_i]_\Sigma = [\widehat{\tau}_{i+1} \,\&\, \xi_{i+1}]_\Sigma$ for all $i \geqslant j$. But then $\widehat{\tau}_i \,\&\, \xi_i \equiv_\Sigma \widehat{\tau}_{i+1} \,\&\, \xi_{i+1}$ for all $i \geqslant j$. In particular, the termination condition of the fixpoint iteration is fulfilled after the j-th iteration.

We can conclude $\mathcal{R}(\widehat{\Gamma};\Sigma;t) = \mu x : \widehat{\tau}_{j+1} \,\&\, \xi_{j+1}.\widehat{t'}_{j+1} : \widehat{\tau}_{j+1} \,\&\, \xi_{j+1}$. Clearly,

$$\lfloor\mu x : \widehat{\tau}_{j+1} \,\&\, \xi_{j+1}.\widehat{t'}_{j+1}\rfloor = \mu x : \lfloor\widehat{\tau}_{j+1}\rfloor.\lfloor\widehat{t'}_{j+1}\rfloor = \mu x : \tau.t'.$$

As a corollary from the foregoing theorems, our analysis is a conservative extension of the underlying type system.

Corollary 2 (Conservative Extension). *Let t be a source term, τ be a type and Γ a type environment such that $\Gamma \vdash_{\mathrm{t}} t:\tau$. Then there are Σ, $\widehat{\Gamma}$, $\widehat{t}$, $\widehat{\tau}$, ξ such that $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t} : \widehat{\tau} \,\&\, \xi$ with $\lfloor\widehat{t}\rfloor = t$, $\lfloor\widehat{\tau}\rfloor = \tau$ and $\lfloor\widehat{\Gamma}\rfloor = \Gamma$.*

Proof. Construct $\widehat{\Gamma}$ from Γ by replacing every binding $x : \tau'$ with a binding $x{:}\bot_{\tau'}\&\bot_\star$. Set Σ to be the empty sort context. By lemma 34, $\bot_{\tau'}$ is conservative, $\Sigma \vdash_{\mathrm{wft}} \bot_{\tau'}$ and $\lfloor \bot_{\tau'} \rfloor = \tau'$ for all such τ'. The latter also implies $\lfloor \widehat{\Gamma} \rfloor = \Gamma$ and therefore $\lfloor \widehat{\Gamma} \rfloor \vdash_{\mathrm{t}} t : \tau$. By lemma 33, $\Sigma \vdash_{\mathrm{s}} \bot_\star : \star$. Hence, $\widehat{\Gamma}$ is well-formed under Σ.

This allows us to apply theorem 6, so there must be $\widehat{t}$, $\widehat{\tau}$ and ξ such that $\mathcal{R}(\widehat{\Gamma}; \Sigma; t) = \widehat{t} : \widehat{\tau} \,\&\, \xi$ with $\lfloor \widehat{\tau} \rfloor = \tau$ and $\lfloor \widehat{t} \rfloor = t$. Since $\widehat{\Gamma}$ is well-formed under Σ, we can conclude from theorem 5 that $\Sigma \mid \widehat{\Gamma} \vdash_{\mathrm{te}} \widehat{t} : \widehat{\tau} \,\&\, \xi$ holds.

F More examples

Construction and Elimination We start with a few simple examples demonstrating that the dependency annotations inferred for constructors and eliminators are sensible. We use the terms *constructors* and *eliminators* deliberately here, because the basic concept generalizes over product types, sum types and function types.

Whenever something is constructed, be it a product, a sum or a lambda abstraction, the outermost dependency annotation that is assigned is $\bot$. This is because the analysis aims to produce the best possible and thereby least annotations for a given source program.

On the other hand, when eliminating through projection, case distinction or function application, the dependency annotation of the term that is eliminated is included in the result. This is a necessary requirement for the analysis results to be sound.

Consider the case of binding-time analysis, and suppose we have a variable of function type $f : \forall\beta.\mathrm{int}\langle\beta\rangle \to \mathrm{int}\langle\beta\rangle \,\&\, \mathbf{D}$. We can see that it preserves the annotations of its arguments, i.e. if we apply f to a static value, the return annotation is also instantiated to be static. The function itself, however, is dynamic. And therefore, the whole result of the function application must also be dynamic, because we cannot know which particular function has been assigned to f.

As elimination always introduces a dependency in the program, and this can uncover subtleties arising when functions only differ in their termination behavior. For example, compare $\lambda p : \mathrm{int} \times \mathrm{int}.p$ with $\lambda p : \mathrm{int} \times \mathrm{int}.(\mathrm{proj}_1(p), \mathrm{proj}_2(p))$.

In a call-by-value language, these two functions would be (extensionally) equivalent. However, with non-strict evaluation, p might be a non-terminating computation. In that case, applying the former function would diverge, while the latter function at least produces the pair constructor. This is also reflected in the annotated types that are inferred. For the former, we get

$$\forall\beta_0, \beta_1, \beta_2 :: \star.(\mathrm{int}\langle\beta_0\rangle \times \mathrm{int}\langle\beta_1\rangle)\langle\beta_2\rangle \to (\mathrm{int}\langle\beta_0\rangle \times \mathrm{int}\langle\beta_1\rangle)\langle\beta_2\rangle \,\&\, \mathbf{S}, \text{and}$$

$$\forall\beta_0, \beta_1, \beta_2 :: \star.(\mathrm{int}\langle\beta_0\rangle \times \mathrm{int}\langle\beta_1\rangle)\langle\beta_2\rangle \to (\mathrm{int}\langle\beta_0 \sqcup \beta_2\rangle \times \mathrm{int}\langle\beta_1 \sqcup \beta_2\rangle)\langle\mathbf{S}\rangle \,\&\, \mathbf{S}$$

for the latter. In particular, the annotation of the product in the second type signature is $\mathbf{S}$. Therefore, it can not depend on the input of the function.

Polymorphic Recursion One class of functions where the analysis benefits from polymorphic recursion are those that permute their arguments on recursive calls. Our example is a slightly modified version of the motivating example of [5] for polymorphic recursion in binding-time analysis:

$$\mu f : \text{bool} \to \text{bool} \to \text{bool}.\lambda x : \text{bool}.\lambda y : \text{bool}.\ \textbf{if}\ x\ \textbf{then}\ \mathit{true}\ \textbf{else}\ f\ y\ x$$

In an analysis with monomorphic recursion, the analysis assigns the same annotation to both parameters, large enough to accommodate for both arguments. This is due to the permutation of the arguments in the else branch.

On the other hand, an analysis with polymorphic recursion is allowed to use a different instantiation for f in that case. Our algorithm hence infers the following most general type.

$$\forall \beta_1 :: \star.\widehat{\text{bool}}\langle \beta_1 \rangle \to (\forall \beta_2 :: \star.\widehat{\text{bool}}\langle \beta_2 \rangle \to \widehat{\text{bool}}\langle \beta_1 \sqcup \beta_2 \rangle)\langle \bot \rangle \ \&\ \bot$$

We see that the result of the function indeed depends on the annotations of both arguments, as both end up in the condition of the if-expression at some point. Yet, both arguments are completely unrestricted, and unrelated in their annotations.

The instantiation of the recursive call is explicitly visible in the target language term generated by the analysis.

$$\begin{aligned}&\mu f : (\forall \beta_1 :: \star.\widehat{\text{bool}}\langle \beta_1 \rangle \to (\forall \beta_2 :: \star.\widehat{\text{bool}}\langle \beta_2 \rangle \to \widehat{\text{bool}}\langle \beta_1 \sqcup \beta_2 \rangle)\langle \bot \rangle) \ \&\ \bot.\\ &\quad \Lambda \beta_1 :: \star.\lambda x : \widehat{\text{bool}} \ \&\ \beta_1.\Lambda \beta_2 :: \star.\lambda y : \widehat{\text{bool}} \ \&\ \beta_2.\ \textbf{if}\ x\ \textbf{then}\ \mathit{true}\ \textbf{else}\ f\ \langle \beta_2 \rangle\ y\ \langle \beta_1 \rangle\ x\end{aligned}$$

A type that is polymorphic in the annotations of the arguments is assigned to the fixpoint binding and the quantification is explicit on the term level. We can observe that β_2 is passed to the formal parameter β_1.

In contrast, a type system with monomorphic recursion would only admit a weaker type, possibly similar to the following, where the annotations of both arguments must be the same.

$$\forall \beta_1 :: \star.\widehat{\text{bool}}\langle \beta_1 \rangle \to (\widehat{\text{bool}}\langle \beta_1 \rangle \to \widehat{\text{bool}}\langle \beta_1 \rangle)\langle \bot \rangle \ \&\ \bot$$

A real world example where the annotations of the arguments eventually end up in both formal parameters is Euclid's algorithm for computing the greatest common divisor.

Higher-Ranked Polyvariance This section discusses several examples where the analysis benefits from higher-ranked polyvariance, those where arguments of function type are used more than once. We do this for the dependency analysis instance of binding time analysis, in particular by comparing against the constraint-based polyvariant binding time analysis of [29].

A simple example to start with is a function that applies a function to both components of a pair[4]

[4] Note that *both* is an instance of a general traversal $\forall f.\mathit{Applicative}\ f \Rightarrow (\mathit{Int} \to f\ \mathit{Int}) \to (\mathit{Int}, \mathit{Int}) \to f\ (\mathit{Int}, \mathit{Int})$ that we have simplified in order to fit the restrictions of the source language [6,15].

$$\begin{array}{l} \mathit{both} : (\text{int} \to \text{int}) \to \text{int} \times \text{int} \to \text{int} \times \text{int} \\ \mathit{both} = \lambda f : \text{int} \to \text{int}.\lambda p : \text{int} \times \text{int}.(f\ (\text{proj}_1(p)), f\ (\text{proj}_2(p))) \end{array}$$

Suppose in the context of binding-time analysis that *both* is used to apply a statically known function to a pair whose the first component is always computable at compile time, but whose second component is dynamic.

For simplicity's sake, we will use the identity function as an argument to *both*.

$$\begin{array}{l} \mathit{id} : \text{int} \to \text{int} \\ \mathit{id} = \lambda x : \text{int}.x \end{array}$$

The non-higher-ranked analysis would assign the following types to our example functions.

$$\begin{array}{ll} \mathit{id} & : \forall \beta_1\ \beta_2.(\beta_1 \sqsubseteq \beta_2, \beta_3 \sqsubseteq \beta_1, \beta_3 \sqsubseteq \beta_2) \Rightarrow \text{int}\langle\beta_1\rangle \xrightarrow{\beta_3} \text{int}\langle\beta_2\rangle \\ \mathit{both} & : \forall \beta_1\ ...\ \beta_{10}.(\beta_3 \sqsubseteq \beta_1, \beta_3 \sqsubseteq \beta_2, \beta_4 \sqsubseteq \beta_3, \beta_4 \sqsubseteq \beta_8, \beta_7 \sqsubseteq \beta_5, \beta_7 \sqsubseteq \beta_6, \beta_8 \sqsubseteq \beta_7 \\ & \quad , \beta_8 \sqsubseteq \beta_{11}, \beta_{11} \sqsubseteq \beta_9, \beta_{11} \sqsubseteq \beta_{10}, \beta_5 \sqsubseteq \beta_1, \beta_6 \sqsubseteq \beta_1, \beta_2 \sqsubseteq \beta_9, \beta_2 \sqsubseteq \beta_{10}) \\ & \Rightarrow (\text{int}\langle\beta_1\rangle \xrightarrow{\beta_3} \text{int}\langle\beta_2\rangle) \xrightarrow{\beta_4} (\text{int}\langle\beta_5\rangle \times \text{int}\langle\beta_6\rangle)\langle\beta_7\rangle \xrightarrow{\beta_8} (\text{int}\langle\beta_9\rangle \times \text{int}\langle\beta_{10}\rangle)\langle\beta_{11}\rangle \end{array}$$

The limiting factor in the expressiveness of the analysis is the fact that the annotations β_1 and β_2 of the return value of the function argument to *both* are chosen wherever *both* is called. In order for this instantiation to be valid for all occurrences of the argument f in the body of *both*, the constraints $\beta_5 \sqsubseteq \beta_1$, $\beta_6 \sqsubseteq \beta_1$, $\beta_2 \sqsubseteq \beta_9$ and $\beta_2 \sqsubseteq \beta_{10}$ must be fulfilled. The first two constraints state that the annotation of the function's argument must be large enough to be able to take both components of the pair as input. The latter two ensure that the annotations of the returned pair are at least as large as whatever the function returns.

When we look at the partial application *both id*, we have the additional constraint $\beta_1 \sqsubseteq \beta_2$ from the type of *id*. This forces β_9 and β_{10} to be larger than both β_5 and β_6. Consider the call *both id p* for some pair $p : \text{int}\langle\mathbf{S}\rangle \times \text{int}\langle\mathbf{D}\rangle \,\&\, \mathbf{S}$. Then, $\beta_5 = \mathbf{S}$ and $\beta_6 = \mathbf{D}$, but even though we are applying the identity function, the whole call has the type $\text{int}\langle\mathbf{D}\rangle \times \text{int}\langle\mathbf{D}\rangle$.

In our higher-ranked analysis, we can infer the following conservative type for the identity function.

$$\begin{array}{l} \mathit{id} : \forall \beta :: \star.\text{int}\langle\beta\rangle \to \text{int}\langle\beta\rangle \,\&\, \bot \\ \mathit{id} = \Lambda\beta :: \star.\lambda x : \text{int} \,\&\, \beta.x \end{array}$$

Because *id* is not a higher-order function, we have not yet gained more precision at this point. However, we obtain the following target term and annotated type for the function *both*.

$$\begin{array}{l} \mathit{both} : \forall \beta_1 :: \star.\forall \beta_2 :: \star \Rightarrow \star.(\forall \beta :: \star.\text{int}\langle\beta\rangle \to \text{int}\langle\beta_2\ \beta\rangle)\langle\beta_1\rangle \\ \quad \to (\forall \beta_3, \beta_4, \beta_5 :: \star.(\text{int}\langle\beta_3\rangle \times \text{int}\langle\beta_4\rangle)\langle\beta_5\rangle \\ \quad \to (\text{int}\langle\beta_2\ (\beta_3 \sqcup \beta_5) \sqcup \beta_1\rangle \times \text{int}\langle\beta_2\ (\beta_4 \sqcup \beta_5) \sqcup \beta_1\rangle)\langle\mathbf{S}\rangle)\langle\mathbf{S}\rangle \,\&\, \mathbf{S} \end{array}$$

$$\begin{aligned}&\mathit{both} = \Lambda\beta_1 :: \star.\Lambda\beta_2 :: \star \Rightarrow \star.\lambda f : (\forall\beta :: \star.\mathrm{int}\langle\beta\rangle \to \mathrm{int}\langle\beta_2\ \beta\rangle).\\&\quad\Lambda\beta_3 :: \star.\Lambda\beta_4 :: \star.\Lambda\beta_5 :: \star.\lambda p : \mathrm{int}\langle\beta_3\rangle \times \mathrm{int}\langle\beta_4\rangle.\\&\qquad(f\ \langle\beta_3 \sqcup \beta_5\rangle\ (\mathrm{proj}_1(p)), f\ \langle\beta_4 \sqcup \beta_5\rangle\ (\mathrm{proj}_2(p)))\end{aligned}$$

The function parameter f can be instantiated separately for each component because our analysis assigns it a type that universally quantifies over the annotation of its argument. It is evident from the type signature that the components of the resulting pair only depend on the corresponding components of the input pair, and the function and the input pair itself. They do not depend on the respective other component of the input.

If we again consider the call *both id p*, we obtain $\beta_2 = \lambda\beta :: \star.\beta$, $\beta_1 = \beta_3 = \beta_5 = \mathbf{S}$ and $\beta_4 = \mathbf{D}$ through pattern unification. Normalization of the resulting dependency terms results in the expected return type $\mathrm{int}\langle\mathbf{S}\rangle \times \mathrm{int}\langle\mathbf{D}\rangle$.

The generality provided by the higher-ranked analysis extends to an arbitrarily deep nesting of function arrows. The following example demonstrates this for two levels of arrows. Functions with more than two levels of arrows can arise directly in actual programs, but even more so in desugared code, e.g., when type classes in Haskell are implemented via explicit dictionary passing. Essentially, higher-ranked polyvariance avoids paying the price of losing precision when such desugaring takes place. Due to limitations of the source language, our examples are heavily restricted instances of this.

Consider the following function that takes a function argument which again requires a function.

$$\begin{aligned}&\mathit{foo} : ((\mathrm{int} \to \mathrm{int}) \to \mathrm{int}) \to \mathrm{int} \times \mathrm{int}\\&\mathit{foo} = \lambda f : (\mathrm{int} \to \mathrm{int}) \to \mathrm{int}.(f\ (\lambda x : \mathrm{int}.x), f\ (\lambda x : \mathrm{int}.0))\end{aligned}$$

The higher-ranked analysis infers the following type and target term (where we omitted the type in the argument of the lambda term because it essentially repeats what is already visible in the top level type signature).

$$\begin{aligned}&\mathit{foo} : \forall\beta_4 :: \star.\forall\beta_3 :: \star \Rightarrow (\star \Rightarrow \star) \Rightarrow \star.\\&\quad(\forall\beta_2 :: \star.\forall\beta_1 :: \star \Rightarrow \star.(\forall\beta_0 :: \star.\mathrm{int}\langle\beta_0\rangle \to \mathrm{int}\langle\beta_1\ \beta_0\rangle)\langle\beta_2\rangle \to \mathrm{int}\langle\beta_3\ \beta_2\ \beta_1\rangle)\langle\beta_4\rangle\\&\quad\to (\mathrm{int}\langle\beta_3\ \mathbf{S}\ (\lambda\beta_5 :: \star.\beta_5) \sqcup \beta_4\rangle \times \mathrm{int}\langle\beta_3\ \mathbf{S}\ (\lambda\beta_6 :: \star.\mathbf{S}) \sqcup \beta_4\rangle)\langle\mathbf{S}\rangle\ \&\ \mathbf{S}\\&\mathit{foo} = \Lambda\beta_4 :: \star.\Lambda\beta_3 :: \star \Rightarrow (\star \Rightarrow \star) \Rightarrow \star.\lambda f : \cdots.\\&\quad(f\ \langle\mathbf{S}\rangle\ \langle\lambda\beta_0 :: \star.\beta_0\rangle\ (\Lambda\beta_5 :: \star.\lambda x : \mathrm{int}\ \&\ \beta_5.x)\\&\quad, f\ \langle\mathbf{S}\rangle\ \langle\lambda\beta_0 :: \star.\mathbf{S}\rangle\ \ (\Lambda\beta_6 :: \star.\lambda x : \mathrm{int}\ \&\ \beta_6.1))\end{aligned}$$

Since the type of f is a pattern type, the argument to f is also a pattern type by definition. Therefore, the analysis of f depends on the analysis of the function passed to it. This gives rise to the *higher-order effect operator* β_3 [12]. Thus, f can be applied to any function with a conservative type of the right shape. As our algorithm always infers conservative types, the type of f is as general as possible.

This is reflected in the body of the lambda where in both cases f is instantiated with the dependency annotation corresponding to the function passed to it. The result of this instantiation can be observed in the returned product type where β_3 is applied to the effect operators $\lambda\beta_0 :: \star.\beta_0$ and $\lambda\beta_0 :: \star.\mathbf{S}$ corresponding to the respective functions used as arguments to f.

Only when we finally apply *foo*, the resulting annotations can be evaluated. We consider three simple functions that differ in their behavior.

$$
\begin{aligned}
&bar_1 : \forall a_2 :: \star.\forall a_1 :: \star \Rightarrow \star.(\forall a0 :: \star.\mathrm{int}\langle a0\rangle \to \mathrm{int}\langle a_1\ a0\rangle)\langle a_2\rangle \to \mathrm{int}\langle \mathbf{S}\rangle\ \&\ \mathbf{S}\\
&bar_1 = \Lambda a_2 :: \star.\Lambda a_1 :: \star \Rightarrow \star.\lambda f : \cdots .0\\
&bar_2 : \forall a_2 :: \star.\forall a_1 :: \star \Rightarrow \star.(\forall a0 :: \star.\mathrm{int}\langle a0\rangle \to \mathrm{int}\langle a_1\ a0\rangle)\langle a_2\rangle \to \mathrm{int}\langle a_1\ \mathbf{S} \sqcup a_2\rangle\ \&\ \mathbf{S}\\
&bar_2 = \Lambda a_2 :: \star.\Lambda a_1 :: \star \Rightarrow \star.\lambda f : \cdots .f\ 0\\
&bar_3 : \forall a_2 :: \star.\forall a_1 :: \star \Rightarrow \star.(\forall a0 :: \star.\mathrm{int}\langle a0\rangle \to \mathrm{int}\langle a_1\ a0\rangle)\langle a_2\rangle \to \mathrm{int}\langle a_1\ \mathbf{D} \sqcup a_2\rangle\ \&\ \mathbf{S}\\
&bar_3 = \Lambda a_2 :: \star.\Lambda a_1 :: \star \Rightarrow \star.\lambda f : \cdots .f\ (\mathrm{ann}_{\mathbf{D}}(0))
\end{aligned}
$$

For *foo* bar_1 we obtain the type *foo* $bar_1 : \mathrm{int}\langle\mathbf{S}\rangle \times \mathrm{int}\langle\mathbf{S}\rangle\ \&\ \mathbf{S}$: the effect operator β_3 is instantiated to the constant function $\lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\mathbf{S}$. Therefore, the components are annotated with $\mathbf{S}$ regardless of the arguments to β_3.

We obtain the same type for bar_2: *foo* $bar_2 : \mathrm{int}\langle\mathbf{S}\rangle \times \mathrm{int}\langle\mathbf{S}\rangle\ \&\ \mathbf{S}$. Even though $\beta_3 = \lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\beta_1\ \mathbf{S} \sqcup \beta_2$ makes use of its arguments, the particular effect operators it is applied to still cause the result to be static. We have

$$
\begin{aligned}
&(\lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\beta_1\ \mathbf{S} \sqcup \beta_2)\ \mathbf{S}\ (\lambda\beta :: \star.\beta) = (\lambda\beta :: \star.\beta)\ \mathbf{S} \sqcup \mathbf{S} = \mathbf{S}\ \textit{and}\\
&(\lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\beta_1\ \mathbf{S} \sqcup \beta_2)\ \mathbf{S}\ (\lambda\beta :: \star.\mathbf{S}) = (\lambda\beta :: \star.\mathbf{S})\ \mathbf{S} \sqcup \mathbf{S} = \mathbf{S}
\end{aligned}
$$

For bar_3 we obtain *foo* $bar_3 : \mathrm{int}\langle\mathbf{D}\rangle \times \mathrm{int}\langle\mathbf{S}\rangle\ \&\ \mathbf{S}$. In this case, $\beta_3 = \lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\beta_1\ \mathbf{D} \sqcup \beta_2$, because bar_3 applies its argument to a value with dynamic binding time. This causes the first component of the returned pair to be deemed dynamic as well. This becomes evident when we reduce the resulting annotation:

$$(\lambda\beta_2 :: \star.\lambda\beta_1 :: \star \Rightarrow \star.\beta_1\ \mathbf{D} \sqcup \beta_2)\ \mathbf{S}\ (\lambda\beta :: \star.\beta) = (\lambda\beta :: \star.\beta)\ \mathbf{D} \sqcup \mathbf{S} = \mathbf{D}$$

where $\lambda\beta :: \star.\beta$ is the effect operator belonging to the identity function passed to bar_3.

On the other hand, in the second component bar_3 is applied to a constant function. Thus, regardless of the argument's dynamic binding time, the resulting binding time is static.

For comparison, the constraint-based rank-1 system [29] assigns the following type to *foo*:

$$
\begin{aligned}
&\forall\beta_1 \mathinner{..} \beta_8 :: \star.(\beta_1 \sqsubseteq \beta_2, \mathbf{S} \sqsubseteq \beta_2, \beta_3 \sqsubseteq \beta_2, \beta_5 \sqsubseteq \beta_4, \beta_4 \sqsubseteq \beta_6, \beta_4 \sqsubseteq \beta_7)\\
&\quad \Rightarrow ((\mathrm{int}\langle\beta_1\rangle \to \mathrm{int}\langle\beta_2\rangle)\langle\beta_3\rangle \to \mathrm{int}\langle\beta_4\rangle)\langle\beta_5\rangle \to (\mathrm{int}\langle\beta_6\rangle \times \mathrm{int}\langle\beta_7\rangle)\langle\beta_8\rangle
\end{aligned}
$$

The increased precision becomes visible in the example of bar_3:

$$\forall\beta_1\ \beta_2\ \beta_3\ \beta_4 :: \star.(\beta_3 \sqsubseteq \beta_4, \beta_2 \sqsubseteq \beta_4, \mathbf{D} \sqsubseteq \beta_1) \Rightarrow (\mathrm{int}\langle\beta_1\rangle \to \mathrm{int}\langle\beta_2\rangle)\langle\beta_3\rangle \to \mathrm{int}\langle\beta_4\rangle$$

For determining the type of *foo* bar_3, we instantiate the latter accordingly resulting in the following constraints in particular: $\mathbf{D} \sqsubseteq \beta_1$, $\beta_1 \sqsubseteq \beta_2$, $\beta_2 \sqsubseteq \beta_4$,

$\beta_4 \sqsubseteq \beta_6$ and $\beta_4 \sqsubseteq \beta_7$. The resulting type therefore is $\mathrm{int}\langle\mathbf{D}\rangle \times \mathrm{int}\langle\mathbf{D}\rangle$ and not $\mathrm{int}\langle\mathbf{D}\rangle \times \mathrm{int}\langle\mathbf{S}\rangle$ as before.

To further illustrate our support for polymorphic recusion, we can formulate the *gcd* algorithm in our language as follows:

$$\begin{array}{l}\mu gcd : \mathrm{int} \to \mathrm{int} \to \mathrm{int}.\lambda a : \mathrm{int}.\lambda b : \mathrm{int}.\\ \quad (\mathbf{if}\ eq\ a\ b\ \mathbf{then}\ a\ \mathbf{else}\ \mathbf{if}\ gt\ a\ b\ \mathbf{then}\ gcd\ (minus\ a\ b)\ b\ \mathbf{else}\ gcd\ a\ (minus\ b\ a))\end{array}$$

where we assume the context is populated with the following primitive functions with the given annotated types and dependency terms:

$$\begin{array}{ll}minus & : \forall\beta_1.\mathrm{int}\langle\beta_1\rangle \to (\forall\beta_2.\mathrm{int}\langle\beta_2\rangle \to \mathrm{int}\langle\beta_1 \sqcup \beta_2\rangle)\langle\bot\rangle \mathbin{\&} \bot\\ eq & : \forall\beta_1.\mathrm{int}\langle\beta_1\rangle \to (\forall\beta_2.\mathrm{int}\langle\beta_2\rangle \to \mathrm{bool}\langle\beta_1 \sqcup \beta_2\rangle)\langle\bot\rangle \mathbin{\&} \bot\\ gt & : \forall\beta_1.\mathrm{int}\langle\beta_1\rangle \to (\forall\beta_2.\mathrm{int}\langle\beta_2\rangle \to \mathrm{bool}\langle\beta_1 \sqcup \beta_2\rangle)\langle\bot\rangle \mathbin{\&} \bot\end{array}$$

In one branch of the inner if-expression, *gcd*'s first parameter is instantiated with the join of the annotations of both arguments (due to the type signature of *minus*). In the other branch, this is the case for the second parameter of *gcd*.

Again, this becomes visible in the annotated type signature and the elaborated term.

$$\begin{array}{l}gcd : \forall\beta_1 :: \star.\mathrm{int}\langle\beta_1\rangle \to (\forall\beta_2 :: \star.\mathrm{int}\langle\beta_2\rangle \to \mathrm{int}\langle\beta_1 \sqcup \beta_2\rangle)\langle\bot\rangle \mathbin{\&} \bot\\ gcd = \mu gcd : \forall\beta_1 :: \star.\mathrm{int}\langle\beta_1\rangle \to (\forall\beta_2 :: \star.\mathrm{int}\langle\beta_2\rangle \to \mathrm{int}\langle\beta_1 \sqcup \beta_2\rangle)\langle\bot\rangle \mathbin{\&} \bot.\\ \quad \Lambda\beta_1 :: \star.\lambda a : \mathrm{int} \mathbin{\&} \beta_1.\Lambda\beta_2 :: \star.\lambda b : \mathrm{int} \mathbin{\&} \beta_1.\\ \quad\quad (\mathbf{if}\ eq\ \langle\beta_1\rangle\ a\ \langle\beta_2\rangle\ b\ \mathbf{then}\ a\ \mathbf{else}\\ \quad\quad\quad \mathbf{if}\ gt\ \langle\beta_1\rangle\ a\ \langle\beta_2\rangle\ b\ \mathbf{then}\ gcd\ \langle\beta_1 \sqcup \beta_2\rangle\ (minus\ \langle\beta_1\rangle\ a\ \langle\beta_2\rangle\ b)\ \langle\beta_2\rangle\ b\\ \quad\quad\quad\quad\quad\quad\quad\quad \mathbf{else}\ \ gcd\ \langle\beta_1\rangle\ a\ \langle\beta_1 \sqcup \beta_2\rangle\ (minus\ \langle\beta_2\rangle\ b\ \langle\beta_1\rangle\ a))\end{array}$$